Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency

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Abstract

This paper develops two layers of Observer Patch Holography: observer clocks, causal order and spatial geometry read from the same repair process, and Einstein dynamics as the coarse-scale response of that process. Normalized repair response on the twelve-port carrier gives a rank-three metric quotient and, with an adjoined scalar line, a 1+3 Lorentz carrier. Conservative Fibonacci record populations with complete reads inside a shrinking radius and a declared model clock have causal orders and normalized counts converging to flat $1+3$ spacetime; the ordering fraction of interior causal diamonds tends to $1/10$, and interval counts reconstruct proper-time ratios. Boost invariance selects the light cone of the declared quadratic form among closed convex pointed cones. Modular covariance on caps gives Lorentz-frame kinematics, and finite null tomography reconstructs a symmetric tensor from null projections. The gravitational result is an implication: null balance on nine design directions, with Ward and Bianchi identities, gives the Einstein equation up to a cosmological constant, and Jacobson's entanglement-equilibrium argument supplies null balance under six realization premises. Realizing those premises on one refinement tower of the executable model is the open join. The static radial geometry of the executable construction fails null balance for its own scalar excitation, so a responsive geometry is required.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geo-

metric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model

in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results supply finite components for the larger construction. The finite campaign itself does not freeze a carrier/refinement class; that boundary belongs to the separate executable program contract. Its observer services must be constructed or counted on the declared source, and any physical use needs the actual accessible algebra and implemented clock. A defect sector or controlled lifetime also needs its own specification outside the excluded commuting-charge class. The finite campaign adds no independent physical comparison or whole-program exclusion.

Same-action finite evidence. A supplied three-dimensional Weyl/contact CAR action has one isolated two-fermion bound band whose wavepacket moves, carries the conserved CAR number charge and has complete contact/scattering spectral records. Its positive contact section makes the band topologically trivial. An exact rational curvature enclosure disagrees with the rest/kinetic

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

mass relation using the same speed parameter; these are coordinate energies and masses, not identified particle species. One invariant pair and dark-reference subspace is attached by a supplied isometry to a native two-level corner. Its record-derived modular clock and complete herald, continuation and energy records are executed with a finite unconditional error bound. This does not encode the whole spatial field.

On the same local action, signed CAR regional restrictions, exact local energy conservation and finite quantum Gibbs/entropy identities have explicit witnesses. They retain all local number sectors, references and detector outcomes. Regional bare-energy modular identification fails, and the contact interaction leaks from any proper Weyl momentum pocket into its complement. Finite thermal positivity and KMS identities do not remove that coupling or establish a chiral continuum. These results reduce specific finite obligations in the existing premise map; physical geometry, Lorentz stress, common normalization and integrated acceptance are not established. Five construction targets organize the current work: compiled reference observations, particle identity, interacting local fields, conditional mass attachments and actual-data comparison. Common physical energy, clocks and geometric theorem premises remain cross-cutting joins.

A further complete-band execution uses the native record-derived rate with counted local field controls. Its unconditional error bound is below 0.037991 on $|s| \leq 1$, conditional on a supplied global field Gibbs reference. Finite independent copies of the original native reference cannot furnish that same field clock: their integer modular-gap ratios conflict with an exactly bounded noninteger pair gap. Directly on the native carrier, a faithful coherence has an exactly reference-matched exterior through every raw history and a finite contrast lifetime, but zero modular-energy excess and a fixed coherent core. The exact-code exterior and clock restrictions prevent combining these separate properties into an established moving particle.

The fair and endpoint-biased actively corrected memories have separate exact core laws. Positive coin rescaling transfers the sharp-code classification to the biased law, extending its optimal asymptotic reset bound to arbitrary adaptive exact code choices. A separately checked variant removes identity recoveries while preserving the admitted channel. Contact-only field protocols also admit a seven-state cyclic reduction in the declared side-four zero-momentum fiber. These simplifications reduce finite service or evaluation costs. A clock on another factor composes with the memory, but supplies no physical mass or propagation relation.

Complete native histories admit exact scalar-weighted partial-map evaluation, reducing host storage while retaining every raw outcome and the same source costs. A centered spectral bound reduces the full-field clock schedule from 32 to 21 split steps and from 26,238,982 to 17,219,334 supplied source pulses, with complete average half-diamond bound below 0.037991 on $|s| \leq 1$. The original schedule and preparation premises are retained. On the fixed two-particle carrier, an added cN is scalar and cannot determine energy relative to a vacuum sector. Sector-preserving histories with the same block-diagonal preparation likewise cannot identify relative sector offsets, even with a separate clock. An independently established Gibbs preparation law or coherent comparisons between sectors would be additional input. For the stated regional number-preserving field algebra, boundary hopping also excludes an exact retraction preserving the global Gibbs reference: the rational witness is $\|[N_A, H]\|_{\mathrm{HS}}^2 = 117/16$. Regional state restriction and clocks from the actual reduced density remain valid. These finite results sharpen the attachment conditions without establishing a physical mass, continuum or canonical premise discharge.

The finite matrix argument also removes a redundant hypothesis for partition pinching. For $L = \sum_c \gamma_c(E_c - I)$, trace orthogonality gives $\mathrm{Tr}(X^*LX) = -\sum_c \gamma_c \|X - E_cX\|_{\mathrm{HS}}^2$. Nonnegative rates preclude cancellation; strictly positive rates identify $\ker L$ with the intersection of the partition commutants. The Lean corollary requires neither commuting pinchings nor a separate no-cancellation premise. It does not identify the native repair generator with this class or select a physical rate

and clock.

The same supplied Weyl/contact action extends to the complete CAR Fock algebra, with source execution on every number sector. Overlapping CAR supports give a volume-linear Strang error coefficient. On the 2^3-site box at $c = 1/8$, $U = 1$ and model duration $1/32$, the coefficient $629/256$ permits operator error below 10^{-4} in one split step and 10,560 source pulses; the norm-sum bound requires ten steps and 105,600 pulses. An integrated factorial commutator tail and nested buffers bound complete local-history half-diamond error by $\sum_i(\delta_i/2 + \eta_i)$, retaining every outcome and arbitrary references. The schedule is independent of ambient volume; the initial restricted state, local interventions, restriction and readout services remain supplied; their service costs are outside this schedule. An explicit local commutator estimate makes the fixed-spacing thermodynamic dynamics norm Cauchy on local observables; the contact interaction survives in its local derivation. This removes the two-particle and finite-volume restrictions for that mathematical lattice field, without deriving a chiral spatial-continuum limit or selecting the action from passive native repair. External link coupling obeys exact gauge covariance and number-current Ward identities. Its contact-pair Schur complement carries charge two, and its energy-dependent norm metric joins the effective current and contact residue to the same full eigenstate.

On the native binary carrier, an exact noncommutative certificate reduces the original 235 generators and 495 face/square relators to $\langle g \mid g^2 \rangle$ by 234 elementary Tietze eliminations. The surviving generator has odd ordered-cover parity, establishing $\pi_1 = \mathbb{Z}/2$ and the positive/sign character catalogue on this deformation domain. Both characters have complete source-operated instruments and coherent readouts; physical sign selection, spin and species are not assigned.

For either character, a faithful-reference Dirichlet identity forces every universally conserved source observable to be scalar under positive all-seam exploration, including classical record feedback. An exact operator-space involution gives $\Phi_- = S\Phi_+S$ separately on every complete branch, transferring spectra and GNS bounds for diagonal references. This isometry is nonpositive: a three-address pure state yields expectation $-1/3$ after the transformation. It therefore supplies no physical channel or gauge equivalence. A negative-lift step includes its declared phase service; equal native-attempt cost or physical duration is not asserted. Joint feedback charges and actively protected memories retain their distinct contracts.

The supplied support tower with $12, 42, 162, 642$ vertices admits rational chain maps C, S and a homotopy H satisfying $CS = I$, $dH + Hd = I - SC$ and $CH = HS = H^2 = 0$. They commute with all sixty proper actions and compose on all six tower intervals. Midpoint averaging avoids the obstruction to an equivariant endpoint-valued coarsening. This realizes rational homology compatibility, with no change to the source endpoint process and no quantum channel, selected port attachment, responsive geometry or infinite refinement limit.

On one or two two-component sites, the complete-Fock Gibbs reference $\rho = e^{-\beta H}/Z$ is prepared by a source word for its canonical purification, retaining the purifier and every first-failure export branch. At two sites with $\beta = 1$, $c = 1/8$, $U = 1$, preparation uses 10,249 pulses and 20,498 load/export service events. The field marginal is faithful; the joint purification is pure. The same action gives a nonzero Wick defect and the noncommuting identity $\Delta S = \beta\Delta E - D(\sigma\|\rho)$ with a quadratic remainder bound. Signed local CAR restrictions compose and supply faithful factor references for the Petz-transpose observation tower. Its bounded adjoint recovery has a native source compilation: Gray-row QR constructs the Stinespring isometry with every phase, initialized blank and environment retained. The two-site recovery uses 348,697 pulses; regional-reference preparation, recovery and a complete occupation pointer use 350,258 pulses on nine initialized bits. Every complement export branch is retained incoherently by its literal prefix. Same-reference compositions retain both environments. This supplies neither arbitrary lost-state recovery nor regional retraction or reference selection. The separate 8,128-dimensional field reference is not

prepared, and the physical vacuum, temperature and clock remain unidentified. No G1–G6 or canonical physical premise is discharged.

Declared particle response and comparison. The same two-site action at $c = 1/8$, $U = 1$ has a unique half-filled ground of the supplied generator $K = H - N/2$. Its lowest charge -1 hole and $+1$ particle are odd-parity internal $SU(2)$ doublets with Casimir $3/4$, common gap $\Delta = \sqrt{17}/8 - 1/16 \simeq 0.4528882032$ and local probe residue 0.3106339063. An explicit particle–hole symmetry intertwines the two spectral projectors. A lower neutral triplet gap is retained. Four complete source Ramsey settings reconstruct their addition/removal correlations at Hamiltonian model duration $1/4$, using 52,484 pulses including the four preparations. All pointer outcomes, purifiers and first-export failures remain present. This pure ground is distinct from the faithful thermal reference; internal spin and finite frequencies do not establish Lorentz spin, spatial dispersion or a topological defect.

Conditional electron-like and positron-like assignments use one CODATA 2022 electron-mass calibration, giving both mass energies 0.51099895069 MeV with fully correlated calibration error. Neither absolute mass is independently predicted. The zero normalized mass difference satisfies the PDG 2025 conditional positronium bound 8×10^{-9} at 90% confidence, under its stated Rydberg assumption. Exact characteristic polynomials also certify the complete signed charge-one frequency catalogue: none of its 672 ordered/sign assignments fits the electron, muon and tau ratio windows with arbitrary common chemical potential and positive scale. This excludes that finite attachment only. The fixed spectral-alpha and ACT comparisons are retrospective. No result establishes a physical particle dictionary, adds an eligible integrated physical comparison or confirms an OPH universe.

For the separate complete two-fermion vacuum sector, the rank-one contact T operator is $\tau(z) = U/(1 - U g_0(z))$ and obeys the optical identity $-\operatorname{Im}\tau = |\tau|^2(-\operatorname{Im} g_0)$. Its bound-pair pole and contact residue belong to the same resolvent. A complex-strip alias estimate gives uniform volume control at fixed positive broadening: at $L = 128$, $\eta = 1/4$, $c = 1/8$, $U = 1$, the resolvent and T errors are below 1.295×10^{-7} and 3.236×10^{-6}. A separate local source interaction echo uses 22,531 pulses with complete outputs. Its exact signal lies in $[0.046942, 0.046943]$ and remains above 0.0361998 after the $11/1024$ ideal-word error bound; the free control is zero. These are fixed-spacing, supplied-action results, without an on-shell physical S matrix, continuum limit or transfer of the vacuum-sector reference to the half-filled particle candidates.

The supplied dictionary is executable on all sixteen dimer states: $Q = e(N - 2I)$, $S_{\text{phys}} = \hbar S_{\text{internal}}$ and $H_{\text{phys}} = \kappa(K - E_0 I)$, with the single measured mass setting κ. Exact SI constants then assign Hamiltonian time unit $5.83360161713 \times 10^{-22}$ s and lattice length unit $1.39909581423 \times 10^{-12}$ m. The latter is 3.6231 reduced Compton lengths. These are supplied correspondences, without a native incident/modular clock intertwiner. Complete charged-candidate electric response satisfies continuity and the double-commutator sum rule; the dipole/current weights are $1/4$ and $1/256$. A phase on the open bond is pure gauge, so no magnetic moment is measured. The bare contact Fourier kernel differs from the Coulomb kernel; this excludes their direct identification without excluding emergent electromagnetism.

A separately declared periodic four-site extension retains all 256 Fock states, explicit signed translation, complete Fourier-probe residues and free doublers. Its charged gap at $k = 0, \pi$ is 0.4175457854; at $k = \pm\pi/2$ the two spin-resolved gaps are 0.4723570583 and 0.5944851107. They fail the declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$, both with the dimer mass anchor and with the ring's own zero-momentum gap. Independent Jordan–Wigner reconstruction checks the hopping factor, periodic signs and momentum labels. The finite-volume rest-gap shift is separate from this same-volume diagnostic. A lowest-envelope fit gives velocity 0.2208544942, but only two

distinct $\sin^2 k$ values make a two-parameter fit insufficient to test dispersion. No renormalized continuum Lorentz behavior is excluded. These results reject the fixed finite dictionary benchmark and leave physical electron/positron and QED identification unestablished.

Complete field interfaces. On the same complete one/two-site field algebra, the faithful Gibbs reference fixes the modular convention without a separate species scale:

$$\sigma_s(A) = \rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A).$$

The source word therefore implements the Schrödinger unitary $e^{i\beta s H}$. Complete input-column certificates cover arbitrary untouched references. Source-prepared histories retain the purifier, both occupation outputs and every terminating export branch. The two-site prepared read uses 11,660 pulses; the occupation read disturbs the reference. Global modular implementation is consequently distinct from regional modular covariance, stationary observed histories and a calibrated physical clock.

The local-energy decomposition applies to every Fock sector: $H = \sum_x h_x$, $J_{xy} = i[h_x, h_y] = -J_{yx}$ and, in Hamiltonian model coordinate t,

$$\frac{dh_x}{dt} = i[H, h_x] = -\sum_y J_{xy}.$$

Half-bond energy assignment retains interaction contributions and nonzero distance-two energy currents. For an occupation projector n, the complete unread measurement changes mean energy by $-\operatorname{Tr}(\rho[n,[n,H]])$. On the source-prepared two-site reference at $\beta = 1$, $c = 1/8$, $U = 1$, the measured increase is 0.00181249 in model energy units. Export failures retain their probability; no field Hamiltonian is assigned to the distinct abort spaces. This establishes finite field-energy bookkeeping, not a physical maintenance-to-rest-energy law or gravitational stress tensor. For $K = H - \mu N$, the local grand-canonical energy is $k_x = h_x - \mu n_x$, with current $J^K = J^H - \mu J^N$. A charge-changing probe obeys $\Delta K = \Delta H - \mu \Delta N$. At the centered dimer point, equal particle/hole K gaps of 0.4528882032 therefore coexist with H changes of 0.9528882032 and −0.0471117968. This chemical-potential term cannot be omitted when comparing charged excitations with field-energy accounting.

Record feedback also has a complete finite instrument realization. A first occupation output chooses the sign of the next field evolution, followed by a second read. All four joint outputs and prefix-labeled export failures remain present. If the first ideal-word error is ϵ_0 and the two conditional continuation errors are $\epsilon_{10}, \epsilon_{11}$, channel telescoping bounds the complete history half-diamond error by $\min\{1, \epsilon_0 + \max(\epsilon_{10}, \epsilon_{11})\}$. The bound includes arbitrary reference systems and does not divide by an outcome probability. The state preparation, initialized pointers, memory and coherent control services retain their declared costs.

A separate periodic four-site calculation tests a genuine loop phase ϕ, retaining every one of its 256 Fock states. With $J = \partial_\phi H$ and $D = \partial_\phi^2 H$, the isolated ground and eigenvalues of $K = H - (U/2)N$ at fixed chemical potential give

$$E_0' = \langle J \rangle, \qquad E_0'' = \langle D \rangle - 2 \sum_{n \neq 0} \frac{|\langle n|J|0\rangle|^2}{E_n - E_0}.$$

The current and curvature are invariant when the bond phases are redistributed by a gauge transformation, although the two curvature terms separately change. At $\phi = 0.4$, $c = 1/8$, $U = 1$, the current is −0.0003834351 and the curvature is −0.0009061439 in model units. Free and atomic

degenerate-ground controls are excluded from the isolated-ground formula. This is a finite operator calculation, with no additional source execution on the ring or physical photon identification.

Finally, an exact symbolic calculation excludes the entire repulsive centered-dimer linear mass dictionary, rather than a single parameter point. For every $t = c/2 > 0$, $U \geq 0$ and $\mu = U/2$, put $d = \sqrt{U^2 + 16t^2}$. Each charge-one sector above the unique half-filled ground has only two doublets, with gaps $g_\pm = d/2 \pm t$. Since $d \geq 4t$, $g_- \geq t > 0$ and $3g_- - g_+ = d - 4t \geq 0$, so $1 < g_+/g_- \leq 3$. The higher local residue vanishes at $U = 0$; retaining that dark level makes the exclusion conservative. No common positive linear gap-to-mass conversion can produce the charged-lepton hierarchy or three distinct same-charge families on this domain. Changing chemical potential can evade the ratio bound while retaining only two doublets per charge. Larger carriers, other actions and different mass observables require their own derivation and comparison.

This paper's hypothesis and place in the tower

This paper sits at layers 4 and 6 of the tower. Its hypothesis is that the clocks, causal order and spatial geometry an observer reads are produced by the same native repair process that runs the executable observer [26], and that Einstein dynamics is the coarse-scale response of that process to its own matter.

The paper proves finite and analytic theorems under displayed premises. The rank-three source Gram quotient and its $1 + 3$ Lorentz carrier are exact. On supplied conservative Fibonacci populations with a complete local read law and model clock, the causal order and counts converge to flat $1 + 3$ spacetime and a count clock reads proper-time ratios. Cap modular covariance gives Lorentz kinematics on the support-visible geometric subnet. The gravitational theorem is an implication. Null balance on nine design directions, together with Ward and Bianchi identities, gives $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$ (Figure 1); the null-balance step follows from the following realization premises on one refinement tower:

(G1) a Lorentzian event and geometry domain with the required locality, differentiability, causal and count/volume limits;

(G2) geometrically normalized cap modular flow whose directional charges define the same matter stress tensor used in propagation and interactions;

(G3) its Ward identity, independent of the Einstein equation;

(G4) the edge-center entropy split and area normalization, the modular first law, and generalized-entropy stationarity at fixed proper volume;

(G5) controlled small-ball remainders and sufficient observer and null directions;

(G6) universal coupling, a common vacuum reference and one positive Newton constant.

Realizing (G1)–(G6) on one refinement tower of the executable model is the open join of this paper. The following facts bound it. The static radial product, the one geometry tested on the executable construction, has vanishing radial null Einstein contractions while an admitted scalar excitation has positive relative null stress, so it cannot balance both; a responsive geometry is required. Stationary Markov or record entropy is not the fixed-volume generalized-entropy variation that (G4) requires (Section C.5). The native screen generator A has spectral dimension two, and the $1 + 3$ scalar description uses a declared radial extension. The Fibonacci population, read law and model clock of the causal limit are premises until the native process produces them.

A proof that no responsive geometry on the native process balances the null stress of its own admitted excitations, or that its cap modular flow or record entropy cannot meet (G2) or (G4) on

any refinement tower, would refute this gravitational branch. A refinement tower on which every component premise holds while the composed Einstein relation fails would contradict the stated implication.

Derivation map

Normalized repair response gives one positive rank-three source Gram quotient E. Signed cumulative records and conservative seam words embed densely in its Euclidean completion. Its unit directions form S^2; adjoining a scalar line gives $\mathbb{R} \oplus E$ with quadratic form $t^2 - \|x\|^2$. This connects the source metric to a four-dimensional Lorentz carrier without selecting a physical event clock. The conditional implications on this carrier are

$$\begin{aligned}
\text{nonzero closed convex pointed cone}\\ \text{finite irreducible operational rotations}\\ \text{one continuous boost direction} &\Longrightarrow \text{Lorentz displacement cone}\\ &\qquad \text{up to time orientation},\\
\text{conservative Fibonacci populations}\\ \text{complete local reads and } \Delta{=}a/c &\Longrightarrow \text{flat causal-order limit}\\ &\qquad \text{raw count–volume convergence},\\
\text{controlled order/count limit}\\ \text{accessible interval records}\\ \text{one reference interval} &\Longrightarrow \text{proper-time ratios from counts},\\
\text{positive translations and finite null tomography} &\Longrightarrow \text{conditional conserved } \langle T_{ab}\rangle,\\
\text{physical stress and universal coupling}\\ \text{entropy, small-ball and tensor-curvature limits} &\Longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab}\rangle.
\end{aligned}$$

These implications have distinct hypotheses. Operational boost covariance acts on admissible histories and interventions; an algebraic action on the celestial sphere does not establish it. Finite proper carrier rotations and one continuous boost of the declared quadratic form select its light cone with unit speed in the scalar coordinate (Theorem B.9). The signature and the time axis are declared, so the theorem selects the invariant cone of a chosen Lorentz form rather than deriving Lorentz structure.

Causal order and counts. Put $\varphi = (1+\sqrt{5})/2$ and $L = 2/\sqrt{\varphi+2}$. For consecutive Fibonacci numbers $q = F_n$ and $p = F_{n+1}$, $n \geq 4$, the q^3 spatial records have a source-metric assignment to equal-volume cells with error $H_q = 2\sqrt{3}L/q$. Each event reads every preceding-layer record inside radius $a_q = L/\sqrt{q}$, including its own preceding record, and the model layer duration is $\Delta_q = a_q/c$, with $c > 0$. Segment subdivision bounds the reachable radius between $c(1-2h_q/a_q)T$ and cT; since $h_q \leq H_q$ and $H_q/a_q \to 0$, these cones agree in the limit away from their null boundary. Normalized raw counts converge to $dt\, d^3x$ volume (Proposition B.4), and on fixed interior timelike intervals the ordering fraction tends to $1/10$. The finite path construction, the layered order, the count-clock enclosure, the generated-interval and strict-pair count limits and the $1/10$ limit for diamonds inside the sampling window are formalized in Lean; the general moving-tip volume-error estimate is an analytic proof. The family is a discretization of flat $1+3$ space, with the ball read as a light cone and a layer as a time slice, so its limit theorems confirm the discretization rather than select the dimension: the spatial dimension three comes from the rank-three carrier, and the time axis is adjoined. Under the same count limit, the observer readout $(|I|/|J|)^{1/4}$ converges to the proper-time ratio of two interior timelike diamonds; it reads accessible ancestry and counts, with a reference interval fixing units (Theorem B.10, Proposition B.11). Finite readings of the family are in Section B.5.

Exact order embeddings form a separate finite interface. For a source spatial readback and positive height scale, a parent-edge speed bound maps authenticated ancestry into the future cone;

equal-height spatial separation and strict spacelike separation of increasing-height nonancestors give the converse, hence an injective order embedding. The source height is the attained longest authenticated-parent-chain length. Height is not identified with proper time, or element number with volume; a causal-set faithful embedding also requires the appropriate population density[1, 2]. Under the continuum causality hypotheses, causal order and volume determine conformal geometry and scale[14, 15].

Source metric. For the declared twelve-port repair mean, the centered equal-port response kernel has an exact spectral decomposition. Dividing by its slow response before the infinite-step limit selects a rank-three projector whose range, with the induced Gram form, is an intrinsic Euclidean carrier. Every finite-step kernel has rank eleven on the centered space and rank six on the signed sector. On the signed cumulative port-record/load module the limiting form has a three-dimensional radical on the real six-control extension; the integer module has no nonzero metric kernel, embeds densely in the quotient, and completes to the same carrier up to the unique label-preserving isometry. The source load map is onto $\mathbb{Z}^6$, with the antipodally equal loads as its kernel. Conservative seam boundaries have image

$$D_6 = \{z \in \mathbb{Z}^6 : \textstyle\sum_i z_i \equiv 0 \pmod 2\},$$

with Smith invariants $(1, 1, 1, 1, 1, 2)$; the pullback response metric makes this submodule dense in the same completion. Cumulative record addition acts simply transitively on the record carrier and by isometries on its completion. The sixty directed seam labels form one proper carrier-rotation orbit. The thirteen finite covariance and isometry table identities of the proper-action subpacket are checked with Lean's `native_decide`; they trust the native compiler and runtime as well as the kernel and are not kernel-only receipts. The source metric does not choose a preferred Cartesian frame or identify record addition with physical translation.

Einstein shape. The minimal source-direction Einstein-shape theorem places supplied symmetric geometry and stress fields on the same finite event type. It asks for balance on nine fixed algebraic source directions, four discrete step maps, Ward/Bianchi identities and step connectedness. Exact tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and

$$\mathsf{geometry}_{ab} = \kappa\, \mathsf{stress}_{ab} + \Lambda \eta_{ab}.$$

The nine-set is algebraic, not selected by provenance or invariant as a set under Lorentz transformations. Identifying geometry with smooth curvature and stress with physical stress-energy requires the source-causal continuum, count–volume, stress, small-ball, coupling, scale and remainder receipts on one family, together with either a tensor-curvature reconstruction converging to the Einstein tensor or the independent continuum small-ball/null-balance identification. Scalar-curvature convergence alone is diagnostic. In the machine-checked continuum composition `composedEinsteinBranch` the premise is the null-contracted Einstein relation itself, or nine of its null components; the composition adds the constancy of the metric residue, the vacuum reference fixing Λ and the $8\pi G$ label. The small-ball step is Jacobson's entanglement-equilibrium argument, each of whose inputs enters as a receipt.

The Lorentz arrow composes two independent inputs on one refinement tower: the finite cap-normal Bisognano–Wichmann certificate (cap-normal density and nondegeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π-KMS comparison, and wrong-normalization separation), and an independently produced complete multiresolution algebra-state

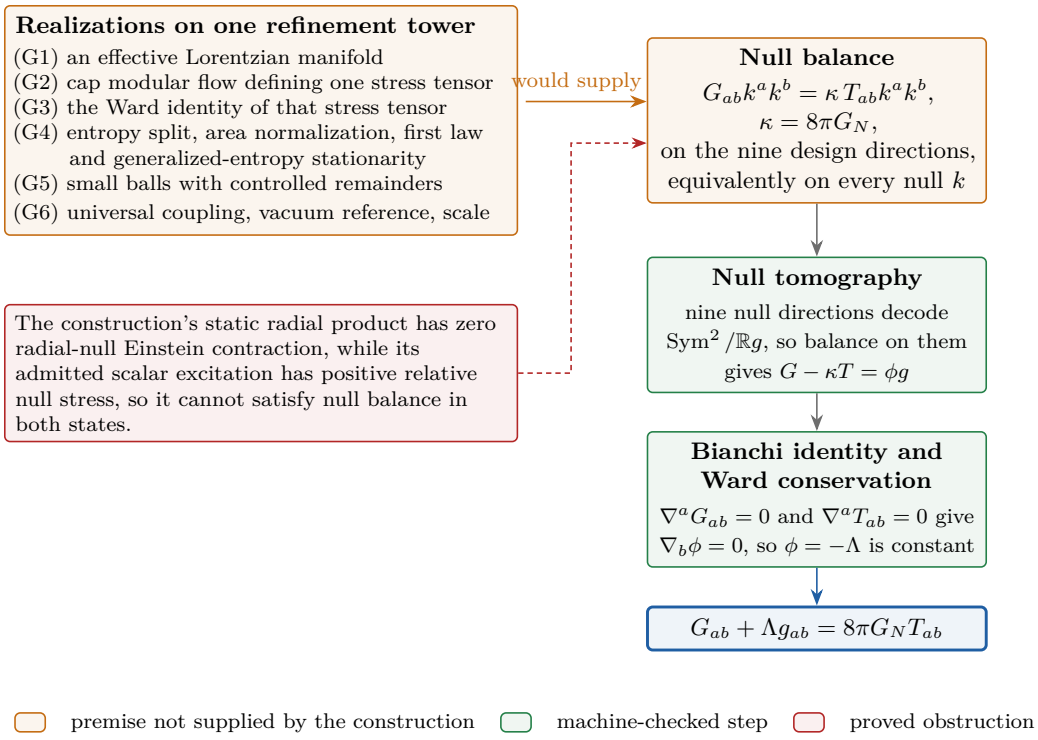


Figure 1: The Einstein equation as tensor completion of a realized null balance. Balance on the nine design directions is equivalent to balance on every null vector, and the tensor step is machine-checked from that premise. Realizations (G1) to (G6) on one refinement tower would supply it; the executable construction's static radial product fails it for its own scalar excitation.

comparison package. Geometry does not manufacture the modular state, and bare finite consensus implies neither input. A finite boundary-circle diagnostic uses half-filled ground states of anti-periodic nearest-neighbour hopping on rings of $16, 32, 64, 128$ sites. Summing every fitting odd hopping range of the half-ring modular Hamiltonian gives median relative profile errors $0.004808, 0.001152, 0.0002850, 0.00007106$ against the conformal profile, while a fixed range-nine extraction increases from 0.0001124 at 64 sites to 0.0003381 at 128. These are finite diagnostics, not a KMS identity on a common cap algebra.

Stress and energy precursors. Stress reconstruction consumes same-source null-stress matching, locality/covariance and the weak Ward balance; finite null tomography does not supply them. Of the local-stress inputs, one finite precursor exists at one energy component: a face-local discrete Poynting law on the committed twelve-port carrier (`LocalEnergyBalance`), exact at every step of the scaled Maxwell evolution, whose face sum is the committed global energy balance. It supplies no momentum or stress components, covariance or continuum limit. The identification of the screen field energy with the modular energy that the composition consumes is read against the clock-and-energy calibration clause, whose reference is the Gibbs state of a declared energy at one declared inverse temperature β (`GibbsReferenceEnergyIdentification`). Against that clause the modular energy ledger of any state is β times its field-energy expectation plus $\log Z$, and the cap first law takes the Clausius form $S(p) - S(\tau) = \beta(\langle H\rangle_p - \langle H\rangle_\tau) - D(p\|\tau)$. A two-point reference outside the Gibbs form has equal energy expectations with unequal ledgers, so the affine reading is specific

to that clause. Nothing derives that the record reference is Gibbs and nothing selects β. The collar modular surface does not supply that inverse temperature (`CollarTemperatureReading`): its reference, with modular Hamiltonian $K = \lambda \cdot S$ on the four joint sectors, equals the Gibbs state of K/β at β for every nonzero β, and is Gibbs for an energy H at β exactly when $K = \beta H + \text{const}$.

On the committed complex, along the scaled Ampère evolution with any current, the Gauss constraint propagates exactly when the sources satisfy the continuity equation; for declared sources a field history solving both exists if and only if the initial load is neutral and the sources are conserved (`SeamChargeContinuity`). Since the closed carrier conserves total load, one hopping charge alone has a Gauss-consistent history only at zero charge, so a worldline joined to the field sector carries a neutral partner or a neutralising background. A neutral pair of crossing charges inhabits the scaled Maxwell bundle in one finite action. At $q = 1$, $h = 1/2$ the Coulomb-started packet is stationary in the field slots, but delaying either crossing changes the action by $5/12$, and both simultaneously by $5/6$; on the window $N = 1$ a second exact rational history with the same crossings is stationary under every in-window closed two-step replacement. These are declared discrete exchanges, not continuum first-variation claims.

The executable construction. On a declared radial product over its refining spherical source, the native wave action of the executable construction has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, and $g = \sqrt{-\det C}\,C^{-1}$ has signature $(-,+,+,+)$ almost everywhere. Smoothing the angular variables converges in the norm-resolvent sense, so every fixed finite menu of observer readouts is approximated by one smooth static $(1+3)$-dimensional Lorentzian metric carrying a free scalar field [26]. The axial counter read through the source frame has a three-dimensional Brownian limit with radial generator $\frac{2}{99}(\partial_r^2 + \frac{2}{r}\partial_r)$, but complete angular histories are identical across distinct radial extensions, so neither that radius nor present observations select the radial product. This description asserts no exact smooth native metric, no responsive geometry and no physical stress, and discharges none of (G2)–(G6). It is the static geometry that fails null balance for its own scalar excitation.

Kinematic and Newtonian precursors are finite identities. On the declared Hermitian Lorentz module, $p = m\,u$ has Lorentz square m^2 and $p_0^2 = m^2 + |\mathbf{p}|^2$ in every oriented chart; no coordinate is identified with a physical energy. For a supplied integer layering, steady source law, shell equidistribution and shell count cn^2 through a finite depth, exact divergence bookkeeping gives outward flux $Q/(cn^2)$ per shell edge, while on a closed finite carrier an n^2 shell count at every positive layer forces $c = 0$, so the depth cutoff is explicit.

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1 A common matter metric and its gravitational response

This section is the interface of the executed construction with the gravitational layer of the tower. The conditional scalar description supplies a concrete dictionary for comparing matter and geometry, and the same dictionary yields the static-product control that any responsive geometry must overcome. The counts, ranks and identities quoted here are retained in the receipts of the matter and common-field contracts of the public repository. Write its four-dimensional coefficient density and metric as

$$C = \mu\,\mathrm{diag}(-1, 1, a), \qquad g = \sqrt{-\det C}\,C^{-1} = N^2\,\mathrm{diag}(-1, 1, a^{-1}), \qquad N^2 = \mu\sqrt{\det a},$$

where a is a positive two-dimensional screen tensor and $\mu > 0$. These identities use the complete action density, including its measure. For the measurable native limit, smooth representatives are understood with the stated finite-menu approximation theorem and smearing transport.

1.1 Common principal symbols and finite field operators

Choose an orthonormal coframe e with $g = e^t \eta e$. Transporting Clifford matrices through its inverse gives

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I, \qquad (\gamma^\mu p_\mu)^2 = (g^{\mu\nu} p_\mu p_\nu) I.$$

The scalar wave symbol, both Weyl principal determinants and the Maxwell symbol therefore have the same characteristic cone. The Maxwell gauge direction is retained, leaving two physical polarizations on a nonzero null covector. A change of clock coordinate rescales all field generators and the reference inverse temperature together, preserving βH. This pointwise dictionary does not select the fields' dynamics or chirality.

For a finite incidence factor C_0 on the declared angular–radial product,

$$D = \begin{pmatrix} 0 & C_0^\dagger \\ C_0 & 0 \end{pmatrix}, \qquad D^2 = \mathrm{diag}(C_0^\dagger C_0, C_0 C_0^\dagger).$$

The node block is the scalar spatial operator $B = C_0^\dagger C_0$. CAR second quantization gives a conditional fermion construction with node propagation $\cos(t\sqrt{B})$ and node-to-edge propagation $-iC_0 \sin(t\sqrt{B})/\sqrt{B}$. The full finite construction retains its extra edge modes. A factor matching the finite angular frequency is also tested; its ideal joint generator is not the exact Floquet generator of the finite angular/radial split word. Their difference retains the specified approximation budget.

Completing the actual tetrahedral cone through faces and tetrahedra gives a separate full-cochain control. Its dimensions are $(13, 42, 50, 20)$, its differential ranks are $(12, 30, 20)$, and its Betti numbers are $(1, 0, 0, 0)$. The complete metric $d + d^\dagger$ construction removes the truncated cone's thirty edge-cycle zero modes, preserves its scalar action block, and retains the scalar constant. This cone is a different global domain from the angular–radial cylinder. Differential-form parity does not establish physical Weyl chirality.

The smooth product admits a spin structure, but scalar radial Dirichlet conditions do not by themselves define a self-adjoint first-order spinor operator. A tested reflecting extension squares to a Neumann component and a Dirichlet component. Spin-connection and curvature terms also enter: a curved spinor square is not a componentwise scalar Laplacian. These domain and connection hypotheses are explicit in the common-field tests.

1.2 Area, volume and the entropy variation

The same product metric gives exact induced measures

$$dA = \mu\, d^2x, \qquad dV = \mu N\, dr\, d^2x.$$

Thus the scalar action measure supplies a whole-screen area dictionary. Fixed counting measure alone does not fix physical three-volume. Radial slices in this product have equal area and are not concentric small geodesic spheres.

Native constrained maximum-entropy completion and its joint counting first law hold on their finite algebraic variation domains. They do not identify fixed cap marginal or fixed total dimension with fixed proper volume. In the exact selected-reference test, forty-five four-port caps have identical counted edge entropy and crossing-seam counts ranging from ten to eighteen. No nonzero universal constant equates that entropy to this boundary proxy. The statement concerns a finite two-dimensional boundary proxy, rather than the whole-screen area above.

The family

$$\mu(z) = \mu_0 e^{-2z/3}, \qquad a(z) = e^{2z} a_0$$

keeps proper volume fixed on the declared radial coordinate interval and decreases its boundary area. At fixed model-coordinate inverse temperature, the cavity thermal entropy also decreases. Hence the trial functional $S_{\text{thermal}} + A/(4G_N)$ is not stationary for positive G_N on these variations. This is a specified thermal-cavity comparison; it is not an identification with local vacuum entanglement or a proof that repair generates those variations.

The distinction between observable algebras matters as well. For an executed symmetric mixture of $\pm\sqrt{\epsilon}$ displacements, the unread work density has positive first-order entropy and energy derivatives at fixed reference Hamiltonian. The retained sign-plus-work density has zero entropy derivative. Both outcomes and the purification are retained. A gravitational entropy identification must specify which region algebra and which records enter its bulk term.

1.3 A fixed-product obstruction and a necessary dynamic response

Proposition 1.1 (Radial null response of the static product)**.** *For a smooth metric*

$$g = f(x,y)\,\mathrm{diag}(-1, 1, h(x,y)), \qquad f > 0,$$

with positive screen metric h, both radial null contractions of its Einstein tensor vanish. With the Gaussian reference and other sources fixed, the same geometry cannot satisfy the Einstein balances for that reference and every coherent scalar excitation at positive gravitational coupling.

Proof. The (t, r) factor has Lorentz symmetry and all metric coefficients are independent of t, r. Its Ricci components obey $R_{tt} + R_{rr} = 0$ and $R_{tr} = 0$, also verified by direct differentiation for arbitrary smooth f, h. For $k = \partial_t \pm \partial_r$, $g(k,k) = 0$, so trace and cosmological metric terms vanish and $G(k,k) = 0$. The restored angular-constant scalar sector admits

$$\phi = A \sin(\pi r/L)\cos(\pi t/L).$$

It obeys the product wave equation and the declared radial Dirichlet condition. Its action-derived coherent relative stress satisfies

$$\Delta T(k,k) = (k\phi)^2 = \frac{A^2\pi^2}{2L^2} > 0 \quad \text{at } t = 0,\ r = L/4.$$

Subtracting the reference and excited Einstein equations contradicts $\Delta G(k,k) = 0$. This uses relative stress and supplies no absolute vacuum-energy prescription. □

A coherent amplitude changes stress quadratically; this argument is not a claimed failure of the linear entropy first law. A symmetric mixture of opposite displacements supplies a first-order state direction, with its observable algebra specified separately. Independently changing compensating sources and geometries outside the static product class are outside the proposition.

For the smooth spherical extension

$$ds^2 = -2du\,dv + R(u,v)^2 d\Omega^2,$$

direct curvature calculation gives $G_{uu} = -2R_{uu}/R$. The null Einstein balance consequently requires

$$R_{uu} = -4\pi G_N R T_{uu}, \qquad \theta_u = -\frac{\theta^2}{2} - 8\pi G_N T_{uu}, \qquad \theta = \partial_u \log(4\pi R^2).$$

Here u is affine along the declared null curves. These are necessary area-response equations in this symmetry class, not a selected repair law or a solution of the full Einstein system. The alternative finite nine-null theorem also requires its independent balances; a pointwise tetrad map alone does not supply them.

The exact block, null-stress and focusing algebra is formalized in the Lean library of the public repository, and the source matrices, curvature identities, complete finite records and approximation controls are executed with the observer-particle model. Their conditional realization establishes neither observed Standard Model masses nor generalized-entropy stationarity on a physical refinement family.

2 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. On the declared screen-first global-support branch, physical data are charted by a horizon screen S^2, and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This support screen is distinct from a local carrier boundary and a federation overlap nerve. On the producer branch, S^2 enters only after the carrier-to-support map satisfies the spherical incidence, mesh, cross-ratio, normalization, and refinement receipts. A finite reference fixture without a composable seam triangle does not supply that map, since its higher-overlap condition is vacuous. The source-derived producer supplies a twelve-chart, thirty-seam, twenty-face incidence nerve, nonvacuous seam-triangle cocycles, confluent phase repair, and a controlled oriented-S^2 support limit on one declared tower. Event, modular, scale, and laboratory attachments require separate premises.

This recovered-core section develops fixed-cutoff overlap repair, collar recovery, overlap gluing and its transport obstructions, and the geometry-readout route to Lorentz frame kinematics and conditional Einstein dynamics.

2.1 Overview

Theorem, lemma and section numbers inside this recovered core, such as Theorem 4.2 or Section 2.3, follow the core's own internal numbering, not the section numbering of the surrounding paper.

Overlap consistency turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers and finite repair dynamics on the declared physical quotient (Sections 2–3). Modular flow on caps supplies Lorentz-frame kinematics and the three-dimensional observer-frame hyperboloid (Section 4). Certified read-from provenance generates a finite informational poset and its longest authenticated-parent-chain height; the rank-three source Gram quotient and a real axis define the ambient $1 + 3$ target carrier, and source height

enters only through the temporal coordinate of a scaled event placement. Positive null translations supply directional charges, finite null tomography reconstructs a symmetric tensor, and fixed-cap generalized-entropy stationarity with the stress, small-ball, coupling, vacuum and scale premises implies the Einstein equation (Section 5). Bare finite consensus stops at quotient normal forms, and the cap-normal H^3 chart stops at observer frames; neither populates a spacetime.

The selected no-G scale certificate enters only the Newton-constant readout. The source-causal continuum and Einstein theorems do not depend on the matter or sector-completion branches.

2.2 Model and Premises

2.2.1 Observers and access model

The support-local algebra-state-record reduct of an observer patch is $O_{\mathrm{red}} = (P_O, \mathcal{A}(P_O), \rho_O, R_O)$, where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O.
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a finite record algebra generated by stable internal record projectors within P_O.

An operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. Observers are internal self-reading patterns in the global state. A patch and its compatible marginal supply a geometric or algebra-state chart. An operational observer requires the interface, record, repair, and checkpoint conditions.

2.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

2.2.3 The premises A1–A3

This paper states the premises A1–A3 of the observer-overlap foundation. The flagship synthesis [25] gives the account of the theory, and the executable observer paper [26] records which of these premises the construction supplies. Branch-local hypotheses are written at the theorem that consumes them.

The support-visible BW scaling theorem handles the scaling step. The source-causal continuum and Einstein theorems use the additional collar-recovery, edge-normalization, generalized-entropy, common-domain, stress, scaling, universal-coupling, vacuum-reference, and scale-readout interfaces stated in their own hypotheses; none of those interfaces belongs to A1–A3.

The screen premise A1 below defines the screen-first support branch. The producer branch reaches the same premise only after its carrier-to-support receipts establish the global support screen.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot

replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

The closure hypothesis. The theorem premises A1–A3 state the library's hypotheses on the carrier and its response, on internal transport and on local state selection. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer

or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

Collar recoverability, generalized-entropy structure, and quantum focusing lie outside A1–A3: they are classified gravity interfaces with their own theorems, countermodels, and required inputs, consumed field by field where the Einstein composition needs them. Sector content is not selected by A1–A3: the conditional rank-45 tensor candidate does not establish physical families or a local Spin carrier, while scalar multiplicity and extra-sector completeness lack physical attachments. Each theorem below lists the declared completions it consumes.

2.2.4 Closure values and theorem-local technical data

The Newton constant of (G6) enters through one selected no-G scale certificate,

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

Dimensionless capacity and pixel coordinates fix dimensionless geometry, not an SI scale. On the de Sitter branch, with capacity $N_\star$ and pixel ratio $P_\star = a_{\mathrm{cell}}/\ell_\star^2$,

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\mathrm{cell}} = \frac{3\pi P_\star}{N_\star},$$

and these relations survive the common rescaling $\ell_\star^2 \mapsto \lambda^2 \ell_\star^2$, $a_{\mathrm{cell}} \mapsto \lambda^2 a_{\mathrm{cell}}$, $\Lambda_\star \mapsto \lambda^{-2}\Lambda_\star$. The SI area $\ell_\star^2$ and G_{SI} therefore require the scale certificate, which supplies $B_\star = \Lambda_\star N_\star$ and $\ell_\star^2 = 3\pi/B_\star$. The branch certificate used for the SI gravity row is

$$B_\star = 3.60787391468\ldots \times 10^{70}\,\mathrm{m}^{-2}, \qquad \ell_\star^2 = 2.61228030237\ldots \times 10^{-70}\,\mathrm{m}^2.$$

Rounded capacity displays such as $N_\Lambda \simeq 3.313 \times 10^{122}$ are not high-precision inputs for the Newton row; with the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the 8.4×10^{-4} level. The capacity normalization uses the de Sitter static-patch entropy [52], $N_{\mathrm{scr}} = S_{\mathrm{dS}} = A_{\mathrm{dS}}/(4\ell_P^2) = 3\pi/(\Lambda \ell_P^2)$. Its selection from the source and the global-capacity closure belong to the cosmology layer [27].

The controlled scaling step for the Lorentz, null-modular, and Einstein statements is the support-visible BW scaling theorem. Fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, and collar decomposition are cited from their preceding theorems.

Local information projection at the regulator scale. At the regulator scale ℓ_{UV}, consider the premise-A3 subbranch carrying a global-extension interface and identity-proportional local references for which the state-determining weighted local objective is represented by one finite density matrix ω. The selected state maximizes the corresponding weighted local entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\le r_0 = O(\ell_{\mathrm{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label a, where x runs over the cells of the UV lattice. The constraints are the N_{con} global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is N_{con} (plus the optional global constraints below) on every lattice, independent of the cell count.

3. Optionally, a finite number of global constraints (total energy, charge).

This is the finite-matrix subbranch needed to derive the local Gibbs form of Lemma 2.6 from premise A3 with a cutoff-independent multiplier count. The global extension and the objective representation are explicit branch premises.

Clarification (conditional maximum randomness and thermal equilibrium). premise A3 is local state selection relative to an exact reference and a complete visible constraint set. It does not state that the universe is in thermal equilibrium. Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers λ_a with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x)\rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms. The refinement-stable branch and the induced maps $R_{\ell\to L}$ below concern the homogeneous N_{con}-parameter branch under the separate optimizer-pushforward interface. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are SO(3)-invariant on S^2.

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. Assume the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. Our comparison-map convention is that U_{ij} maps chart-j column components to chart-i components; comparison products follow the chronological order of path labels. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk}, \qquad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. The defect is data of the implementer lift; writing only $\mathrm{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

Collar double-scaling hypothesis. There is one refinement family with $\delta \to 0$, $\ell_{\mathrm{UV}} \to 0$, and uniform finite-range and strong conditional Gibbs mixing constants such that

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \le c|\partial C|_{\mathrm{UV}} e^{-\delta/\xi}, \qquad \frac{\delta}{\xi} - \log|\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

The second condition controls the complete boundary-prefactored envelope. The weaker ratio $\delta/\ell_{\mathrm{UV}} \to \infty$ does not imply it. Section 2.3 gives the finite-range theorem, explicit constants, a sufficient logarithmic schedule, and the finite-receipt fields. On the exact central-interface branch the CMI is zero without this mixing premise.

Geometric-subnet BW lift. The continuum Lorentz statement is asked only on the support-visible extracted geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level on the emitted geometric cap pair and the special type-I generator form is $K_C = 2\pi B_C + Z_C$ with Z_C central. The spacetime and Einstein paper proves the support-visible BW cap automorphism only when one cofinal branch carries the finite cap-normal certificate and the complete multiresolution algebra-state comparison package. It also proves the cap-normal H^3 chart: $q(\Omega) = (1, \Omega)$, $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, with cap half-spaces but no event-position consequence. Bare confluence underdetermines topology, dimension, and modular

normalization by explicit countermodels. Separately, authenticated provenance generates the finite informational order and canonical source height. The exact rank-three source Gram quotient and an independent real axis define the ambient 1+3 Lorentz target carrier with an S^2 null boundary. After multiplication by a supplied positive `timeScale`, source height supplies only the temporal coordinate of an event placement. It is not an execution rank, intrinsic poset dimension, physical clock, or physical event time. The null-net-standardness packet supplies derived positive translations, while the source-causal continuum certificate keeps physical order, spatial placement, history-family custody, count–volume calibration, independent dimension, manifoldlikeness, topology, uniqueness, and curvature convergence explicit. The closure packets prove finite null tomography and finite entropy/MaxEnt identities; local stress, Ward conservation, generalized-entropy stationarity, smooth curvature, and the composed Einstein reading remain conditional on their named analytic and physical supplies. No source-derived family carrying the complete continuum packet is constructed (Theorems 4.3c–4.3h).

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV}, the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR}: for local operators A, B supported on regions separated by distance d,

$$\|[A(t), B]\| \leq c\|A\|\|B\| \min(|R_A|, |R_B|)e^{-(d-v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the premise-A3 branch stated at regulator scale.

> **Approximate modular covariance on the fixed-cutoff realized branch.** Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:
>
> $$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$
>
> where R^{+s} denotes the s-neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C.

This statement is a regulator-scale control; the tangent-limit package used in Theorem 4.2 is stated there with its own hypotheses.

Refinement-stable information-projection branch. Choose a family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale ℓ and IR scale L satisfying the separately declared optimizer-pushforward and refinement-tail interface. Then on the realized branch one may write

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda)),$$

for an induced map $R_{\ell\to L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell\to L}$ as a self-map of that same family is the load-bearing cross-scale premise. Reusing the density labels a does not establish it. Coarse-graining a finite-range Gibbs family generically generates interactions outside the retained density list. The closure defect, the moment-matching I-projection realizing $R_{\ell\to L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b. Exact vanishing of the defect along the realized branch remains part of the

declared optimizer-pushforward interface. Under that interface, the regulator states lie in a shared multiplier family, and the realized branch is a persistent trajectory or invariant subset under the induced refinement maps. This state-side persistence statement does not upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV}, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the edge-center (EC) theorem.

External mathematical inputs include SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015).

The Einstein route also uses the fixed-volume area-variation identity for small geodesic balls and the local quadratic-polarization argument in the tensor upgrade. The null modular bridge and small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law.

Recovered-core theorem boundary. The Lorentz, null-modular, and Einstein recovered core runs through the support-visible automorphism theorem. Fixed-cutoff statements remain regulator statements.

Full citations appear in the References.

2.2.5 Notation

- ρ_C: reduced state on cap C.
- $K_C := -\log \rho_C^{\omega}$: modular Hamiltonian of the reference state.
- B_C: geometric generator of the cap-preserving conformal dilation.
- $S_{\mathrm{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV}: UV length scale of the refined screen net.
- δ: collar width around a cap boundary.

2.3 Information-Theoretic Tools

2.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC},

$$I(A:C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A:C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

2.3.2 Approximate recovery

If $I(A:C \mid B) \leq \varepsilon$ in nats, there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

2.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \mathrm{dist}(x, \partial C) \le \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ. The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\mathrm{inv}}(R) = \tilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; it does not apply to the entire fixed-point algebra on the unreduced tensor product.

The quotient repair law acts on this overlap-invariant data: the global map is the finite quotient normal-form operator. Its termination, confluence, schedule independence, the separated cofinal inverse-limit bridge and the distributed-run certificate are proved in the consensus paper [19]. Those results establish at most a common finite quotient metric or pseudometric. Its identification with Lorentzian spacetime goes through the source-causal continuum certificate of Theorem 4.3e and the modular branch below.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_δ around a cap boundary Σ, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

and the sector-preserving collar algebra

$$A_{\mathrm{EC}}(B_\delta) := \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha}))$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\mathrm{EC}}(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ. QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the Hayden–Jozsa–Petz–Winter (HJPW) structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors b_L^α, b_R^α of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \big(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\big), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits A, B_L, B_R, D and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2\ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; The spacetime and Einstein paper states MSA formally and records this counterexample. MSA is checkable state by state: The spacetime and Einstein paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on Ab_L^α and $b_R^\alpha D$, to the existence of a ρ-preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [61]), and to blockwise vanishing one-sided mutual information $I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the local information-projection branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** The spacetime and Einstein

paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit named branch premise, so the literal Markov-modular identities hold on the declared branch without a further state hypothesis. This premise is independent of A1–A3: The spacetime and Einstein paper exhibits a finite regulator package that satisfies observer agreement, schedule-independent transactional repair, and an exactly closed refinement-channel family while its retained constraint family carries a boundary-invariant noncentral cross-cut coupling. Observer agreement and repair do not force the premise. Off the declared branch MSA is carried as an explicit hypothesis.

On the separately declared rate-controlled family obtained by pushing the optimizer forward, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and theorem supply the quantitative decay route when the approximation is kept explicit.

Lemma 2.6 (interior local information projection implies local Gibbs form). Under premise A3 on the subbranch with identity-proportional local references, state-determining aggregation equivalent to the declared entropy objective, homogeneous global-sum constraints $\langle \sum_x O_a(x) \rangle = C_a$, and the finite-dimensional regulator realization above, assume also that the constraint vector lies in the relative interior of the attainable moment set, so the selected state is faithful on the retained sector. Then the selected state has the finite-multiplier Gibbs form

$$\omega = \frac{e^{-H_{\mathrm{eff}}}}{\operatorname{Tr} e^{-H_{\mathrm{eff}}}}, \qquad H_{\mathrm{eff}} = \sum_x \sum_a \lambda_a O_a(x) + H_{\mathrm{sec}},$$

where the sum runs over UV cells x and density labels a, with one cell-independent multiplier λ_a per label. For the finite-range collar theorem, H_{sec} is either a sum of uniformly bounded finite-range charge densities or is central and fixed on the chosen superselection sector. A genuinely nonlocal noncentral term falls outside the theorem. Under this condition the effective Hamiltonian is finite-range in UV graph units, and the multiplier count matches the constraint count N_{con} plus the number of retained sector terms on every lattice.

Proof. On a finite-dimensional algebra, an interior entropy maximizer subject to linear constraints $\operatorname{Tr}(\rho O_i) = c_i$ is given by finite Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\operatorname{Tr} e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the N_{con} global sums $O_a := \sum_x O_a(x)$ plus the retained sector terms, so the exponent has one shared multiplier per density label. MaxEnt gives the Gibbs form on this faithful/interior branch; the displayed locality restriction on H_{sec} is separate and load-bearing. A boundary constraint can instead force a rank-deficient optimizer (for example $\operatorname{Tr}(\rho|1\rangle\langle 1|) = 0$ on $\mathbb{C}^2$); it has no finite full-algebra multiplier and must be treated on its support or by infinite penalties. Such a boundary branch does not supply the stagewise faithfulness used below. Had the constraints instead been imposed per cell, $\langle O_a(x) \rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family

would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

Definition 2.6a (closure defect and induced refinement map). Fix successive regulator scales ℓ (finer) and L (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale L, and let $\Phi_{\ell\to L}$ be an admissible coarse-graining channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell\to L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell\to L}(\lambda) := \inf_{\lambda'} D(\sigma \,\|\, \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^*, set $R_{\ell\to L}(\lambda) := \lambda^*$.

Lemma 2.6b (I-projection residual bound). If the constrained operators at scale L together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma\|\omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the minimizer $\lambda^* = R_{\ell\to L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c\rangle_{\omega_L(\lambda^*)} = \langle S_c\rangle_\sigma$ for every constrained operator S_c; and

$$\|\sigma - \omega_L(R_{\ell\to L}(\lambda))\|_1 \le \sqrt{2\,\varepsilon_{\ell\to L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell\to L}(\lambda)}$. The separately declared exact optimizer-pushforward interface is the statement $\varepsilon_{\ell\to L}(\lambda) = 0$ along the realized branch. In that case $\sigma = \omega_L(R_{\ell\to L}(\lambda))$, and the induced refinement map is this moment-matching I-projection.

Proof. Writing $\log\omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c\langle S_c\rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful σ is standard finite-dimensional exponential-family duality [59]. The displayed residual bound is the quantum Pinsker inequality $D(\rho\|\tau) \ge \frac{1}{2}\|\rho-\tau\|_1^2$ [60] evaluated at λ^*, and $\varepsilon_{\ell\to L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

Acceptance check. A finite two-lattice test implementing this package (the constraint/multiplier count against the displayed N_{con}-dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell\to L}$ acts as the identity) is included with the repository sources.

Numerical projection and closure are distinct. For faithful σ on $\mathbb{C}^d$, choose a Hilbert–Schmidt orthonormal basis of the observable space V modulo identity, and write its traceless Hamiltonian as $H(\theta) = \sum_j \theta_j Q_j$, with $\rho_\theta = e^{-H}/\operatorname{Tr} e^{-H}$. The invariant moment residual is $\eta = \|P_V(\sigma-\rho_\theta)\|_{\mathrm{HS}}$; a raw residual in caller units is not a convergence criterion. Coercivity and convexity give

$$D(\rho_*\|\rho_\theta) \le B := \eta\left(\|\theta\|_2 + \frac{\log d}{\lambda_{\min}(\sigma)}\sqrt{\frac{d-1}{d}}\right), \qquad \|\rho_* - \rho_\theta\|_1 \le \min\{2, \sqrt{2B}\},$$

where ρ_* is the unique exact projection. This optimizer error is separate from the closure defect $D(\sigma\|\rho_*)$; the Pythagorean identity adds the two. The implementation evaluates these bounds in floating point, without outward-rounded enclosures or an exact-zero certificate. Invertible changes of observable coordinates and identity offsets preserve the state; scalar energy origins shift $\log Z$ and must be retained in partition and free-energy calculations. Compensated scalar accumulation and a stable relative-entropy remainder avoid cancellation of small finite effects; neither supplies a spectral support certificate or recovers input differences lost to rounding.

Strong conditional mixing premise. Let G_ℓ be the UV cell graph. Assume local dimension at most q, interaction degree at most Δ_0, interaction diameter at most r_0 graph layers, and

$$\sup_x \sum_{X \ni x} \|\beta \Phi_\ell(X)\| \le J_0$$

with constants independent of the cut and refinement stage. Let m_ℓ be the resolved collar width in graph layers and set $\delta = m_\ell \ell$. For the faithful Gibbs state, define

$$\mathbf{J}_\rho(A : D \mid B) := \log \rho_{ABD} + \log \rho_B - \log \rho_{AB} - \log \rho_{BD}.$$

The strong mixing premise is a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z \in \partial_{r_0}^{\mathrm{UV}} C} E_{z,\ell,\delta}, \qquad \|E_{z,\ell,\delta}\|_\infty \le \kappa e^{-(m_\ell - r_0)/\zeta},$$

with κ, ζ uniform in the cut, boundary condition, and stage. This conditional or matrix mixing condition is stronger than ordinary two-point exponential clustering. Local Gibbs form, a Lieb–Robinson bound, or a Hamiltonian spectral gap does not imply it for a general noncommuting Gibbs state.

Theorem 2.5 (finite-range conditional Gibbs mixing gives collar recovery). Under Lemma 2.6 and the strong conditional mixing premise,

$$\boxed{I_\omega(A_\delta : D_\delta \mid B_\delta) \le \kappa |\partial C|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta} \le c |\partial C|_{\mathrm{UV}} e^{-\delta/\xi_\ell}}$$

with

$$\xi_\ell = \zeta \ell, \qquad c = \kappa e^{r_0/\zeta}.$$

Proof. Embedding the marginal logarithms in the tripartite algebra gives

$$I(A : D \mid B)_\rho = \mathrm{Tr}[\rho_{ABD} \mathbf{J}_\rho(A : D \mid B)]\,.$$

The boundary expansion and trace-norm duality bound this expectation by the sum of the boundary term norms. The second form follows from $\delta = m_\ell \ell$. QED.

For one uniform family, the sharp sufficient limit condition is

$$\frac{\delta}{\xi_\ell} - \log |\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

If

$$|\partial C|_{\mathrm{UV}} \le a(\ell_0/\ell)^p, \qquad \delta(\ell) \ge \zeta(p+\eta)\ell \log(\ell_0/\ell), \quad p \ge 0, \quad \eta > 0,$$

then $\delta/\ell \to \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \le ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

This lower bound on δ does not imply $\delta \to 0$. Physical collar shrinking remains the separate double-scaling hypothesis stated above. It is realized, for example, by choosing $\delta(\ell) = \zeta(p+\eta)\ell \log(\ell_0/\ell)$, or by rounding its graph-layer width upward with an additional error at most ℓ. For a counterexample to inferring shrinking from the lower bound alone, take $\ell_n = 1/(4n)$, $\ell_0 = \delta_n = 1/4$, $p = 0$, and $a = \zeta = \eta = 1$. Then $\delta_n \ge \ell_n \log n$ and $\delta_n/\ell_n = n$, but $\delta_n = 1/4$ does not shrink. A zero-Hamiltonian product Gibbs chain has zero conditional matrix defect and CMI and satisfies all the local finite-range and mixing bounds in this example; it fails precisely the separately declared shrinking-collar hypothesis. The ratio $\delta/\ell \to \infty$ alone is insufficient. For $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_n|_{\mathrm{UV}} = e^{n^2}$, that ratio diverges while the upper envelope grows as e^{n^2-n}.

A finite receipt records the interaction range and norm bound, treatment of global sector terms, boundary-cell count, graph separation, regional CMI in nats, matrix-defect norm, predeclared κ, ζ, log-envelope slack, Fawzi–Renner error, held-out cuts, and the rate margin $\delta/(\zeta\ell) - \log|\partial C|_{\mathrm{UV}}$. Fitted constants and local random-triplet CMI remain finite proxies. They do not certify a cofinal limit.

This bound is the quantitative hinge for constructive gluing.

2.3.4 Conformal-modular fixed-point microphysics

On the faithful/interior local finite-constraint branch of premise A3 with identity-proportional reference, the logarithm of the selected state is a quasi-local UV generator. A separately declared optimizer-pushforward and refinement-tail interface places the branch inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves the local-Gibbs result and propagation. Endpoint-Lipschitz control of *reduced* modular Hamiltonians requires the additional exact-Markov/central-interface and endpoint-control branch stated below; it does not follow from finite-range locality alone. The cross-scale interface supplies refinement stability. The geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-$*$/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

Local finite-constraint information-projection branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$. premise A3 selects the fixed-stage information projection. Retention of the same finite label set under refinement belongs to the separate optimizer-pushforward and refinement-tail interface.

Theorem 2.6 (Interior local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV}, and the prescribed moment vector lies in the relative interior of the attainable moment set so that the optimizer is faithful on the retained sector, then the entropy maximizer is

$$\omega \propto \exp\left(-\sum_a \lambda_a O_a\right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log\omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. The finite-dimensional interior exponential-family theorem gives finite Lagrange multipliers λ_a. Boundary optima require restriction to their support or extended infinite multipliers and are not included in this faithful branch. QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [41] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y,

$$\|[\tau_t(A_X), B_Y]\| \le C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\mathrm{LR}}|t|)/\xi}.$$

Additional endpoint-control branch. Locality of the global Gibbs generator does not by itself make reduced interval modular Hamiltonians local; the four-qubit counterexample in Theorem 4.3d rules out that inference. On the exact-Markov/central-interface branch, assume in addition a specified endpoint-to-regulator-cell map and a uniform local norm/site-density bound such that, after subtracting the central endpoint block, the operator-norm sum of terms changed between allowed intervals I, I' is at most $L|I'\Delta I|$. On this explicitly named branch, the local split and that geometric norm bound give bounded-interval endpoint-Lipschitz matrix elements,

$$\big|\langle \psi, (K[I'] - K[I])\phi \rangle\big| \le C_{\psi,\phi,I_{\max}} \, |I'\Delta I|,$$

used later in the null-modular bridge. Thus quasi-local propagation follows on the faithful local-Gibbs branch, while reduced-modular endpoint control is a separate exact-Markov/central-interface receipt with the stated endpoint geometry and uniformity.

Refinement-stable multiplier branch. The separately declared optimizer-pushforward and refinement-tail interface preserves the same finite constraint family under coarse-graining and places the regulator states in one common finite-dimensional multiplier family. This is a substantive renormalization condition because coarse-graining a finite-range Gibbs family generically generates interactions outside a fixed density list. The "refinement-stable" language used later means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. FixedCutoffBosonicSectorCategory constructs the fixed-stage tensor-generated category and its forgetful fiber. RefinementFunctorAndFiberDescent constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt. The separate finite Standard Model route uses complete reversible response and endogenous overlap transport to force the local Standard Model gauge Lie algebra. Incidence and target-blind readback derive the carrier module and $R = -J$. The finite conditional fixture supplies the charge-conjugate rank-15 projector pair, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source derivation of that current, matter action, and global quotient is not supplied. Under the complete-band and operational-cost premises, the exact screen theorem selects the rank-three response band. Physical family attachment, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own premises. premise A3 supplies neither the finite current realization nor the sector-colimit data or laboratory attachment.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental

modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Corollary 2.6 (Geometric modular action on paired-certified caps). For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ emitted on a branch carrying both the finite cap-normal certificate and the independently complete same-branch multiresolution algebra-state comparison package, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

Proof. This is The spacetime and Einstein paper's finite cap-net BW theorem applied to the extracted geometric cap pair. The finite cap-normal input supplies BW framing, support-order faithfulness, held-out oriented cross-ratio rigidity, the geometric support flow, independently normalized geometric 2π-KMS comparison, and wrong-normalization controls. The separate multiresolution algebra-state comparison input supplies the complete algebra-state reference tower, cap-family-uniform mixed-GNS convergence, comparison maps, and modular support covariance. QED.

Support envelope from net regularity (not an alternative derivation). The outer-regularity / minimal-support condition used later lets one read a support envelope for the modular image from the scaling-limit net. By itself it does not establish modular covariance; that equality comes from the paired certificate in Corollary 2.6, or must be assumed as an additional two-sided net-automorphism property.

(NR) Outer regularity / minimal support. For any operator O, the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

Proposition 2.7 (Modular support-envelope inclusion). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\Big(\sigma_t^{\omega,C}(O)\Big).$$

Then

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(f_t^C(R)).$$

Equality does not follow from (NR). It requires, for example, a two-sided order-isomorphism of the local net compatible with σ_t and σ_{-t}, together with the relevant additivity/local-generation property. Those are substantive covariance data, not consequences of knowing minimal operator supports.

Proof. For every $O \in \mathcal{A}(R)$, (NR) assigns $\sigma_t^{\omega,C}(O)$ a minimal support contained in $f_t^C(R)$; isotony therefore places that operator in $\mathcal{A}(f_t^C(R))$. Taking all such operators gives the stated inclusion. Surjectivity onto the entire envelope algebra is not supplied by the fact that σ_t is an automorphism of the global algebra. QED.

Null-surface modular structure. On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type

tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, and renormalized endpoint-Lipschitz control up to the weak tail generator. On the scaling branch the geometric action supplies the half-sided relation; the separately registered common-vector and relative-commutant standardness receipts make it a standard half-sided modular inclusion whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. The same half-line family fixes the generator's endpoint-derivative identity internally; its local null-stress reading consumes the separately supplied effective-locality and same-source bridge.

- **Derived half-sided modular relation.** Nested null half-line algebras satisfy the half-sided covariance and containment relation on the declared geometric scaling branch by Corollary 5.2e. To obtain a *standard* half-sided modular inclusion, the common reference vector must additionally be cyclic and separating for both half-line algebras and cyclic for their relative commutant. On that separately registered standardness receipt, Borchers–Wiesbrock identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [45].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follows on the exact-Markov/central-interface branch only after the specified endpoint-to-cell map and uniform norm/site-density bound of the endpoint-control clause. It then defines the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.

2. **Gauge charges:** fix boundary flux or charge operators.

3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

Focusing boundary. QNEC has rigorous QFT proofs in broad settings [55, 57], with hypotheses (null-surface locality and analyticity, modular-flow/causality/defect-OPE structure) that have not been verified on the reconstructed net, and QFC is strictly stronger than QNEC [56]. Quantum focusing is an optional gravity interface and is not consumed by the Einstein theorem:

- EC plus the declared state, edge-reference, and entropy-split premises identify $S_{\mathrm{gen}} = S_{\mathrm{bulk}} + \langle L_C \rangle$ on the finite branch (Section 5.4).

- The null-stress and Einstein theorem branch supplies the kinematic inputs required by an internal QNEC/QFC proof; that derivation is not supplied.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Fixed-stage and cross-scale consequences:** premise A3 with the declared local constraint and reference package gives the local-Gibbs form and quasi-local propagation on the faithful/interior branch. Bounded-interval reduced-modular endpoint-Lipschitz control additionally consumes the exact-Markov/central-interface split and the specified endpoint-to-cell uniform bound. The separate optimizer-pushforward interface supplies the realized state-side refinement branch along which later transport questions are asked.
- **Support-visible scaling statement:** the scaling limit emits the support-visible geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization follows. The later null half-sided-inclusion bridge additionally consumes its registered common-vector and relative-commutant standardness receipts.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-$*$ extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel). Exact or asymptotically exact Markov recovery at finite cutoff does not by itself imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1 - e^{-n} \end{pmatrix}.$$

Each ρ_n is faithful at finite n, and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \to 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit E_{12}, the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n\, E_{12} + O(1) E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-$*$/GNS extraction.

Theorem 2.6b (regularized support-visible modular transport). Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\widehat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \widehat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable O in the support-visible algebra,

$$\left|\operatorname{Tr} \rho_n O\big(K_a(\rho_n) - K_a(\widehat{\rho}_n)\big)\right| \le \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta} a + 4\Delta_n |\log a| \right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n / a_n \to 0, \qquad d_{m,\delta} a_n \to 0, \qquad \Delta_n |\log a_n| \to 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof sketch. On the spectral interval $[a,\infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above a, the a-tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

Definition 2.6c (explicit multiresolution reference branch). The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where m is the refinement depth, L is the finite support or volume scale, and b is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of S^2; Euclidean branches use dyadic mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing a_t when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel j carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j. At regulator r,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \to M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s\setminus r}, \qquad \tau_{s\setminus r} = \bigotimes_{j \in \mathcal{J}_s \setminus \mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s\Big[\Gamma_r^{-1}(A) \otimes \mathbf{1}\Big],$$

and

$$Q_{sr}(X) = \Gamma_r\Big[(\mathrm{id} \otimes \tau_{s\setminus r})(\Gamma_s^{-1}(X))\Big].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs} Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

Theorem 2.6d (exact modular-compatible reference tower). The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r \preceq s}$$

form a directed tower. The maps ι_{rs} are unital injective $*$-homomorphisms with $\iota_{st}\iota_{rs} = \iota_{rt}$. The Q_{sr} are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr}Q_{ts}$. The maps E_{sr} are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \qquad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \qquad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

The finite reference modular groups are also exactly compatible:

$$\sigma_t^{\widehat{\omega}_s}(\iota_{rs} A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in r; in bare coordinates it is only the identity $A \mapsto A \otimes \mathbf{1}$, partial expectation against faithful detail states, and conjugation by Γ_r, Γ_s.

Theorem 2.6e (canonical renormalization and correlation Cauchy bound). Let $\mathfrak{A}$ be the C^*-inductive limit with embeddings $j_r : M_r \to \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega} \circ j_r = \widehat{\omega}_r$, and the maps Q_{sr} induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \to j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \to 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \le M$,

$$\left| \widehat{\omega}_r\left(\prod_{k=1}^{n} \mathcal{R}_r(A_k)\right) - \widehat{\omega}\left(\prod_{k=1}^{n} A_k\right) \right| \le M^{n-1} \sum_{k=1}^{n} \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

Theorem 2.6f (compact-time modular convergence on fixed local algebras). Let $\rho_{s\downarrow r}^{\mathrm{phys}}$ be the density matrix of the physical state at fine stage s, restricted to the embedded fixed local algebra M_r, and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\mathrm{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\mathrm{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t|\le T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \le 2T\|A\| \left[\frac{\epsilon_{s,r}}{\eta_s} + \log\left(1 + \frac{\eta_s}{\lambda_r}\right) \right].$$

Hence $\eta_s \to 0$ and $\epsilon_{s,r}/\eta_s \to 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \to 0$ admits a diagonal subsequence and one cutoff schedule. The phrase "finite-stage modular action" below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

Cap-family uniformity clause. For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \le \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric 2π-KMS, and wrong-scale controls.

Theorem 2.6g (finite positive transfer and Lorentzian handoff condition). On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar C. Ref. [16] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \ge 0$, and set

$$L_r^{\mathrm{rep}} = \sum_C c_{C,r}(I - P_{C,r}), \qquad T_r(a_t) = e^{-a_t L_r^{\mathrm{rep}}}.$$

Then $L_r^{\mathrm{rep}} = L_r^{\mathrm{rep}\,*} \geq 0$, $0 < T_r(a_t) \leq I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \geq 0$$

for positive-time cylinder functions F. A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent substitute, including vacuum-projection and intertwiner convergence. Weak-$*$ state convergence alone is not a transfer theorem.

BW-side closure boundary. Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

2.4 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the finite-dimensional overlap algebras and their allowed chart-change groupoid are put in the block-complete spatial presentation specified below, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this spatialized compact boundary action, not as an automatic consequence of an arbitrary algebra representation. On the genuinely noncentral branch, the corresponding declared weak gluing data are encoded by a compact crossed-module change system. Block permutations and other groupoid data are retained rather than silently treated as inner automorphisms. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

2.4.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k. If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_k B_k C_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_k B_k})\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k},$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

Proof. Induct on k. The recovery error contracts under CPTP maps, so the errors add. QED.

2.4.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose block-complete faithful finite-dimensional spatial presentations of the overlap algebras on a connected cut Σ, and require the allowed overlap-chart isomorphisms to be represented by their unitary normalizers. A finite-dimensional C^*-algebra is a direct sum of full matrix blocks: its star automorphisms permute isomorphic central blocks and act inner within each block. They are spatial in the canonical block-complete presentation, but need not be inner in the algebra itself (an abelian two-block swap is the elementary counterexample). Fixing a reference chart and retaining any allowed block permutations, the compact closure of the subgroup generated by the spatial recharting unitaries is the boundary gluing group K_Σ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Spatialized gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ, together with the block-complete spatial presentations and unitary-normalizer realization of the allowed chart-change groupoid just stated. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, after a reference chart is fixed, this transition system reduces to a compact boundary group K_Σ, and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. On this realized presentation, gauge names the overlap redundancy; the spatialization data are explicit hypotheses rather than consequences of finite dimensionality alone.

Proof. The structure theorem for finite-dimensional C^*-algebras decomposes each chart algebra into full matrix blocks. An allowed star isomorphism permutes isomorphic central blocks and is inner on each matched block; the chosen block-complete presentation implements the combined operation by a unitary normalizer. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

transforms by conjugation at the chosen basepoint under local frame changes. Its conjugacy class, and therefore every class-function value (including every one-dimensional character value), is invariant. In particular, nontriviality is invariant and obstructs global trivialization. QED.

Proposition 3.2ba (Finite path ratio, character phase, and cyclic quantization). For reversal-compatible group-valued edge labels on a typed finite path system, the ordered transport ratio of two paths with common endpoints is exactly the holonomy of the loop obtained by following the first and reversing the second. Every one-dimensional unitary character sends this identity to an exact relative phase. A vertex rechart conjugates based holonomy, so holonomy itself is invariant

for an abelian structure group and the character phase is invariant for an arbitrary group. There is an explicit four-vertex path/face control whose two declared triangular faces have trivial holonomy while a separate undeclared, unfilled loop and its two-arm relative character phase are nontrivial. Finally, if the loop holonomy factors through a supplied $\mathbb{Z}_n$ sector, every character phase is an nth root of unity.

Scope. The result is machine checked for supplied finite paths, edge labels, character, faces, and cyclic-sector factorization. The path/face control declares exactly two triangular faces; it neither claims flatness on undeclared triangles nor proves a puncture or noncontractibility statement. No theorem here constructs the connection from observer records or identifies the phase with a physical gauge flux, spacetime loop, charge, laboratory arm, detector, or interference fringe.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ, and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

2.4.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group Z_Σ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects z_{ijk} at the implementer level by

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\mathrm{Ad}(z_{ijk})$ alone would be the identity and would not determine z_{ijk}.

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K}_\Sigma$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

2.4.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{ijk}h_{ikl} = (g_{ij} \triangleright h_{jkl})h_{ijl}.$$

The order follows by comparing $(g_{ij}g_{jk})g_{kl}$ with $g_{ij}(g_{jk}g_{kl})$: their defect factors are respectively $\partial(h_{ijk}h_{ikl})$ and $\partial((g_{ij} \triangleright h_{jkl})h_{ijl})$. Coherence of the chosen 2-morphisms requires the displayed equality in H. Equality after applying ∂ is a weaker condition.

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Write

$$q_\Sigma = [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair (g, h) but does not in general determine a unique ordinary G_Σ-valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \\ [g] &\longmapsto [(g, 1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0, 1\}$-valued strictification obstruction by

$$\begin{aligned} o^{(2)}_\Sigma(g, h) = 0 \quad &\iff \quad q_\Sigma \in \mathrm{im}(\iota_\Sigma) \\ &\iff \quad [(g, h)] \text{ has a representative } (g^{\mathrm{str}}, 1) \\ &\qquad\qquad \text{with } g^{\mathrm{str}}_{ij} g^{\mathrm{str}}_{jk} = g^{\mathrm{str}}_{ik}. \end{aligned}$$

and set $o^{(2)}_\Sigma(g, h) = 1$ otherwise.

Theorem 3.4 (non-central higher-defect strictification). The higher associator defect of (g_{ij}, h_{ijk}) is removable iff $o^{(2)}_\Sigma(g, h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine G-valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class.

Proof sketch. Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary G-valued 1-cocycle locus, which is the definition of $o^{(2)}_\Sigma = 0$. The map ι_Σ need not be injective: H-valued edge changes can relate strict representatives and alter

their ordinary holonomy through ∂. Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ, the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_\lambda \mathbb{C}\,\mathbf{1}_\lambda,$$

and the full orbit q_Σ is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o_\Sigma^{(2)} = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary G_Σ-valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to q_Σ.

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$-modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

Theorem 3.4b (TransportabilityFromOverlapGluing). Fix a connected cut Σ and a finite regulator charting nerve N_Σ. Comparison transport of a collar charge is the path composite in the overlap recharting groupoid: along $p = (i_0 \to \cdots \to i_m)$, $U_p = U_{i_0 i_1} \cdots U_{i_{m-1} i_m}$ compares terminal-chart column components to initial-chart components. Forward column transport is $U_p^\dagger$. Write $p * q$ for first following p, then q, and $\overline{p}$ for path reversal; thus $U_{p*q} = U_p U_q$ and $U_{\overline{p}} = U_p^{-1}$. The residual comparison for paths p, p' with common endpoints is $U_p U_{p'}^{-1} = U_{p*\overline{p'}}$, a loop at their common initial chart. A frame change $U_{ij} \mapsto V_i U_{ij} V_j^\dagger$ gives $U_p \mapsto V_{i_0} U_p V_{i_m}^\dagger$, and hence conjugates closed-loop holonomy at its base chart. These conventions agree with $U_{ij}U_{jk} = U_{ik}$ on a strict triangle. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let U^{str} denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\mathrm{Hol}_{\alpha, U^{\mathrm{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \qquad [\gamma] \longmapsto U_\gamma^{\mathrm{str}}\big|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle z_{ijk}. Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\mathrm{Hol}_{\alpha, U^{\mathrm{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is path dependent.

On the genuinely noncentral branch, path comparisons are H_Σ-valued 2-morphisms in the crossed module $H_\Sigma \to G_\Sigma$. Strict ordinary transport exists iff $o_\Sigma^{(2)} = 0$ and at least one crossed-module strictification gives a genuine G_Σ-valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$. If $o_\Sigma^{(2)} \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o_\Sigma^{(2)} = 0$ but every allowed strict representative has nontrivial represented G_Σ-holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g, h)]$ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus "zero obstruction" below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

Triangle-free cycle check. Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \mathrm{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o_\Sigma^{(2)} = 0$. Its ordered cycle holonomy is $V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\mathrm{U}(2)/\mathrm{SU}(2) \cong \mathrm{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An $\mathrm{SU}(2)$-valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

2.5 Modular Flow and Lorentz Kinematics

2.5.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the EC-aligned exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Four quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error

$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \leq \varepsilon \right\},$$

where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

4. the EC-aligned subset and a family-specific aligned replacement modulus,

$$\mathfrak{M}^{\mathrm{EC}}_{A:B:D} := \{\sigma \in \mathfrak{M}_{A:B:D} : \sigma \text{ satisfies MSA on the preselected EC split}\},$$

$$\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma\in\mathfrak{M}^{\mathrm{EC}}_{A:B:D}} \|\rho-\sigma\|_1 : \rho\in\mathcal{F},\ I(A:D\mid B)_\rho \le \varepsilon \right\}$$

for the declared controlled collar family $\mathcal{F}$. Convergence of the generic modulus in item 3 does not imply convergence of this aligned modulus; $\delta^{\mathrm{M,EC}}_{\mathcal{F}} \to 0$ is a separate family hypothesis.

Lemma 4.1a (EC-aligned exact Markov implies exact additivity up to a central term). If $I(A:D\mid B)_\omega = 0$ and ω satisfies the Markov-split alignment hypothesis of Section 2.3, then the state takes the EC-aligned HJPW block form

$$\omega_{ABD} = \bigoplus_\alpha p_\alpha\, \omega^{(\alpha)}_{Ab^\alpha_L} \otimes \omega^{(\alpha)}_{b^\alpha_R D},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

Proof. By alignment, the HJPW blocks are the EC blocks. On each block the modular Hamiltonians of ABD, AB, BD, and B are the logarithms of tensor-product states with the same classical block weight p_α. The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central in $Z(\mathcal{A}(B))$. Without alignment, exact Markovity cancels the four logarithms on the state's own HJPW blocks, but the resulting block constants need not lie in $Z(\mathcal{A}(B))$, so the central form used here is unavailable. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A:D\mid B)$ directly to an exact Markov state. The limit points are exact Markov states in the CMI-zero sense; identifying their factorization with the preselected EC factors is the Markov-split alignment hypothesis of Section 2.3, which is carried separately and is not produced by small CMI (the Bell-pair counterexample of Section 2.3 has zero CMI at fixed distance from every EC-aligned state).

Theorem 4.1b' (Finite-collar Markov replacement stability). On one fixed faithful finite-dimensional collar model, the qualitative modulus above can be made collar-local and quantitative. If all relevant marginals stay above a chosen floor $\lambda_* > 0$, then there are constants $C_{A:B:D,\lambda_*} > 0$ and $\theta_{A:B:D,\lambda_*} > 0$, depending only on that collar model and floor, such that

$$d_{\mathrm{M}}(\rho) \le C_{A:B:D,\lambda_*}\, I(A:D\mid B)_\rho^{\theta_{A:B:D,\lambda_*}}.$$

This is a fixed-collar Lojasiewicz-type rate for the analytic function $I(A:D\mid B)$ on the faithful state manifold. It is not a dimension-free stability theorem for arbitrary tripartite quantum systems.

Corollary 4.1c (Carried EC-aligned collar defect operator). Let $\omega_\varepsilon \in \mathcal{F}$ satisfy $I(A:D\mid B)_{\omega_\varepsilon} \le \varepsilon$, assume the declared family has an EC-aligned replacement modulus tending to zero, and choose $\sigma_\varepsilon \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ with

$$\|\omega_\varepsilon - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon) := \Delta K(\omega_\varepsilon) - \Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A\cup B$,

$$|\mathrm{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \le \|X\|_\infty\, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

Independently, Fawzi–Renner recovery guarantees a channel acting on the collar and a comparison state $\omega_\varepsilon^{\mathrm{rec}}$ with squared Uhlmann fidelity at least $e^{-\varepsilon}$, for CMI ε in nats, and hence

$$\|\omega_\varepsilon - \omega_\varepsilon^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term bounds observable errors for that recovered state; it is not a guarantee for an arbitrary unrotated Petz map. The generic modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ justifies replacement only by some exact Markov state on its state-dependent HJPW split. The separately assumed $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon)$ is the quantity that justifies the EC-aligned normal form used in later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov and EC-aligned on the relevant collar; or
2. a controlled collar family for which $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon_\delta) \to 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

Theorem 4.1d (Finite-stage modular-defect propagation and dimension quarantine). For any fixed downstream branch calculation that uses finitely many collar or strip modular-additivity identities, replacing each exact identity by its controlled finite-stage form changes every bounded downstream modular observable O by at most

$$\mathcal{P}_O\Big(\{r_{\mathrm{FR}}(\varepsilon_j)\}, \{\delta^{\mathrm{M,EC}}_{\mathcal{F}_j;j}(\varepsilon_j)\}, \{\eta_j^{\mathrm{reg}}\}\Big),$$

where $\mathcal{P}_O \to 0$ as all listed errors vanish. The polynomial-continuity modulus depends only on the declared fixed collar models, support-visible dimensions, bounded observable class, faithful floors or regularization schedules, and bounded modular-parameter intervals. No BW/Lorentz, null-modular, or Einstein scaling step uses a dimension-free trace-norm stability theorem from small CMI directly to exact Markov normal form.

2.5.2 Theorem: BW_{S^2} on the extracted geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity and not the slogan "finite cells imply Lorentz invariance." Its target is the support-visible refinement-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$. The branch theorem is the conditional implication

$$\mathsf{FiniteCapBWCertificate} \;+\; \mathsf{ModularAlgebraStateData} \Longrightarrow \sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The finite cap-normal certificate consists of cap-normal density and nondegeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π-KMS comparison, and wrong-normalization separation. The independent multiresolution algebra-state comparison package supplies the algebra-state reference tower, common comparison maps, cap-family-uniform regularized modular transport, mixed-GNS convergence, and modular support covariance. The support-visible theorem below uses both inputs; it does not require a type-I continuum algebra or a full-algebra unregularized common spectral floor. Bare finite consensus implies neither input.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M,EC}}_{\mathcal{F}_\delta; A_\delta : B_\delta : D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound for comparing modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at EC-aligned exact Markovity or along a controlled collar family satisfying

$$\delta^{\mathrm{M,EC}}_{\mathcal{F}_\delta; A_\delta : B_\delta : D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

For each cap C, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 4.2 (Finite BW_{S^2} cap-net scaling). For a nondegenerate BW-framed cap $\widehat{C} = (C, n_C, p_C^-, p_C^+)$, consider an OPH realization on an observer-supporting refinement branch that satisfies premises A1–A3 and carries the named fixed-cutoff regulator, collar, consensus, and optimizer-pushforward interfaces. Assume the finite cap-normal BW certificate above and, independently, the complete multiresolution reference tower and algebra-state comparison package of Theorems 2.6c–2.6f on the same branch. The support-visible geometric cap net then admits a weak-$*$/GNS scaling-limit cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$, whose modular automorphism group is

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

is confined to the shrinking collar up to the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$, the EC-aligned fixed-collar replacement bound η_δ^{M}, and the regularized support-visible modular transport bound. The support-visible modular limit is independent of the EC-aligned exact-Markov replacement sequence after projective compatibility and GNS quotienting, and support-readable modular covariance turns the limiting modular group into a cap-local support map before the certificate's framed-cap and oriented cross-ratio rigidity clauses are applied. No separate cap-isotropy/SO(2)-equivariance selector, finite-cell Lorentz-invariance premise, or full-algebra unregularized common floor is used. Record/pointer and interface-inert auxiliary registers are outside the extracted geometric subnet. If the scaling-limit cap algebra is type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}};$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, under the paired same-branch finite cap-normal and complete multiresolution algebra-state comparison inputs, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-parameter intervals to the geometric cap-dilation action. Without both inputs, this theorem makes no BW or finite-to-continuum modular claim. Even with them, it does not assert compact-time convergence of finite-regulator modular groups beyond the declared support-visible certificate.

Proof. Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. Proposition 2.6a shows why a full-algebra unregularized floor is unavailable in general. Theorem 2.6b supplies the matrix-element replacement on fixed collar models, and Theorem 2.6f supplies the stronger compact-time group estimate only for transported regularized local actions with the stated reference-tower certificate. Projective replacement compatibility and the support-visible convergence theorem ensure that the emitted modular limit is independent of the chosen EC-aligned exact-Markov comparison family. Support-visible cap-pair extraction on the local GNS support quotient uses weak-$*$ compactness of finite-stage state spaces and the consensus refinement mechanism to supply local limit states, then applies GNS construction to obtain the scaling-limit cap pair. Support-readable modular covariance reads the modular group as a cap-local support map. The finite cap-normal certificate supplies support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and compact-time equicontinuity, so the support flow is the framed cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$, with the certificate-selected frame understood in λ_C. The certificate then passes the independently normalized geometric 2π-KMS condition to $s \mapsto \alpha_{\lambda_{\widehat{C}}(s)}$. KMS uniqueness, together with wrong-β separation and nontriviality, fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit algebra is type I, this automorphism identity is represented by $K_C = 2\pi B_C + Z_C$ with Z_C central; otherwise the automorphism identity is the full statement and the action is outer. QED.

Definition 4.2a (BW-branch observer-relative modular ordering). On the branch satisfying the hypotheses of Theorem 4.2, the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ supplies a dimensionless ordering for that observer's accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.

Scope note. The theorem fixes the 2π normalization and the carried refinement limit on the support-visible extracted geometric cap pair under the paired finite cap-normal and independently complete same-branch multiresolution algebra-state comparison inputs. The UV-side scaffold is the realized transported geometric cap-local system, regularized support-visible modular transport on fixed local collars, support-visible cap-pair extraction on the local GNS support quotient, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric KMS normalization. No separate cap-isotropy input is used, and the conformal support-map clause is the explicit scaling-limit branch condition rather than a finite-regulator Lorentz premise. Compact-time convergence of finite-regulator modular groups is available only through the transported-local certificate of Theorem 2.6f; a black-box conformal-net BW reconstruction would be an additional validation route, not a hidden premise.

2.5.3 Theorem: cap-normal H^3 reconstruction

Theorem 4.3 (Cap-normal H^3 reconstruction, imported from the spacetime and Einstein derivation). On the support-visible round-cap BW branch, write the celestial null section as

$$q(\Omega) = (1, \Omega), \qquad \Omega \in S^2.$$

For a nondegenerate oriented round cap

$$C(\mathbf{c}, \alpha) = \{\Omega : \mathbf{c} \cdot \Omega \geq \cos\alpha\}, \qquad 0 < \alpha < \pi,$$

define

$$n_C = (\cot\alpha, \csc\alpha\, \mathbf{c}).$$

Then, for the time-first Lorentz form $\eta = -dt^2 + d\mathbf{x}^2$,

$$\eta(n_C, n_C) = 1, \qquad \eta(n_C, q(\Omega)) = \frac{\mathbf{c} \cdot \Omega - \cos\alpha}{\sin\alpha}.$$

Thus $\eta(n_C, q(\Omega)) = 0$ exactly on ∂C, with positive sign in the cap interior and negative sign outside. For $g \in \mathrm{Conf}^+(S^2)$ with Lorentz representative Λ_g,

$$\Lambda_g q(\Omega) = \omega_g(\Omega) q(g\Omega), \qquad n_{gC} = \Lambda_g n_C.$$

Consequently

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1),$$

oriented round caps form the de Sitter normal space

$$\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2) \cong dS^{\mathrm{cap}}_3 \cong \mathrm{SO}^+(3,1)/\mathrm{SO}^+(2,1),$$

and future unit timelike observer frames form the distinct hyperbolic space

$$H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \qquad \dim H^3 = 3.$$

Each cap normal determines the geodesic half-space

$$\mathcal{H}^+_C = \{u \in H^3 : \eta(n_C, u) \geq 0\},$$

not a preferred observer point.

Proof. This is the cap-normal reconstruction theorem in The spacetime and Einstein paper, which supplies the projective-null-cone model, the cap-normal bijection, Lorentz equivariance, the H^3 observer-frame hyperboloid, the ideal-boundary identification, and the cap/half-space duality. The BW theorem supplies the cap-preserving conformal subgroup and its 2π modular normalization; the cap-normal theorem supplies the explicit projective Lorentz representation and does not rederive BW from bare finite consensus. QED.

Corollary 4.3a (Frame-space boundary). The theorem proves the exact canonical Lorentz/H^3 observer-frame hyperboloid on the stated branch. Its points are future unit timelike frames, not event positions. It does not populate a physical spacetime, construct a chart-blind neutral bulk, derive a physical curvature radius R_H, produce stress-energy, or close the Einstein branch-entry umbrella. Record-conditioned cap responses can estimate frame data under the certificate imported next; event location, object families, neutral bulk, scale, stress, and Einstein branch entry are separate receipts. Theorems 4.3c–4.3h below import The spacetime and Einstein paper's geometry-producer, null-net-standardness, source-causal precursor and continuum, stress, entropy-bridge, and composed branch-entry packets. Their composition is conditional on one source-derived common-domain tower with certified cofinal tails, universal coupling, a vacuum reference, and independent physical scale readouts. No source-derived tower carrying that complete packet is constructed here.

Theorem 4.3b (Conditional record-conditioned H^3 frame estimate). Fix a quotient-visible record token i and clock slice t. On the frame space of Corollary 4.3, suppose its record-conditioned modular cap responses

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad \omega_{i,O}(A) = \frac{\omega_O(P_{i,O}AP_{i,O})}{\omega_O(P_{i,O})},$$

admit the calibrated frame-local factorization

$$y_{i,j} = F_j(X_i(t)) + e_{i,j}, \qquad F_j(X) = \psi_j\Big(\frac{\eta(X,n_j)}{R_H}\Big),$$

on a compact frame region $\Omega \subset H^3_{R_H}$. Assume the finite response map has quantitative observability

$$\alpha\bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad \alpha > 0,$$

and the total error obeys $|e_i|_W \le \sigma_i$. If $\mathcal{N}_\varepsilon$ is an ε-net of Ω and $\widehat{X}_i(t)$ is a residual minimizer up to tolerance τ_i, then the conditioned feasible frame support obeys the conservative enclosure

$$S_i(t) \subseteq B_H\Big(\widehat{X}_i(t), R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i + \frac{1}{\alpha}\tau_i\right]\Big).$$

The theorem does not assert that every point of this output ball fits the data. Exact continuous data have a unique minimizer. Finite noisy data produce the displayed frame ball; a unique finite frame value is licensed only when the certified residual gap $\Delta_{\mathrm{loc},i}(t) > 0$. If that gap fails, the fail-closed protocol returns AMBIGUOUS, meaning that no unique frame value is certified; it is not a proof that the mathematical data have multiple minimizers. This theorem does not locate an event, populate the event base, resolve unlabeled mixtures, identify particle species, produce stress-energy, or establish chart-blind neutral bulk.

Theorem 4.3c (Quotient-intrinsic geometry producer and dimension-selection boundary). On a refinement tower of repaired quotient normal forms, form at each level the support-visible incidence complex: vertices are the quotient-visible patch classes and simplices include every jointly supported patch set, including all higher simplices. A filled tetrahedron

is not its spherical boundary. Each level is computable from its normal form and is invariant under repair schedule and gauge. Refinement generally changes the literal complex. Topology preservation requires a certified subdivision or homeomorphism: a complete midpoint subdivision has such a realization, whereas its endpoint-valued simplicial projection is only homotopic to it. Simplicial maps or CPTP transport alone do not establish topology preservation. If this compatible tower carries the computable receipts (i) spherical incidence (connected closed orientable combinatorial surface with Euler characteristic 2), (ii) disk/mesh nondegeneracy, (iii) modular cross-ratio Cauchy convergence, and (iv) independently normalized geometric 2π-KMS comparison with wrong-normalization separation, then the refinement limit produces the round S^2, its conformal structure, the oriented round-cap family with cap normals, and every clause of the finite cap-normal BW certificate, with quantitative error envelopes. Conversely, six finite repair systems with one rewrite signature (three patches in the seed record, one repair step) have incidence complexes realizing S^2, T^2, the real projective plane, the Klein bottle, the 2-skeleton of $\partial\Delta^4$, and a wedge of two spheres, and the scalar KMS receipt admits any modular temperature β: repair confluence alone does not select topology, orientability, dimension, or modular normalization. On a connected closed surface the orientability clause of (i) follows from Euler characteristic 2, so the two nonorientable systems also fail (i) at that clause. Orientation framing enters as a separately typed receipt: the two coherent orientations of the S^2 system share every confluence-level datum, and a mirror pair of embeddings carries complex-conjugate cross-ratios. Finite cap reconstruction also requires a resolved spacelike boundary-circle plane and an inside witness or equivalent oriented boundary data to distinguish the two sides, including major caps. Cross-ratios determine only conformal data up to Möbius transformations; a declared observer frame is needed for angular radii, and a physical scale remains separate. The MGNS state enters as a separately typed receipt, and no countermodel artifact is exhibited for it. The screen S^2 is therefore *produced on the receipt branch*; the receipt set is one sufficient certificate, but no semantic-minimality theorem or full-tower countermodel matrix is supplied. Producers that reach an oriented S^2 support limit by another computable function of the normal forms, such as the source-derived incidence nerve of the carrier-support bridge, are untouched by these countermodels. (Spacetime and Einstein paper, geometry-producer packet; machine receipts accompany the public geometry code.)

Theorem 4.3d (Null-net standardness, derived translations, and the four-translation assembly). On the producer branch, the null half-line/interval blow-up net at a produced cap boundary point lives on one common GNS space, and: (i) the reference vector is cyclic and separating for every half-line used, given faithful finite-stage states on the interior MaxEnt branch, the mixed-GNS Cauchy clause, and a cofinal cyclicity receipt uniform over every half-line in the construction; boundary MaxEnt solutions require a support-restricted treatment and do not supply faithfulness by themselves; (ii) the produced geometric cap flow derives the half-sided covariance relation without assuming translation unitaries, but the Borchers–Wiesbrock step additionally consumes standardness for every nested pair—cyclicity and separation for both half-line algebras and cyclicity for their relative commutant; interval cyclicity is therefore a registered standard-relative-commutant receipt, not a consequence of one half-line receipt; (iii) weak additivity of the null net is a separate cofinal receipt used for the generated net, rather than a consequence of finite-stage faithfulness; (iv) on the exact-Markov/central-interface branch, with a specified endpoint-to-cell map and a uniform local norm/site-density bound, the reduced interval modular Hamiltonians split into local terms plus uniformly bounded interface blocks (endpoint-Lipschitz), while an explicit four-qubit nearest-neighbour Gibbs witness shows finite-range locality alone does *not* imply this (machine-verified); (v) on the standard half-sided-inclusion and modular-intersection receipt branches, the derived positive null translations and modular dilations generate Möbius covariance and the projective bounded-interval action; and (vi) the per-direction positive generators commute,

transform with the null-vector weight under the produced Lorentz action, satisfy the dependent-family linearity relation, and assemble into one four-parameter translation representation with joint spectrum in the closed future cone: a positive-energy Poincaré representation. (Spacetime and Einstein paper, null-net standardness packet.)

Theorem 4.3e (Source-derived finite $1+3$ precursor and effective-continuum criterion). An authenticated semantic-event log supplies its finite event carrier and strict generated read-from order. Its canonical source height is computed from that parent relation: roots have height zero and every nonroot has one plus the maximum height of its direct parents. Lean proves strict increase on parent edges and generated ancestry, bounds every authenticated direct-parent chain ending at an event by that event's height, and constructs an attaining chain. Thus source height is exactly the longest authenticated-parent-chain length. Let V_{src} be the exact rank-three positive source Gram quotient. Then $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ has real dimension four, and $Q(t,x) = t^2 - g_{\mathrm{src}}(x,x)$ has inertia $(1,3)$. Every source-unit direction gives the future-null vector $(1,u)$, realizing the exact equivalences $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$. For a supplied spatial readback $x(e)$ and positive height scale τ, the placement $F(e) = (\tau h_L(e), x(e))$ carries authenticated parenthood, and hence generated ancestry, into the future cone whenever every parent displacement obeys the source-metric speed bound. If every future-cone comparison between placed events is supported by reflexive generated ancestry, order and cone agree exactly. Lean derives this support from two concrete finite checks: spatial readback separates each equal-height layer, and every increasing-height pair unsupported by ancestry obeys the strict spacelike inequality. Antisymmetry then derives injectivity, every source interval agrees with the corresponding cone interval on the placed image, and composition with the source-carrier equivalence gives a separated one-chart finite causal-chart interface. This global finite chart is not an open manifold atlas. Thus the ambient target-carrier dimension and Lorentz signature are constructed without a freely declared rank-four chart, fitted cone, or supplied placement rank; the intrinsic poset dimension, spatial placement, speed/converse conditions, physical interpretation, and clock remain separate data. The exact-embedding route to a smooth effective manifold uses one cofinal source-selected family carrying physical event/link identification, exact two-way cone/order agreement, refinement compatibility, dense and isotropic source-S^2 link directions, a calibrated $\#I/\rho \to \mathrm{Vol}(I)$ law, independent dimension/manifoldlikeness and height/profile tests, stable thickened-antichain topology or homology, and a specified distinguishing Lorentzian limit with uniqueness control. On that certificate, causal order fixes conformal geometry and the calibrated measure fixes scale. Smooth Einstein promotion additionally requires either same-family tensor-curvature reconstruction converging to the Einstein tensor or the independently premised continuum small-ball/null-balance identification, together with physical stress, Ward/Bianchi, coupling, and remainder data on the same family. Scalar-curvature convergence alone is diagnostic. No physical instance of this certificate is supplied. (Spacetime and Einstein paper; Lean theorems and independently verified simulation evidence.)

A source-record order and count limit. In the continuous rank-three carrier, conservative Fibonacci records give q^3 distinct spatial sites. Repeat this population at every layer and let each event read all preceding-layer sites inside a radius a_q, including itself. With spatial fill distance $h_q = o(a_q)$, $a_q \to 0$, and model duration a_q/c, actual read paths approach the speed-c causal cone. Equal-volume cell assignments with vanishing displacement error make normalized raw counts converge to $dt\, d^3x$; the ordering fraction in interior timelike diamonds tends to $1/10$. This is a flat effective spacetime theorem for a specified population, read law and model clock (Spacetime and Einstein paper). It supplies neither their native dynamical selection nor their identification with a matter action or laboratory clock.

The same histories admit a retrospective clock. For intervals approaching fixed interior timelike diamonds, the fourth root of their inclusive count ratio converges to their proper-time ratio. A

reference interval fixes the unit, and complete ancestry access supplies the count. This readout needs no oscillator or numerical timestamps. Operational covariance gives a separate cone criterion: a nonzero closed convex pointed cone invariant under the source icosahedral rotations and all boosts along one axis must be one of the two Lorentz cones. A time orientation selects the future cone. The covariance must act on possible influences and readouts. Algebraic direction maps alone do not establish that operational premise[17].

For Einstein promotion there are two distinct sufficient routes at this step: an explicit same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or a separately premised continuum small-ball/null-balance identification. Convergence of a scalar-curvature estimator is an independent diagnostic and is insufficient by itself.

Every finite semantic log also admits an explicit auxiliary separated one-way placement in the source (1+3) carrier. Enumerating events along one source axis and taking height scale one larger than the event count makes the event map injective, sends generated precedence into the future cone, and makes distinct same-height events spacelike. The enumeration is arbitrary and generally introduces cone comparisons unsupported by ancestry; it is not an order-reflecting, source-selected, or physical placement.

The provenance grammar has the complementary exact expressivity theorem. Every finite decidable irreflexive transitive relation R compiles into an abstract authenticated semantic log whose direct parent relation and generated strict order are both exactly R; predecessor cardinality supplies the rank. The generic parent relation is all of R rather than its Hasse reduction alone. The relation is supplied to the compiler, and the per-event snapshots are not proved to arise from one threaded executor. This is a universal finite-poset representation result, not a dynamics, manifoldlikeness, or continuum theorem.

An exact four-event Boolean diamond inhabits the order-faithful finite interface: its parent edges are null and its two independent branches are spacelike. This is a machine-checked non-chain causal-set-like witness, not a physical continuum or an empirical dimension measurement.

The finite provenance order is causal-set-like rather than a physical causal set. Lean packages reflexive authenticated ancestry as a partial order and proves every interval finite at each finite cutoff, exactly realizing the abstract order-theoretic causal-set axioms. Standard causal set theory starts from a locally finite order interpreted as proto-causality [1, 2]. The exact-embedding continuum route requires an order embedding whose image has the approximately Poisson count density needed for a causal-set faithful embedding, together with manifoldlikeness and dimension [3, 4, 5]. Under the relevant distinguishing hypotheses, continuum causal order fixes the conformal geometry [14, 15]; the volume datum fixes the conformal factor. The Lorentz-invariant causal-set approximation uses Poisson sprinkling [13]; the OPH commit log is not a Poisson sprinkling, and its event count is not a spacetime volume. The finite source-causal theorem supplies the (1+3) precursor, not a continuum geometry. It derives neither a physical order embedding nor the count-to-volume density law that together constitute a causal-set faithful embedding. It also establishes neither continuum existence nor the causal-set Hauptvermutung from the finite OPH order. The finite causal/Lorentz precursor instead constructs the ambient carrier from an independent real axis and the exact rank-three source carrier. A spatial event readback, canonical source height, and positive `timeScale` then give a concrete placement; a local speed bound proves one-way cone compatibility, while equal-height spatial separation plus strict spacelikeness of unsupported increasing-height pairs supply converse cone support and exact two-way agreement. Antisymmetry then derives injectivity, while the two-way equivalence preserves every source interval exactly on the placed image. The event-count measure is distinct from public capacity N. The finite join does not constitute spacetime. Physical event/link identification, refinement, manifoldlikeness, topology, independent dimension tests, count–volume calibration, and uniqueness are required before the con-

tinuum can be read as emergent rather than fundamental. The exact source-direction bridge gives $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$ for the rank-three source Gram quotient and fixes the finite (1+3) carrier and cone after adjoining an independent real axis. Positive `timeScale` times canonical source height enters only through event placement; supplied event-local Lorentz frames preserve the rest metric. This is an algebraic direction and causal-carrier result, not a physical sky or signal theorem. The effective reading additionally requires the physically faithful images of authenticated local read-from links to cover this S^2 densely and isotropically; that coverage is not supplied. The measure calibration separately requires $\#I/\rho \to \mathrm{Vol}(I)$. Refinement must preserve order, directions, and count density while soldering rank-three neighborhoods with $|B_{\mathrm{car}}(r)| \asymp r^3$ over a scaling window. This is compatibility of the four inputs, not a fifth one. Neither the local rank-three carrier nor the seeded global finite wiring determines event-base dimension.

The theorem-status composition is exact at the finite interface and conditional thereafter. The source-native causal-chart interface uses semantic commits as the event carrier and the generated reflexive order definitionally; it uses no separate population map or duplicate precedence field. The finite compatibility theorem constructs the ambient (1+3) carrier, proves its rank and source-unit/null identities, and proves the one-way or two-way cone result under the explicit placement conditions. Supplied chart transports retain their bounded algebraic covariance role. No physical placement, count-density field, or refinement family is constructed. Physical faithfulness, link-direction coverage, manifoldlikeness, topology, volume, and refinement conditions remain independently testable.

The exact same-event-type source-order Einstein theorem, in a separately supplied $3+1$ tensor interface, replaces an all-null balance hypothesis by nine supplied balances on fixed algebraic source-direction representatives. Given supplied symmetric geometry/stress fields indexed by the finite event type, four discrete step maps, Ward/Bianchi identities, step connectedness, a base event, and an algebraic coupling, exact nine-direction tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and the Einstein-form relation. The minimal premise bundle contains no normalization, vacuum-calibration, Newton-constant, or universal-source field. The generated order and the independently derived $1 + 3$ carrier use the same event type, but no provenance link selects a tensor field, step, or balance. The theorem does not identify tensor-coordinate differences with those of the constructed carrier or identify the supplied fields with smooth curvature or physical stress-energy. That reading requires the source-causal continuum, count–volume, convergent curvature/stress, small-ball, coupling, and scale receipts on one family.

Proposition 4.3e0 (Bounded algebraic event-frame soldering). Let raw record representatives carry an actual coincidence setoid and a readback into the intrinsic Hermitian Lorentz module that is constant on coincidence classes. The readback descends uniquely. Given one time-oriented affine Lorentz overlap cocycle and, separately, a supplied chart-coordinate family satisfying its affine overlap law, coordinate displacements transform through the cocycle's linear part, their Lorentz intervals are chart independent, and the reverse transition and triple-overlap laws follow from the same cocycle. Future-null displacements descend to covariant celestial directions. One base-chart future-unit frame selection induces compatible frames in every chart, and Lorentz-orthogonal rest projections transport by exact rest-metric isometries. The rank-three source Gram quotient is linearly and isometrically equivalent to the standard internal rest fiber, as a candidate local readout. The standalone quotient-descent theorem does not construct this atlas coordinate family or identify the descended readback with it. A finite two-event/two-chart control has nonzero affine translation and a nonzero future-null displacement. A separate three-record control proves that a reflexive symmetric pairwise-overlap relation need not be transitive, so it cannot replace the coincidence setoid. This bounded algebraic adapter assumes the supplied chart-coordinate overlap law, coincidence setoid, separation, physical cone attachment, refinement naturality, and

operational clock. It can be used after a source-causal continuum certificate has selected an effective atlas, but it does not construct the event carrier and does not supply physical causality, manifoldlikeness, topology, count–volume calibration, or smooth curvature.

Proposition 4.3ea (Finite signed-seam operator realization). A declared sign-twisted derivative on a finite seam complex with unit-counting measure gives a positive semidefinite kinetic operator. Its kernel consists of parallel sections; on a connected frustrated component that kernel is zero. The finite realization with 8,662 sites and 11,816 seams has 38 frustrated triangles. Exact boundary-rank elimination gives lift-ambiguity rank 3,117, and deterministic integer sections check adjoint, kinetic, covariance, gluing and boundary identities. The separate rank-three response and declared fifteen-state charge assumptions define a conditional 45-dimensional internal factor. The declared operator $D_\sigma \otimes I_{45}$ retains the gap of D_σ with multiplicity 45. These are properties of the specified finite operator and matter table. They select neither a spacetime chart nor a physical matter action, and do not identify the domain with the twelve-port Spin construction. The microphysics paper gives the finite algebra and independent replay [28].

Theorem 4.3f (Finite null tomography and conditional local-stress upgrade). A symmetric bilinear form on $(\mathbb{R}^4, \eta)$ is determined by its null-null values up to a multiple of η, and a directional scalar family arises from a tensor iff it satisfies the finite dependent-family linearity relations; nine null directions with resolved rank nine reconstruct the trace-free part. Nine readings always interpolate and therefore do not test tensoriality. Additional dependent directions test consistency within a declared total measurement and numerical error budget; they do not constrain unsampled rays without a family-level premise. For a full-rank design using rays normalized to $k^0 = 1$ and a Frobenius-orthonormal trace-free basis, the tensor Frobenius error from charge noise is bounded by $\delta/s_{\min}$, separately from geometry and solve errors; floating-point singular values are estimates, not interval certificates. The released code verifies this finite linear algebra on declared and synthetic charge families. Values violating a dependent-family relation admit no exact rank-two reconstruction, and scalar collar CMI alone cannot define a rank-two source.

The physical upgrade is conditional. If one common source-derived endpoint-charge family on the event region satisfies the linearity relations, admits the declared same-source identification with local null-stress charges, carries one common normalization and Poincaré-covariant local transformation law, and satisfies the weak Ward identity, then tomography defines a symmetric semiclassical $\langle T_{ab} \rangle$, with null projections equal to those charges and ambiguity ϕg_{ab} plus the declared improvement terms. Locality, covariance, the charge/stress identification, and weak conservation are separate weak-Ward and null-stress matching supplies; they are not consequences of finite tomography or of the synthetic reconstruction receipt. Universal coupling to the geometric branch is a further named premise. (Spacetime and Einstein paper, stress packet; finite tomography receipts accompany the public geometry code.)

Theorem 4.3g (Repaired entropy bridge and absolute Einstein equation). On the central-interface branch, write the state as a direct sum of bulk states tensored with maximally mixed edge factors. Its entropy splits exactly into $S_{\text{bulk}} = H(p) + \sum_\alpha p_\alpha S(\rho_\alpha^{\text{bulk}})$ and $S_{\text{edge}} = \sum_\alpha p_\alpha \log d_\alpha$. At a faithful reference σ, the tangent first law is $\delta S = 2\pi\delta\langle B_C \rangle + \delta\langle Z_C \rangle$; for a finite change at fixed $K_\sigma = 2\pi B_C + Z_C = -\log\sigma$, the exact identity is $S(\rho) - S(\sigma) = 2\pi\Delta\langle B_C \rangle + \Delta\langle Z_C \rangle - D(\rho\|\sigma)$, with $Z_C = \sum_\alpha (\log d_\alpha) P_\alpha$. Hence $\delta S_{\text{bulk}} = 2\pi\delta\langle B_C \rangle$ and $\delta S_{\text{edge}} = \delta\langle Z_C \rangle$ throughout the declared central-interface class, including variations of the sector weights. The supplied nonconstant Gibbs family has the analytic MaxEnt envelope identity $dS/dt = \lambda$; setting $\lambda = 2\pi$ and identifying t with physical energy require the declared branch normalization, giving the Clausius balance $\delta S_{\text{edge}} = \lambda\delta t - 2\pi\delta\langle B_C \rangle$ on the coupled variation class. Reaching the geometric balance additionally requires the declared fixed-cap generalized-entropy stationarity interface, the conditional same-source local-stress and weak-Ward bridge, the continuum diamond kernel, and the fixed-volume

small-ball area identity. One common scaling family must carry the conditional-mixing, boundary-count, fixed-collar, sharp-rate, and $o(\ell^4)$ remainder bounds. Coverage of all local timelike directions follows from population plus the produced Lorentz orbit. The vacuum-reference premise fixes the first-variation integration constant. Independent scale readouts with trivial positive-rescaling stabilizer identify the structural coupling. Under all of these premises, $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$ per connected component. Without the vacuum reference, the result fixes the residue tensor only up to one covariantly constant metric term. (Spacetime and Einstein paper, entropy-bridge and small-diamond packets.)

Theorem 4.3h (Conditional Einstein branch composition). Bare finite consensus is not Einstein-complete: the same finite-consensus reduct admits extensions in which the Einstein equation holds and extensions in which it fails. The valid gravity result is conditional. Let one source-derived repaired quotient tower carry deterministic geometry, modular, causal-order, stress, entropy, and scale readouts on a common domain, with commuting refinement diagrams. Assume the finite source-causal precursor, geometry and null-net receipts; cap-interior modular data; one physical source-causal continuum certificate on a cofinal family with either same-family tensor-curvature reconstruction or the independently premised continuum small-ball/null-balance identification, for which every relevant remainder is $o(\ell^4)$; the universal-coupling and vacuum-reference receipts; and two independent scale readouts with trivial positive-rescaling stabilizer. The stress-tomography, entropy, curvature-identification, timelike-polarization, Ward, and Bianchi results then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$$

on each connected event component. At the curvature-identification step, the causal-set route requires a same-family tensor-curvature reconstruction converging to G_{ab}; the alternative continuum route instead premises the small-ball/null-balance identification independently. Convergence of a scalar-curvature estimator alone establishes neither route. The exact finite layer includes the nine-direction tomography frame with determinant $8192/27$ and the entropy and MaxEnt identities. Conditional on the continuum diamond-kernel formula and the smooth fixed-volume small-ball area identity, exact arithmetic gives the coefficients $8\pi^2\ell^4/15$ and $-4\pi\ell^4/15$; the continuum formulas themselves remain explicit analytic premises. The finite operator of Proposition 4.3ea does not supply the geometric modular, stress, entropy, scale, or asymptotic data of this theorem. No source tower satisfying the joint continuum antecedent is constructed here, and no semantic-minimality theorem for that antecedent is claimed. (Spacetime and Einstein paper, composed branch-entry theorem.)

The register surface of this composition is inhabited. For every declared repair law and every horizon record over it, an explicit value of the typed premise bundle is machine-checked: its modular flow supply is the finite thermodynamic first-law datum on the simplex tangent space at the declared reference, with the boost and central charges of the record, so the first-law clause of the composed conclusion at this value is the first-order entanglement first law of the horizon-thermality surface on the same variation space. The other supplies are synthetic witnesses at the demo tier: the two-patch consensus tower with six typed readouts extending the committed event and geometry fragment, a sixteen-point four-direction torus chart with a nonconstant divergence-free stress field and the geometry defined as $\kappa T_{ab} + \Lambda\eta_{ab}$, a strict audit manifest evaluated on that chart, a shrinking family with ℓ^5 error channels, and small-ball reals solved from the three small-ball identities. The Einstein clause, the small-ball arithmetic, the universal-source clause, and the vacuum reference hold at this value by definition of those witnesses and select nothing. A nonvacuity receipt exercises the clauses the data can fail on nonzero data, and a counterexample shows that the split of the horizon record is the load-bearing input: with a uniform reference and zero central charge in place of the split, the same boost charge fails the modular-generator clause on the same variation. The value certifies

joint satisfiability of the premises and places the composed branch on the thermodynamic variation space; it attaches no physical metric, stress tensor, horizon, temperature, or Newton constant, and realized nonemptiness on a source-generated tower is not established.

2.6 Gravity from Fixed-Cap Generalized-Entropy Stationarity

2.6.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted geometric subnet, the scaling-limit cap modular flow is geometric with the standard 2π normalization,

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to be type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\mathrm{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\mathrm{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\mathrm{geo},C}$ the first-law identity reads

$$\delta S_C = \delta\langle K_C\rangle = 2\pi\,\delta\langle B_C\rangle.$$

This displayed variation is the special type-I/fixed-central-sector form of $K_C = 2\pi B_C + Z_C$, where the central term has no first-order contribution on the chosen sector. In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity; the theorem there is the geometric modular automorphism statement, so the cap first-law formula is restricted to this special realization.

2.6.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage, the exact endpoint-derivative identity for its half-line generator, and a conditional local-charge reading. At fixed cutoff one first transfers cut-center data to regulated null strips, then imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore the half-sided covariance relation. Together with the separately supplied common-vector and relative-commutant standardness receipts, this is a standard half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. Bounded-interval formulas are discharged by the E0.5 affine/projective kernel below, and the tensor upgrade carries the null-invisible metric ambiguity. Accordingly the null modular bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- the geometrically derived half-sided relation plus the registered standardness receipts on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density-upgrade template is kept below as an explicit downstream template rather than as a fixed-cutoff theorem proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit, and Theorem 5.2g records the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, and the exact half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$. Identifying that endpoint derivative with a local null-stress charge additionally consumes an effective-locality and same-source matching premise. Bounded-interval formulas are downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W_{\alpha_-}^* \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W_{\alpha_+}^* \otimes \mathcal{H}_{i_+^{\alpha_+}},$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma\in\mathfrak{M}_{I_-:J:I_+}} \|\rho-\sigma\|_1 : I(I_-:I_+\mid J)_\rho \le \varepsilon \right\}.$$

Also define the EC-aligned subset $\mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}$ by requiring the strip MSA on the inherited split. For a declared controlled strip family $\mathcal{F}_J$, let

$$\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma\in\mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}} \|\rho-\sigma\|_1 : \rho\in\mathcal{F}_J,\ I(I_-:I_+\mid J)_\rho \le \varepsilon \right\}.$$

The generic exact-Markov modulus need not control this aligned one; convergence of the latter is a separate controlled-family hypothesis. For any strip state η, let

$$\Delta K_J(\eta) := K_{I_-\cup J\cup I_+}(\eta) - K_{I_-\cup J}(\eta) - K_{J\cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov and EC-aligned on the inherited split (the strip form of the Markov-split alignment hypothesis of Section 2.3), then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,J}(\omega)\in Z(\mathcal{A}(J))$ such that

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $\omega\in\mathcal{F}_J$ and $I(I_-:I_+\mid J)_\omega\le\varepsilon$, assume the aligned modulus tends to zero and choose $\widetilde{\omega}_J\in\mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}$ with

$$\|\omega-\widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega,\widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega,\widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J) = \Delta K_J(\widetilde{\omega}_J)\in Z(\mathcal{A}(J))$. Every bounded observable on $I_-\cup J$ or $J\cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty\,\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_*>0$, then

$$\|\mathfrak{D}_J(\omega,\widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\,\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at EC-aligned exact Markovity or in a controlled strip family on one fixed inherited strip model with $\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon_J)\to 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the EC-aligned exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: alignment identifies the HJPW

blocks with the inherited blocks, the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. Without alignment, HJPW factorizes the state only over its own state-dependent split, and the centrality of ΔK_J in $Z(\mathcal{A}(J))$ is not available. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω, let $K_\partial(I,\Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I,\Omega] := K[I,\Omega] - K_\partial(I,\Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a,\infty),\Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). Work on the exact-Markov/central-interface inherited-strip branch, and assume the endpoint-to-regulator-cell map and uniform local norm/site-density bound of the endpoint-control clause in Section 2.7, uniformly on the allowed compact endpoint window. For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I,\Omega] := K[I,\Omega] - K_\partial(I,\Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a,\infty),\Omega],$$

the additional branch-internal endpoint-control gives:

$$\big|\langle\psi, (\widetilde{K}[(a',b'),\Omega] - \widetilde{K}[(a,b),\Omega])\phi\rangle\big| \le C_{\psi,\phi,\Omega}\big(|a'-a| + |b'-b|\big)$$

for bounded intervals in a compact endpoint window, and

$$\big|\langle\psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\big| \le C_{\psi,\phi,\Omega}|a'-a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle\psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle\psi, q(a,\Omega)\phi\rangle := -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle\psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi\int_a^b \langle\psi, q(v,\Omega)\phi\rangle\, dv.$$

$q(a,\Omega)$ is a weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The assumed endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a'-a| + |b'-b|$; for half-lines it is exactly $|a'-a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an L^∞_{loc} derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a,\Omega)$ is weakly differentiable in a and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle\psi, p(a,\Omega)\phi\rangle := -\partial_a\langle\psi, q(a,\Omega)\phi\rangle = \frac{1}{2\pi}\,\partial_a^2\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle$$

and obtain the half-line formula

$$\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi\int_a^\infty (v-a)\,\langle\psi, p(v,\Omega)\phi\rangle\,dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v=0$. For $a \geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a \leq b \quad\Longrightarrow\quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular relation on null half-lines). On the scaling-limit geometric-cap branch of Theorem 4.2, assume that the common reference vector is cyclic and separating for every half-line algebra used. The blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v \mapsto e^{-2\pi t}v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \leq 0),$$

so the inclusion $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ obeys the half-sided modular relation. After the harmless convention change $t \mapsto -t$, this has the standard positive-time orientation. The relation is derived from the null-interval structure, isotony, and the scaling-limit geometric action. It becomes a *standard* half-sided modular inclusion only when the reference vector is also cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$; that relative-commutant standardness is a separate receipt, not a consequence of the geometric action.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c,d) \subset H_a$, this sends $\mathcal{A}((c,d),\Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d),\Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t \leq 0$, then $e^{-2\pi t}a \geq a$, so $H_{e^{-2\pi t}a} \subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). Fix $a > 0$ and assume that the common reference vector is cyclic and separating for both algebras below and cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$. For the resulting standard half-sided modular inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

In addition,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas are downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the half-sided modular relation, and the hypotheses of this lemma supply standardness of the inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator P_Ω, and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (Half-line generator and conditional EFT null-stress reading). In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. If that same renormalized half-line family admits an effective local-stress description and a same-source identification, its null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The OPH half-line theorem fixes P_Ω; reading it as a local null-stress charge consumes the effective-locality and same-source identification premise rather than deriving that premise. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ is discharged by E0.5 below, and reconstruction of a full tensor from the directional charges is subject to the null-invisible metric ambiguity.

Boundary of the null bridge. The null bridge is an explicit theorem branch rather than an automatic fixed-cutoff identity. Its required data are:

- transferred cut-center data alone give the central sector decomposition of the null strip rather than the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise uses a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the exact-Markov/central-interface branch, together with its specified endpoint-to-cell map and uniform norm/site-density bound, gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines; the scaling-limit geometric-cap branch of Theorem 4.2 then derives the half-sided relation on the half-line blow-up net, while common-vector and relative-commutant standardness remain separate receipts;
- with those standardness receipts, the Borchers–Wiesbrock theorem supplies $P_\Omega \geq 0$, its Stone domain, and $K_a = K_0 - 2\pi a P_\Omega$;
- the half-line generator and its endpoint-derivative identity are fixed inside the bridge itself; the local null-stress reading requires the separately supplied effective-locality and same-source bridge, while bounded-interval transport and the later tensor upgrade are downstream.

This is the route from null-strip factorization to the Einstein branch, with the positive half-line generator proved inside the null modular bridge and its physical stress reading conditional on the named EFT bridge.

2.6.3 Conditional modular-energy stress-tensor reading on null half-lines

The effective-theory input here is the standard stress-tensor representation that names the operator fixed by the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

If that same half-line family admits the supplied effective local-stress description and same-source identification, its local null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator has the displayed local null-stress reading conditional on an effective description that admits a local stress tensor and identifies the endpoint derivative with the same source family. The algebraic half-line construction alone does not supply that effective-locality or source-identification bridge. In a UV CFT regime, or another local Lorentzian regime with local modular Hamiltonians, the conditional reading is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab}\zeta^b \, d\Sigma^a,$$

so, under that EFT premise, the conditional identification is the null-half-line version of the same modular-energy locality.

The downstream boundary is narrower. The E0.5 bounded-interval kernel transports this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade determines T_{ab} only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

The generator is fixed inside the null bridge; its physical local-stress interpretation remains conditional on the stated EFT bridge.

2.6.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition, exact Markovity, and MSA, the declared central-interface branch has the following edge-aligned Markov normal form. On the nonexact branch, Theorem 2.5 supplies a recovered comparison state and a vanishing CMI envelope only under strong conditional mixing; convergence to an exact Markov state is a fixed-collar statement, and alignment with the displayed edge factors remains a separate MSA condition. Under those conditions MaxEnt selection within each edge sector gives

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^\alpha_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^\alpha_{\mathrm{edge}}}{d_\alpha} \right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\mathrm{bulk},C}) + \sum_\alpha p_\alpha \log d_\alpha.$$

Convention: Throughout this paper, "log" denotes the natural logarithm (ln), so entropies are measured in **nats** (1 nat $= 1/\ln 2 \approx 1.443$ bits). This is standard in thermodynamics and QFT; the

Bekenstein-Hawking formula S = A/4G uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\mathrm{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\mathrm{bulk},C}),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\mathrm{gen}}(C) := \mathrm{Tr}(\rho L_C) + S_{\mathrm{bulk}}(C).$$

Newton coupling as an edge-entropy area-law readout.

In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\mathrm{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \qquad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where N_Σ is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (the fixed-cutoff edge-sector law of the gauge-structure paper [18]). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\mathrm{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.

Matching these expressions gives the geometric area-law coupling:

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\,\bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\mathrm{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

On the OPH gravity branch this formula does not back-solve the scale. The selected no-G scale certificate supplies $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. The local cell readings are

$$a_{\mathrm{cell}} = P\ell_\star^2, \qquad \bar{\ell}_{\mathrm{shared}} = P/4,$$

so the Newton area-law readout gives

$$G_{\mathrm{geom}} = \frac{P\ell_\star^2}{4(P/4)} = \ell_\star^2.$$

The SI display is $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$.

2.6.5 Finite MaxEnt identities and the fixed-cap generalized-entropy stationarity interface

The proved finite MaxEnt envelope identities hold within the declared state family. The calculation does not, by itself, identify a variation of the state constraints with a geometric area variation or prove stationarity of their sum. The Einstein branch therefore assumes the following joint state–geometry interface: for its admissible variations preserving cap labels (fixed size and charges),

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Once this balance is supplied, the split above and the tangent first law for the bulk term give

$$\delta S_{\mathrm{gen}}(C) = \delta\langle L_C\rangle + \delta\langle K_{\mathrm{bulk}}\rangle.$$

2.6.6 No-smuggling dependency discharge for the Einstein bridge

Theorem E0 (Einstein bridge dependency discharge). Write E for the Einstein-branch premise package: the premises A1–A3 together with the collar-recovery, edge-normalization, and generalized-entropy interfaces consumed by this branch, each an explicit hypothesis outside A1–A3. Let one source-derived cofinal OPH regulator tower carry the typed common-domain geometry, modular, event, stress, entropy, and scale readouts of Theorem 4.3h, together with its finite, asymptotic, and physical-identification premises. On that branch, E and the additional receipts supply the inputs used by the Einstein bridge:

- geometry readout through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;
- Lorentz/H3 readout from round cap pairs, since $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$;
- 2π-normalized support-visible BW flow on the extracted geometric subnet;
- the positive null generator on the same branch and, under the separately supplied effective-locality and same-source bridge, its local null-stress charge reading;
- the bounded-interval kernel of Lemma E0.5;
- fixed-cap generalized-entropy stationarity from the separately declared stationarity interface;
- the standard fixed-volume small-ball area identity after geometry readout;
- $o(\ell^4)$ remainder control by Lemma E0.6; and
- all local timelike directions by the cap-pair coverage of Lemma E0.7.

Thus the finite consensus reduct supplies quotient normal forms. The typed branch receipts supply the geometric, modular, stress, entropy, coverage, tail, coupling, vacuum, and scale inputs needed by the Einstein bridge. No source-derived tower carrying all of those inputs is constructed here. Bare finite consensus does not imply the Einstein equation.

Lemma E0.5 (Conditional bounded-interval kernel). On the null blow-up of the support-visible geometric branch, assume the effective-locality and same-source bridge that identifies the proved half-line endpoint derivative with a local null-stress charge. Together with endpoint-Lipschitz renormalized interval control and affine/projective covariance, this gives

$$K^{\mathrm{null}}_{(a,b)} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a}\, T_{kk}(v)\, dv + R_{(a,b)}.$$

The remainder $R_{(a,b)}$ is the carried endpoint/collar term of the same refinement family. In the small-ball limit this is the ball modular kernel used below.

Lemma E0.6 (Uniform remainder control with its transfer bounds). Let (r, ℓ_r, δ_r) be one declared cofinal family with $\ell_r \to 0$ and faithful Gibbs states ρ_r, on which the strong conditional matrix-mixing premise of Theorem 2.5 holds uniformly over the event region and cap family. Set

$$\varepsilon_r = c|\partial C_r|_{\mathrm{UV}} e^{-\delta_r/\xi_r}.$$

The boundary-anchor expansion directly controls the operator norm:

$$\|\Delta K_r(\rho_r)\|_\infty = \|\mathbf{J}_{\rho_r}(A_r : D_r \mid B_r)\|_\infty \le \varepsilon_r.$$

Here the four canonical marginal logarithms define ΔK, so $\Delta K = -\mathbf{J}$. Exact faithful Markov references have zero canonical four-log defect. This direct bound requires no uniform spectral floor; it does not identify an EC splitting or control each individual modular Hamiltonian separately.

Assume separately that the needed EC-aligned replacements obey one modulus $d_r^{\mathrm{M,EC}} \le C_{\mathrm{M}} \varepsilon_r^\theta$, with $C_{\mathrm{M}} < \infty$, $\theta > 0$, and that for some

$$s > \max\{8, 4/\theta\}, \qquad \omega_r \longrightarrow +\infty,$$

the same family satisfies

$$\frac{\delta_r}{\xi_r} - \log\bigl(c|\partial C_r|_{\mathrm{UV}}\bigr) \ge s \log(\ell_0/\ell_r) + \omega_r.$$

Then

$$2\sqrt{\varepsilon_r} = o(\ell_r^4), \qquad d_r^{\mathrm{M,EC}} = o(\ell_r^4), \qquad \|\Delta K_r(\rho_r)\|_\infty = o(\ell_r^4).$$

Indeed $\varepsilon_r \le (\ell_r/\ell_0)^s e^{-\omega_r}$, and the exponents are $s/2 > 4$, $s\theta > 4$, and $s > 4$, respectively. For a downstream remainder bounded by a stage-independent constant times these three quantities, plus a separately controlled $o(\ell_r^4)$ kernel residual, the remainder is $o(\ell_r^4)$. Any growing transfer constant or further remainder must instead be included in the same-family tail receipt; the rate calculation does not bound it by itself.

On a weaker route that has only a trace-norm replacement estimate, the matrix-mixing bound above is unavailable. The logarithm comparison instead requires the complete rate

$$4\lambda_r^{-1} d_r^{\mathrm{M,EC}} = o(\ell_r^4),$$

where $\lambda_r > 0$ bounds the relevant marginals of both states below. For example, $\lambda_r \ge \lambda_0(\ell_r/\ell_0)^p$, with $\lambda_0 > 0$ and $p \ge 0$, gives the sufficient strengthened margin $s > \max\{8, (4+p)/\theta\}$, under uniformly bounded downstream transfer constants. Stagewise faithfulness alone supplies no such margin. This weaker route is not a replacement for the stronger hypotheses used by Theorem 2.5. Mere pointwise collar convergence also supplies neither common-family rate, and no per-radius diagonal choice is used.

Lemma E0.7 (Cap-pair timelike coverage). On the cap-normal H^3 branch, every future timelike vector v has $r = \sqrt{-\eta(v,v)}$ and $u = v/r \in H^3$. For any unit $s \in u^\perp$, the vectors

$$q_\pm = u \pm s$$

are future null and

$$v = \frac{r}{2}(q_+ + q_-).$$

Thus normalized future null directions in observer skies cover every local timelike direction once the observer-frame chart is supplied. This does not say that two unnormalized projective rays alone

select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required.

Proposition E2 (Conditional recovered-core Ward closure). Suppose the finite tomography hypotheses reconstruct a tensor from one compatible family of null and timelike charges on the recovered core. Assume separately that its same-source local charge family obeys the weak Ward balance: signed fluxes on opposite faces of infinitesimal recovered-core diamonds cancel with a controlled remainder, and this balance converges in the recovered charts to the distributional divergence law. Then $\nabla^a \langle T_{ab} \rangle = 0$ weakly, and pointwise on a smooth branch. Charge compatibility and tensor tomography alone imply neither the face cancellation nor its continuum limit.

2.6.7 Einstein equation from fixed-cap generalized-entropy stationarity

This subsection is a branch theorem. The finite consensus reduct supplies quotient normal forms; Theorem E0 discharges the recovered-core premise package E needed by the Einstein bridge: geometry readout, support-visible BW modular covariance, the conditional local null-stress reading under effective-locality and same-source matching, bounded-interval transport, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area variation, carried-remainder control, and the timelike scalar-to-tensor upgrade.

In the d=4 scaling regime, the small-ball bridge follows conditionally once the effective-locality, same-source, endpoint-control, and affine/projective hypotheses of E0.5 are supplied. The additional standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The kernel shape then follows on the extracted geometric subnet: a sufficiently small cap is assumed to admit the tangent causal-diamond description D_ℓ, whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\, \partial_t - 2t\, x^i \partial_i \Big).$$

On the t=0 slice its lapse is

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel combines the geometric cap generator with the affine/projective transport of the half-line stress bridge, so the cap-modular theorem and the stress bridge together yield

$$\delta S_{\mathrm{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00} \rangle\, d^3x + \delta\langle E^{(\eta)}_{C,\ell} \rangle.$$

If $\delta\langle T_{00} \rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00} \rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta\Big[(G_{ab} + \Lambda g_{ab}) u^a u^b \Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab} \rangle$$

for the local diamond rest-frame four-velocity u^a. In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00} \rangle.$$

2.6.8 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u. Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G \langle T_{ab} \rangle.$$

The local-diamond rest-frame relation says

$$u^a u^b \, \delta Y_{ab} = 0$$

for every such u. In local inertial coordinates,

$$u^a = \gamma(1, v^i), \qquad |\vec{v}| < 1,$$

so

$$0 = u^a u^b \, \delta Y_{ab} = \gamma^2 \Big(\delta Y_{00} + 2 v^i \delta Y_{0i} + v^i v^j \delta Y_{ij} \Big)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, Y_{ab} is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the geometric-cap theorem, the null bridge, and the overlap-complete timelike family are in place.

2.6.9 Numerical constants of the gravity chain

The gravity chain carries specific numerical constants. One is supplied by a certificate clause; the other is an identity.

The 2π KMS normalization. In Theorem 4.2 the standard 2π normalization of cap modular flow on the support-visible scaling-limit geometric cap pair is fixed by the certificate's geometric 2π-KMS comparison clause, a supplied receipt, with the same normalization as the Unruh and Hawking temperatures. In the Bisognano–Wichmann setting Lorentz covariance and the spectrum condition force it; here the certificate checks it.

The geometric coefficient $\Omega_{d-2}/(d^2-1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell} \, d^{d-1}x = \frac{\Omega_{d-2} \, \ell^d}{d^2 - 1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2 - 1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\mathrm{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Fixed constants**: 2π (KMS period, supplied by the certificate clause), $\Omega_{d-2}/(d^2-1)$ (geometric coefficient, an identity), the existence of the Einstein form under the stated premises.
- **Global closure constants**: G is the conversion between edge entropy and geometric area on the emitted local gravity branch. Λ is not fixed by the local reference-state variation data alone; stable correctable-record closure plus identification of the record carrier with de Sitter horizon entropy fixes $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, while the SI curvature scale also uses the selected OPH scale certificate.

2.6.10 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information
$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$
2. the constructive recovery error
$$r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta};$$
3. the exact-Markov replacement error on one fixed faithful collar model,
$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$
where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:
$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$
with
$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later "exact" collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with
$$\|\omega_\delta - \sigma_\delta\|_1 \le \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$
and a constructive recovered comparison state $\omega_\delta^{\mathrm{rec}}$ with
$$\|\omega_\delta - \omega_\delta^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon_\delta).$$
These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,
$$|\mathrm{Tr}[X(\omega_\delta - \omega_\delta^{\mathrm{rec}})]| \le \|X\|_\infty\, r_{\mathrm{FR}}(\varepsilon_\delta),$$
and
$$|\mathrm{Tr}[X(\omega_\delta - \sigma_\delta)]| \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$
The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \le \eta_\delta^{\mathrm{M}}.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^{M} of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\mathrm{FR}}(\varepsilon_\delta) + \eta_\delta^{\mathrm{M}},$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ-controlled approximations and then lets $\eta_\delta \to 0$ in the controlled collar limit.

Finite-cutoff Einstein remainder bound. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00}\rangle + \delta\langle E_C^{(\delta)}\rangle + O(\ell^5 \partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the declared fixed-cap generalized-entropy stationarity interface, together with the proved finite MaxEnt identities, gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi \ell^4}\, \delta\langle E_C^{(\delta)}\rangle + O(\ell\, \partial T),$$

and, by the bounds above,

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\, \eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model. Equivalently, before the controlled collar and small-ball limits, one may write

$$|\mathcal{E}_{\ell,\delta}| \le C_1 r_{\mathrm{FR}}(\varepsilon_\delta) + C_2 \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) + C_3 \eta_\delta^{\mathrm{reg}} + C_4 \ell\, \|\partial T\| + o_\delta(1) + o_\ell(1),$$

with constants depending only on the fixed collar model, the bounded observable class, and the small-ball chart. Thus the exact Einstein relation is the controlled limit, not a finite-cutoff identity inherited from approximate Markovity.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2 \ell^4}\, \delta\langle E_C^{(\delta)}\rangle.$$

Then

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta(\langle T_{00}\rangle + \langle T_{00}^{\mathrm{anom}}\rangle) + O(\ell\, \partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le \frac{15\, C_C}{8\pi^2 \ell^4}\, \eta_\delta.$$

On a separate conditional dark/anomaly branch, this carried scalar remainder may be identified with a repair stress only after a quotient-invariant source-localization receipt identifies the same collar operator with the small-ball Einstein remainder, and after a positive causal packet lift supplies the full tensor. In SI units that continuation has rest-energy normalization

$$T_A^{ab} u_a u_b = \frac{15\hbar c}{8\pi^2 \ell^4} R_{r,\ell} + O(\epsilon_{\mathrm{loc}}),$$

where $R_{r,\ell}$ is measured in nats and ℓ is the proper geodesic small-ball radius. The scalar quantity $I(A : D \mid B)$ alone is not a stress tensor: packet factorization, causal directions, an allocation rule, a rest frame, source-localization map, and transport law are additional continuation data.

The framework carries the Markov error through three distinct controls:

1. $r_{\mathrm{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η^{M}_δ controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from "premises about screens" to "precision GR predictions plus a disciplined carried remainder".

2.6.11 Focusing/QNEC boundary on the null-modular branch

The focusing content does not follow from premises A1–A3. The two focusing statements on the null-modular branch require the additional hypotheses stated below. Quantum focusing is an optional gravity interface and the Einstein theorem does not consume it.

What the branch supplies. The fixed-cutoff null bridge of §5.2 supplies transferred cut-center data and exact-or-controlled strip additivity on the inherited strip model. The geometric scaling limit supplies the half-sided relation; on the separately registered common-vector and relative-commutant standardness receipts, Borchers–Wiesbrock supplies its translation generator and the affine relation $[K, P] = -i2\pi P$. These are kinematic inputs a QNEC statement consumes; they are not a proof of it or an identification of (P) with a local stress charge.

Why relative-entropy monotonicity does not yield QNEC. Monotonicity under restriction orders the relative entropies $S(\rho_{R(\lambda)} \| \omega_{R(\lambda)})$ along nested null cuts; it fixes the sign of the first λ-derivative and says nothing about the second. A monotone function of the cut need not be convex, so the step from monotonicity to $d^2 S_{\mathrm{rel}}/d\lambda^2 \geq 0$ is unavailable. The entanglement first law $\delta S = \delta\langle K \rangle$ is a first-order state variation at fixed algebra and reference state; the QNEC second null shape variation is a different object, and expanding the first law to second order in the cut does not produce it. The published QNEC proofs supply exactly the missing machinery: null-surface locality and analyticity hypotheses in the free and super-renormalizable settings [55], and modular-flow, causality, and defect-OPE inputs in the general QFT setting [57]. None of those hypotheses is verified on the reconstructed patch net.

Why QFC is not derived from QNEC. The Quantum Focusing Conjecture is strictly stronger than QNEC: the foundational QFC statement obtains QNEC as its locally parallel limit [56], and no implication runs the other way. QNEC together with the Einstein branch (Theorem 5.1) and the Raychaudhuri identity controls the diagonal, stationary-cut part of the generalized-expansion variation and leaves the off-diagonal, shear, counterterm, and general nonstationary-cut content of QFC untouched. The optional quantum-focusing interface therefore carries no independent internal support from this branch.

Claim boundary. The null-modular branch supplies the half-sided modular structure but not an internal QNEC proof. Such a proof additionally requires the QFT-side hypotheses named above on the reconstructed net and the off-diagonal QFC content. Any theorem that consumes focusing must name the optional quantum-focusing interface explicitly.

Theorem 5.1 (OPH Einstein-branch closure in the scaling regime). Under the premise package E on a recovered-core, support-visible scaling branch satisfying the E0 dependency discharge above, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, the timelike scalar-to-tensor theorem yields the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity. The capacity branch closes that metric term globally under stable whole-fiber correctable-record closure, the unique finite-size slack zero, and the horizon–record identification.

Proof sketch. The recovered-core branch supplies the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, 2π-normalized BW cap flow, the conditional same-source local null-stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area identity, and all timelike directions. The cap first law identifies δS_C with $\delta\langle K_C\rangle$, while the null bridge together with its effective-locality and same-source premise and the bounded-interval kernel relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the declared joint state–geometry balance for admissible fixed-cap variations, combined with the proved MaxEnt envelope identities on the cap-label-preserving state family, yields the displayed rest-frame relation. The timelike-coverage lemma supplies all local timelike directions, the scalar-to-tensor theorem upgrades the relation to the tensor equation modulo Λg_{ab}, and the capacity branch closes the remaining metric term under its separately typed stable-capacity and horizon-record premises. QED.

2.6.12 Black-hole structural statements and continuation-level spectroscopy

The structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum with the leading semiclassical Schwarzschild first-law relation for an actual sector change. Finite same-boundary records do not supply physical radiation entropy, exterior time, evaporation flux, a geometric island or an asymptotic waveform readout.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha,$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α. With the normalization $\mathrm{Tr}(\rho L_C) = \langle A\rangle / 4G_{\mathrm{geom}}$, the area eigenvalues are

$$A_\alpha = 4G_{\mathrm{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

where $\ell_\star^2 = \hbar G_{\rm SI}/c^3$ is the scale-readout area displayed as the Planck area. Since d_α is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2/c^4$, define the signed change of the hole for a transition from $d_{\rm before}$ to $d_{\rm after}$ by

$$\Delta_{\rm BH} A = 4\ell_\star^2 \ln(d_{\rm after}/d_{\rm before}).$$

The differential first law, evaluated on the initial background, gives the leading semiclassical relation

$$\Delta_{\rm BH}(Mc^2) = k_B T_H \ln(d_{\rm after}/d_{\rm before})$$

with $T_H = \hbar c^3/(8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2. This is not an exact finite-difference identity: using the exact Schwarzschild mass–area relation, an emission with entropy loss $\ln k$ has $E_{\rm emit} = Mc^2[1 - \sqrt{1 - \ln k/S_{\rm BH}}]$, whose expansion is $k_B T_H \ln k[1 + O(\ln k/S_{\rm BH})]$.

Continuation templates. Spectroscopy needs an additional selection rule. If dominant emission steps reduce the edge dimension by an integer factor k, then $d_{\rm before} = k d_{\rm after}$, the signed black-hole entropy change is $-\ln k$, and to leading semiclassical order the emitted spectrum is the comb $\Delta E_k = k_B T_H \ln k$, $\Delta f_k = c^3 \ln k/(16\pi^2 GM)$. For Kerr remnants, observed hertz must use $M_{\rm det} = (1+z)M_{\rm source}$, and the $(k-1)/k$ detailed-balance factor is a within-line net-response factor, not a derived hierarchy of transition probabilities. The integer-transition rule, the burst and ringdown templates and their frozen tests are continuation premises of the cosmology layer and are not part of this paper's chain.

2.6.13 Classical consequences and precision tests on the conditional branch

If the physical source-causal continuum, curvature/stress, scale and convergence premises promote the finite Einstein-form identity to the smooth Einstein equation, its ordinary consequences apply: the contracted Bianchi identity makes stress-energy conservation automatic, pressureless matter follows geodesics, and the weak-field limit is Newtonian gravity. Light bending $4GM/(c^2 b)$, perihelion advance $6\pi GM/(a(1-e^2)c^2)$ per orbit and gravitational redshift are inherited consequences of that identification, not independent predictions of the informational order. On the additional pure two-derivative Einstein–Hilbert action branch the two transverse-traceless modes propagate on the invariant null cone of the Maxwell modes, so $(c_{\rm GW} - c_\star)/c_\star = 0$ on that branch, and the quadratic operator has no Fierz–Pauli hard-mass parameter, $\mu^2_{\rm FP,hard} = 0$; this is an action-content statement, not a graviton-pole prediction. The published multi-messenger bound $-3 \times 10^{-15} < (c_{\rm GW} - c)/c < +7 \times 10^{-16}$ and the model-dependent dispersion bound $m_{\rm grav,disp} \le 1.76 \times 10^{-23}\,\mathrm{eV}/c^2$ are comparisons with that branch. The pure Einstein linearization has two tensor polarizations and no dipole channel; it does not prove that every broader diffeomorphism-invariant effective theory lacks extra modes.

2.6.14 State–observable error propagation and the dynamical bridge

On the additional pure-Einstein action/background branch, the classical equalities of Section 2.6.13 hold. Finite Markov/recovery control supplies a separate and more limited quantitative statement. Two error classes must remain separate. The *state–observable error class* concerns expectation values of specified bounded local observables in a recovered state; the Markov/recovery machinery controls this class. The *dynamical-parameter error class* concerns spectral and dispersion properties

of an evolution generator: propagation speeds, particle-pole masses, and correlator poles. The trace-distance chain alone does *not* control those quantities.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \le \varepsilon_k,$$

then, writing ρ for the target state and σ for its recovered comparison state, the recovery map gives the trace-norm bound

$$\|\rho - \sigma\|_1 \le \delta_k := 2\sqrt{1 - e^{-\varepsilon_k}} \le 2\sqrt{\varepsilon_k}.$$

State–observable bound. Trace distance gives immediate bounds on expectation-value errors for the declared operator class: bounded operators supported in the recovered region, compared at the time of recovery. Using the standard dual norm inequality:

$$|\langle O\rangle_\rho - \langle O\rangle_\sigma| \le \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance. This inequality is exactly as strong as it looks and no stronger: the norm $\|O\|_\infty$ is the Lipschitz constant only for bounded O, and the comparison is an equal-time statement about states.

Exponential decay from the mixing hypothesis. Writing w for collar width, so it is not confused with the recovery error δ_k, the mixing assumption (Section 2.3) provides

$$I_\omega(A_w : D_w \mid B_w) \le C_{\mathrm{mix}} |\partial C|_{\mathrm{UV}} e^{-w/\xi}.$$

Combining these gives an explicit precision dial for bounded local observables:

$$\delta_{\mathrm{step}} \lesssim 2\sqrt{C_{\mathrm{mix}} |\partial C|_{\mathrm{UV}}} \cdot e^{-w/(2\xi)}.$$

For an operator normalized by $\|O\|_\infty \le 1$, an independently chosen absolute expectation-error target τ_O is guaranteed at one step if $\delta_{\mathrm{step}} \le \tau_O$; the theorem therefore gives the sufficient CMI condition $\varepsilon \le (\tau_O/2)^2$. The target τ_O applies only to the state–observable bound and cannot be set by importing a gravitational-wave speed tolerance.

What the dial does not certify: dynamical parameters. The gravitational-wave propagation speed and any graviton pole or dispersion-mass parameter are not bounded local observables. They are spectral/dispersion properties of the dynamical generator (poles of retarded correlators, or long-time wave-packet phase and group velocities), so the trace-distance inequality supplies neither their norm nor their Lipschitz constant. Two states can be close in trace distance at one time while being vacua of generators with different dispersion relations, and a small equal-time expectation error does not control phase accumulation over the $\sim 10^8$-year propagation baseline of GW170817. Consequently, $\delta \lesssim 10^{-15}$ is *not* a theory-side certificate for the multimessenger bound $|c_{\mathrm{GW}}/c - 1| \lesssim 10^{-15}$ of Section 2.6.13, and the recovery error must not be presented as the precision dial for that published bound. The ideal classical relation $c_{\mathrm{GW}} = c_\star$ and the vanishing pure-Einstein hard parameter $\mu^2_{\mathrm{FP,hard}} = 0$ belong to the additional action-level branch recorded there; neither stability away from that branch nor a quantum graviton pole follows from the trace-distance chain.

Required dynamical-parameter bridge. A sufficient conditional theorem can be stated precisely, but its hypotheses have not been derived from the Markov remainder in this paper. Let $\mathcal{B} = [k_{\min}, k_{\max}]$, with $0 < k_{\min} < k_{\max}$, be the declared experimental wave-number band. For each tensor helicity, assume a C^1 self-adjoint finite-regulator generator $H^{\mathrm{TT}}_\nu(k)$ and the GR reference generator $H^{\mathrm{TT}}_0(k)$, expressed in angular-frequency units ($\hbar = 1$) on one specified common finite-dimensional helicity fiber or after a declared C^1 unitary identification of the two fibers. Assume

that the reference TT tensor-mode eigenvalue $\omega_0(k) = c_\star k$ is simple in the helicity block and is separated from the rest of the spectrum by a uniform gap $g_{\mathcal{B}} > 0$. Define

$$M_{\mathcal{B}} := \frac{1}{c_\star} \sup_{k\in\mathcal{B}} \left\| \partial_k H_0^{\mathrm{TT}}(k) \right\|_\infty$$

and the dimensionless dynamical error

$$\eta_{\mathrm{dyn},\nu} := \max\left\{ \frac{1}{g_{\mathcal{B}}} \sup_{k\in\mathcal{B}} \left\| H_\nu^{\mathrm{TT}}(k) - H_0^{\mathrm{TT}}(k) \right\|_\infty , \frac{1}{c_\star} \sup_{k\in\mathcal{B}} \left\| \partial_k H_\nu^{\mathrm{TT}}(k) - \partial_k H_0^{\mathrm{TT}}(k) \right\|_\infty \right\}.$$

For $\eta_{\mathrm{dyn},\nu} < 1/4$, let $\omega_\nu(k)$ be the continued isolated eigenvalue and define its group velocity by $c_{\mathrm{GW},\nu}(k) := \partial_k \omega_\nu(k)$. The standard isolated-eigenvalue projector estimate and the Hellmann–Feynman formula then give, uniformly over $\mathcal{B}$,

$$\sup_{k\in\mathcal{B}} \left| \frac{c_{\mathrm{GW},\nu}(k)}{c_\star} - 1 \right| \le L_{\mathcal{B}}\, \eta_{\mathrm{dyn},\nu}, \qquad L_{\mathcal{B}} := 1 + 4M_{\mathcal{B}}.$$

Indeed, if $P_\nu(k)$ and $P_0(k)$ are the rank-one spectral projectors, then $\|P_\nu - P_0\|_\infty \le 2\eta_{\mathrm{dyn},\nu}$, hence $\|P_\nu - P_0\|_1 \le 4\eta_{\mathrm{dyn},\nu}$. Differentiating the isolated eigenvalue therefore bounds the group-velocity change by the derivative perturbation plus $4M_{\mathcal{B}} c\, \eta_{\mathrm{dyn},\nu}$. This displays a dimensionless constant derived from a specified generator and uniform over the declared band. If $q := L_{\mathcal{B}}\eta_{\mathrm{dyn},\nu} < 1$, then stationary propagation at fixed group velocity over path length D obeys the group-delay bound

$$|\Delta t| \le \frac{D}{c_\star} \frac{q}{1-q}.$$

The same bound applies to varying propagation only if it holds pointwise along the entire path; on a cosmological path, the redshifted wave number must remain inside $\mathcal{B}$. A wave-packet phase bound likewise requires uniform control of the inverse dispersion $k_\nu(\omega)$ on the corresponding frequency band. A model-dependent particle dispersion-mass bound is a separate spectral estimate at $k = 0$; it is not a corollary of the band-limited speed estimate.

The OPH-specific bridge requires three additional conditions:

1. H_ν^{TT}, or an equivalent retarded two-point kernel, is derived from the finite and scaling OPH dynamics;
2. a norm-resolvent, quadratic-form, or correlator estimate has the form $\eta_{\mathrm{dyn},\nu} \le C_{\mathcal{B}} r_N$, where $r_N := \min(2, \sum_{k=2}^{N} \delta_k)$ is the accumulated trace-norm recovery bound of Theorem 3.1 for the associated N-step construction and $C_{\mathcal{B}}$ is derived rather than assumed; and
3. the band-uniform dispersion estimate applies to the detector band and propagation baseline, including phase accumulation where used by the likelihood.

Only after the second item is proved may one combine the estimates to obtain

$$\sup_{k\in\mathcal{B}} \left| \frac{c_{\mathrm{GW},\nu}(k)}{c} - 1 \right| \le L_{\mathcal{B}} C_{\mathcal{B}} r_N.$$

No estimate $\eta_{\mathrm{dyn},\nu} \le C_{\mathcal{B}} r_N$ is proved here. Substituting δ_k, δ_{step}, or any equal-time trace-distance error for $\eta_{\mathrm{dyn},\nu}$ without that bridge is invalid.

Precision summary. The framework provides:

1. The ideal pure-Einstein linearization statements $\mu^2_{\mathrm{FP,hard}} = 0$ and $c_{\mathrm{GW}} = c_\star$, without a quantum graviton-pole conclusion.
2. Translation of those ideal equalities into the specific observables experiments constrain (Section 2.6.13).
3. Explicit bounds on how far expectation values of bounded local observables in the recovered state can drift, using the conditional mutual information → trace distance → observable error chain.

It therefore provides quantitative error control on recovered states and bounded observables. No theory-side certificate is supplied for the published GW-speed or model-dependent particle dispersion-mass bounds, and no derived dynamical-parameter certificate is supplied.

2.6.15 The cosmological constant and the dark-sector slot

The Einstein implication leaves two cosmological objects to the cosmology layer. A candidate anomaly-tensor slot T^{anom}_{ab} becomes available only after a finite covariant source packet supplies source localization, rank-two reconstruction, conservation, normalization and universal coupling; its rest-frame normalization would be $\langle T^{\mathrm{anom}}_{00}\rangle = (15/8\pi^2)\,\delta\langle K_C^{(\mathrm{anom})}\rangle/\ell^4$. The conditional dark-sector continuation built on that slot is developed in [22]; raw scalar collar CMI supplies only a recovery diagnostic. A compact neutral coherent source has no monopole and therefore cannot close the declared single-monopole source ansatz. Bound neutral composites, dipole or higher-multipole sources, and topological or solitonic architectures are not excluded.

The mechanism: Λ as global capacity, not local physics. Once the separately supplied effective-locality and same-source bridge identifies the directional null charges with one symmetric local stress tensor, finite null tomography determines that tensor only up to an additive metric term. Vacuum-energy or cosmological-constant shifts are therefore invisible to this conditional local null-data route. The conditional Einstein equation obtained under the declared fixed-cap generalized-entropy stationarity interface is correspondingly fixed only up to Λg_{ab}. Accordingly, this null-data route cannot fix Λ without independent trace-sensitive or global input. A source-selected vacuum potential or non-null local equations could supply such input.

On the static-patch branch each timelike observer carries a horizon screen S^2 rather than a boundary at infinity, and Λ is read as a global capacity through $N_{\mathrm{scr}} = 3\pi/(\Lambda\ell^2_\star)$ of Section 1.4. The source selection of that capacity is a cosmology-layer join.

Fixed-capacity shock sign. The pure de Sitter shock parameter obeys

$$\mu^2 = (d-2)\kappa r_c = d - 2 = \lambda_{\ell=1}(S^{d-2}),$$

so the cosmological constant cancels and the smooth isometry modes lie at zero shock eigenvalue. For finite sector dimensions, maximizing

$$S_{\mathrm{gen}}(p,d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i$$

over p gives $p_i = d_i/M$ and $S^{\max}_{\mathrm{gen}} = \log M$. The associated logarithmic area observable $\mathcal{A} = \sum_i (d_i/M)\log d_i$ has symmetric-point Hessian

$$\mathrm{Hess}\,\mathcal{A}|_{d_i=d} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right)$$

on the positive-real relaxation of the integer dimensions. Its curvature is positive on fixed-M tangent directions and negative in the homogeneous direction; the gradient at the symmetric point

is nonzero. The physical sign comes from a one-sided transfer. Assigning a fraction f of a fixed total horizon–observer capacity to the observer uniformly depletes the horizon and gives

$$\Delta S_{\mathrm{gen}}^{\max} = \Delta \mathcal{A} = \log(1-f) < 0.$$

The finite identity applies to admissible integer depletions, and its positive-real interpolation is strictly decreasing. Under the independent horizon-area and fixed-budget dictionaries, this is the de Sitter time-advance sign studied in Ref. [53]. It is a boundary maximum along the transfer direction, rather than an interior maximum at fixed horizon capacity. Identifying the finite capacity transfer with observer mass is an additional physical dictionary.

On the icosahedral carrier, the spectrum

$$\{-2,\ 0,\ 1+3/\sqrt{5},\ 1+\sqrt{5}\}$$

is conditional on two further statements: the A_5 rotation triplet is an exact gauge sector, and the shock kinetic term is the scaled nearest-neighbour port or edge-sector Laplacian. The regular line-graph identity makes the port and edge-sector low spectra equal. It adds the eigenvalue 10 with multiplicity 18 on the edge carrier. No shock coefficient, physical attachment, or generator for sector-dimension shocks is supplied. The finite heat-bath repair generator acts at fixed sector dimensions and therefore supplies none of these objects. The sign mechanism does not rescue the static-patch trace conjecture [27]. That obstruction tests the cited positive cyclic trace interpretation. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and source-derived kinetic operators beyond the scaled nearest-neighbour graph remain outside these classes. None of these boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

2.7 Einstein-chain boundary and entropy-attachment tests

The finite support and algebraic packets establish repair-invariant incidence, the spherical branch receipts, cap-normal identities, the Lorentz action on the celestial screen, exact finite entropy splitting on the central-interface branch, and the finite tomography identities that reconstruct a trace-free symmetric tensor from null projections. Authenticated read-from provenance supplies the finite event carrier and its generated order; every finite decidable strict partial order compiles into an abstract authenticated log with exactly that order, which proves grammar expressivity, not dynamical or physical selection. The ambient carrier $\mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t,x) = t^2 - g_{\mathrm{src}}(x,x)$ has Lorentz inertia $(1,3)$ independently of the poset, and source height enters only as the temporal coordinate of a scaled event placement.

On the exact-embedding route, an effective Lorentzian manifold requires the source-causal continuum certificate of Theorem 4.3e: one physical refinement family with exact two-way order/cone agreement, calibrated count–volume, independent dimension and manifoldlikeness tests, stable topology, and a unique distinguishing Lorentzian limit. The smooth Einstein reading additionally requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, plus modular and scaling control, physical stress, Ward/Bianchi, universal coupling, a vacuum reference, and physical $8\pi G$ scale identification on that same family. A convergent causal-set scalar-curvature estimator is an independent diagnostic and is insufficient by itself. No source-derived family carrying the full continuum antecedent is constructed here.

The following finite calculations test proposed attachments of native entropy and stress to premise (G4). Each is a fact about the stated finite law.

The finite native-screen Gaussian vacuum provides a direct test of one proposed entropy attachment. On a three-site Dirichlet radial slab with the actual twelve-port angular operator, a fixed-proper-volume uniform metric variation gives regional bare-entropy derivative -1.1004136186 and two-face-area derivative $-4/3$. They have the same sign for every nontrivial variation in this uniform family, so no positive coefficient of bare area makes their sum stationary. Equal port-local canonical stress variations can also give distinct regional entropy derivatives. The slab is not a geodesic small ball; its finite bare entropy has no renormalized generalized-entropy or physical stress dictionary. This is a counterexample to that finite attachment, not to the stated small-ball Einstein theorem.

A separate three-dimensional proper-ball Gaussian calculation reconstructs the regional modular kernel from the restricted vacuum covariance. Momentum noise obeys the entropy first law; coherent displacement gives the same relative-entropy charge. At the retained fine lattice cut, two charge profiles differ by 2.124% and 5.716% from the independently fixed conformal stress integrals. Grid and box controls retain their errors and precision refusals without asserting a convergence rate. An independently specified space-form metric gives $G_{00} = 3k$ and the fixed-volume coefficient $\delta A/\delta G_{00} = -4\pi R^4/15$. A frozen native chart tensor defines this constant comparison frame; it is not identified with the native continuum limit. These calculations supply neither a renormalized area coefficient nor joint state–geometry stationarity.

The calibrated fermion action supplies a separate same-matter regional test. A local canonical particle-hole map identifies its 21 Dirac blocks, counting quark color, and three massless Weyl blocks. One antiperiodic Fourier box and ball define the restricted vacuum projector, modular kernel and entropy first-law variation. At the retained fine cut the massless comparison differs from the conformal stress integral by -0.1953%; the running-top comparison differs by $+22.83\%$. The conformal formula is not assumed for massive fields. A 32-CAR source control prepares the corresponding small-grid vacuum in 13,616 native pulses; the larger entropy grids are not prepared source histories. This calculation retains the calibrated masses and spectral regulator, and supplies neither an interacting gauge-region algebra nor native vacuum selection or a renormalized area coefficient.

A native geometry-bit control also executes conditional vacuum preparation and a small complete reset dilation. Compatibility is independently stated as a uniform geometry bit and low excess field energy on the fixed-volume slab family. Exact Rayleigh enclosures certify the retained band preparations against the unrestricted oscillator vacuum. Fresh blanks and entropy export remain apparatus premises. Switching between distinct geometry vacua costs strictly positive stationary cycle work,

$$W_{\mathrm{mode}} = \frac{\omega_+ + \omega_-}{16}\frac{(a_+ - a_-)^2}{a_+ a_-}, \qquad a_\pm = \mu_\pm \omega_\pm.$$

The constant angular mode ensures strict positivity; an exact bound also holds for the literal retained source preparations. A static geometry potential has zero mean change in the stationary cycle. Thus this compatible joint repair requires a work reservoir or reciprocal geometric response; it does not derive autonomous cooling or entropy–area stationarity.

A separate reciprocal thermal control uses the same two geometries $r = 9/10, 11/10$ and the three-oscillator, two-level Fock compression in the native angular band of eigenvalue six. Its two conditional Hamiltonians need not commute. At the supplied inverse temperature $\beta = 1/\sqrt{14}$, a bank of fixed-gap thermal qubits implements energy-conserving exchanges between opposite-

geometry energy levels. For weights $w_i = e^{-\beta E_i}$ and a symmetric proposal probability q,

$$w_i \frac{q w_j}{w_i + w_j} = w_j \frac{q w_i}{w_i + w_j}.$$

The collision commutes with the system-plus-bath Hamiltonian, so the full quantum Gibbs state is stationary. Auxiliary native proposals retain their records, rejected proposals and bath outputs. Geometry responds through $p_g = Z_g/(Z_0 + Z_1)$, with probabilities approximately $0.57509, 0.42491$; forcing uniform geometry breaks this stationary law. The corresponding finite free-energy variation in p_g vanishes and has positive curvature. This probability-space stationarity supplies no continuous metric variation or entropy–area identity. Temperature, Hamiltonians, energy readout and fresh thermal bath preparation are specified. Endpoint energy conservation of the compiled collision does not certify the work cost of a physical controller or reservoir maintenance. The same collision law applies to the native correlation/exchange fermion chain with both bonds scaled by the declared geometry. Its fixed-one-particle sector preserves charge and has a finite thermal geometric response, without attraction across charge sectors or a vacuum-gravity identification.

A zero-temperature construction uses the free native-coefficient staggered field and its filled negative-energy vacuum. The cubic action derives from specified inverse-exchange, phase and parity controls; it retains two Dirac tastes per native sector. Transverse Fourier reduction preserves a planar cut and gives four scalar chains per reduced transverse momentum, with effective masses $\pm M_\perp$, each twice. Independently assigned site and bond positions define $H[N] = \{N, H\}/2$ and the boost energy $B = \{X, H\}/2$. Mapping the exact staggered half-chain theorem of Eisler, Eqs. (64)–(72), to model-coordinate speed $c = 2J$ gives

$$K_{\mathrm{half}} = \gamma B, \qquad \gamma = \frac{4\kappa K(\kappa')}{c}, \qquad \kappa = (1+\mu^2)^{-1/2}, \quad \mu = M_\perp a/c.$$

Directed AGM intervals imply $0 \le 1 - \gamma/(2\pi/c) \le \mu^2/4$, with no temperature fit. Here H, B, K_{half} are one-particle kernels. For a finite faithful region, $-\log \rho_A = d\Gamma(k_A) + \log Z_A\, I$, with the actual kernel $k_A = \log[(I - C_A)/C_A]$; finite truncation does not imply $k_A = \gamma B$. The infinite-half-chain quadratic operator uses the filled-sea reference, or particle/hole excitation normal ordering. Scalar normalization cancels in trace-zero first-law variations. A separate Legendre sign-polynomial bounds local vacuum covariance error by $\Lambda r^{k+1}/m$, $r = (\Lambda - m)/(\Lambda + m)$, when degree-$2k+1$ walks agree on the observation window. This gives a finite-volume preparation schedule, not a global state-distance bound. An eight-CAR source preparation has an independent exact binary Rayleigh/Fermi-gap fidelity certificate and a full-Fock entropy first-law check. Exact staggered entanglement levels and transverse factorization give a bare planar entropy density. Finite-interval modular discrepancies and purity refusals are retained with the calculation. The result supplies the specified free half-space modular/stress attachment; it supplies no interacting matter limit, renormalized Newton coefficient, massive small-ball identity or joint curved-metric stationarity. The larger vacuum grids are not source-executed histories.

The implication requires its receipts on one cofinal branch, a rank-three source Gram quotient, tensor-linearity of the directional charges, the stated curvature or small-diamond remainder rate, and scale readouts with trivial common rescaling. The chosen causal route must meet its exact order–cone criterion or its controlled causal and volume error bounds.

A Technical Details for the Specialist Derivations

This appendix carries the gravity details that support the spacetime and Einstein derivation: the port-record metric completion, the Lorentz and Einstein bridge, and the blindness of local null data to the cosmological constant.

A.1 Port-record metric completion

Theorem A.1 (Exact repair-response three-dimensional carrier)**.** *Let A be the adjacency matrix of the declared twelve-port incidence, $L = 5I - A$, $T = I - L/60$, and let Q remove the constant port mode. Write P_{slow}, P_5, and P_{fast} for the exact nonconstant spectral projectors of ranks $(3, 5, 3)$. For the complete equally weighted port-probe census,*

$$C_n = QT^{2n}Q = r_{\mathrm{slow}}^{2n}P_{\mathrm{slow}} + (9/10)^{2n}P_5 + r_{\mathrm{fast}}^{2n}P_{\mathrm{fast}},$$

where

$$r_{\mathrm{slow}} = \frac{55+\sqrt{5}}{60}, \qquad r_{\mathrm{fast}} = \frac{55-\sqrt{5}}{60}, \qquad 0 < r_{\mathrm{fast}} < \frac{9}{10} < r_{\mathrm{slow}} < 1.$$

Trace-twelve normalization therefore gives

$$\frac{12C_n}{\operatorname{tr} C_n} \longrightarrow 4P_{\mathrm{slow}} =: G.$$

The intrinsic space $H = \operatorname{ran} P_{\mathrm{slow}}$ has dimension three, and the labeled vectors $v_p = 2P_{\mathrm{slow}}e_p$ satisfy $\langle v_p, v_q\rangle = G_{pq}$.

Let

$$M_{\mathbb{Z}} = \mathbb{Z}[P]/\langle e_{-p} + e_p\rangle \cong \mathbb{Z}^6$$

be the cumulative signed port-record module. The limiting form on its real six-control extension has a three-dimensional radical. Its real quotient is isometric to H. The integer module has no nonzero metric kernel, its image is dense in H, and its metric completion is an abstract continuous three-dimensional Euclidean vector group. The construction is unique up to port-label-preserving isometry and selects no Cartesian axes.

Proof. The spectral polynomial of A gives the four exact projector ranks $(1, 3, 5, 3)$ and the displayed decomposition of C_n. The strict response order makes the two faster ratios vanish after normalization by the slow response. Vertex transitivity fixes the unit diagonal and hence $G = 4P_{\mathrm{slow}}$. Projector symmetry and idempotence give the intrinsic Gram identity. The source Gram calculation identifies its real radical with the coordinate kernel. Irrationality of the golden ratio makes the integer map injective, and the raw coordinate image has finite index eight in $\mathbb{Z}[\varphi]^3$. Its density in $\mathbb{R}^3$ gives the completion. These finite identities, the limit, the radical calculation, injection, and density are machine checked. □

Corollary A.2 (Conditional local position carrier)**.** *Suppose cumulative signed port load is the carrier-position readout, the complete probe census and T belong to that same source record, and the normalized limiting Gram is operational distinguishability. Then the finite record architecture reconstructs one abstract continuous local three-dimensional Euclidean carrier instead of consuming a Cartesian three-space.*

Boundary. For finite n, C_n has rank eleven on the centered port space and rank six on the signed antipodal sector. The normalized limit must precede the radical quotient and metric completion. The theorem does not select the repair law, the cumulative record semantics, or the Gram topology from the premises A1–A3. It supplies no faithful isometric completion action, overlap/refinement gluing, global or physical spacetime, preferred frame, physical scale, clock, field sector, or observable. Ordered record history and repair cost remain a separate fiber.

A.2 Lorentz and Einstein Bridge

Theorem A.3 (Conformal group isomorphism)**.** *The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

Theorem A.4 (Support-visible geometric modular flow on caps on the extracted geometric subnet)**.** *Assume the premises A1–A3, the derived fixed-cutoff collar package, and the support-visible BW scaling theorem on the extracted geometric subnet. That theorem requires the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Then the scaling-limit modular automorphism group of the cap pair is geometric conformal dilation on that subnet, with the standard 2π normalization supplied by the finite cap-normal certificate. If the emitted scaling-limit cap algebra is type I, this may be written as $K_C = 2\pi B_C + Z_C$ with Z_C central; in the generic continuum case the theorem is the automorphism statement on a non-type-I cap algebra with outer geometric modular action. The collar replacements used in this branch are exact only at EC-aligned exact Markovity or in controlled fixed-collar families with $\delta_{\mathcal{F}}^{\mathrm{M,EC}} \to 0$; small CMI supplies a Fawzi–Renner recovered comparison state, not a dimension-free one-shot exact Markov normal form.*

Definition A.5 (BW-branch observer-relative modular ordering)**.** *On the branch satisfying the hypotheses of the preceding geometric modular-flow theorem, the modular automorphism parameter t of the extracted cap pair supplies a dimensionless ordering for that observer's accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.*

BW-side UV scaffold. The UV data on this branch are not a separate cap-isotropy or Euclidean-regularity selector. The fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may leave that class, and MaxEnt alone does not select the BW / canonical cap phase. The UV package is the realized transported geometric cap-local system together with the carried-collar schedule derived from the transported fixed-local-collar Markov/faithfulness datum on each fixed local collar model, followed by support-readable modular covariance and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Null-strip completion boundary. The fixed-cutoff strip package is more specific than the generic inherited-strip summary. The null cuts first transfer the same cut-center data as the spatial collar branch, which fixes the central sector-pair decomposition of the strip algebra. The stronger left/right tensor decomposition used by the null modular bridge then requires the extra inherited strip-split condition on the multiplicity spaces, together with the exact-or-controlled Markov hypotheses on one fixed inherited strip model. On that same exact-Markov/central-interface model, a specified endpoint-to-cell map and uniform local norm/site-density bound make

the renormalized half-line family endpoint-Lipschitz; this is additional endpoint-control data, not a consequence of finite-range locality alone. It hence defines the weak tail generator, and on the scaling-limit geometric-cap branch the half-line blow-up net carries the derived half-sided modular pair. Borchers–Wiesbrock then supplies the positive null-translation generator on its Stone domain together with the affine half-line modular relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$, and the same half-line family fixes the generator and its endpoint-derivative identity internally. Its reading as a local null-stress charge additionally requires the effective-locality and same-source identification premise. Bounded-interval formulas are downstream of the E0.5 affine/projective kernel in the Einstein bridge.

Lemma A.6 (Null data ambiguity)**.** *If a symmetric tensor X_{ab} satisfies*

$$X_{ab}k^a k^b = 0$$

for every null vector k, then $X_{ab} = \phi\, g_{ab}$ for some scalar ϕ.

Corollary A.7 (Conditional null-stress data leave the metric term invisible)**.** *If the supplied effective-locality and same-source bridge identifies the directional generators with one symmetric local stress tensor, their null contractions determine that tensor only up to ϕg_{ab}. Under the remaining Einstein-branch premises, the local equation consequently retains the corresponding Λg_{ab} ambiguity.*

Theorem A.8 (Jacobson-type rest-frame relation)**.** *In the local Lorentzian scaling regime, assume the declared fixed-cap generalized-entropy stationarity interface for admissible fixed-cap variations, together with the proved finite MaxEnt identities on the realized cap-label-preserving family. If the effective-locality and same-source bridge gives the half-line generator its local null-stress reading, the conditional bounded-interval transport and small-ball premises yield the rest-frame first-variation relation that drives the spacetime and Einstein paper's local Einstein branch; on the same scaling branch, if that rest-frame relation holds for all local observer four-velocities and all reference states, the spacetime and Einstein paper upgrades it to the full tensor equation by the explicit local quadratic-polarization argument in the main derivation.*

Theorem A.9 (Newton coupling from the scale certificate and edge entropy density)**.** *The geometric coupling read by the Newton area law is*

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

On the OPH gravity row the scale is supplied first by $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, as an independent scale certificate rather than a consequence of $P_\star$ and $N_\star$ alone. With $a_{\mathrm{cell}} = P\ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P/4$, this gives

$$G_{\mathrm{geom}} = \ell_\star^2, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

Here $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density. It is not the logarithm of an independent cell Hilbert-space dimension and does not select a fixed primitive observer capacity.

A.3 The cosmological constant and local null data

Lemma A.10 (Vacuum energy blindness)**.** *For any null vector k,*

$$T^{\mathrm{vac}}_{kk} = T^{\mathrm{vac}}_{ab}k^a k^b = -\rho_{\mathrm{vac}}\, g_{ab}k^a k^b = 0.$$

Vacuum-energy contributions therefore lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

Proposition A.11 (Structural separation)**.** *Within the structural split used here:*

1. *local modular/null data fix T_{ab} only up to ϕg_{ab}; and*
2. *the dimensionless Λ-capacity relation is fixed by global screen capacity, not by local null data.*

So the large vacuum-energy bookkeeping of EFT is not itself the local quantity determining curvature on this branch.

Theorem A.12 (No Λ prediction from the declared local null data)**.** *Within the null-modular reconstruction used here:*

1. *all T_{kk} data are unchanged under $T_{ab} \to T_{ab} + \phi g_{ab}$;*
2. *therefore Λ cannot be fixed by these local null-modular data alone; and*
3. *the dimensionless Λ-capacity relation requires the global screen-capacity closure beyond the local null data. The stable closure target is $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_{\mathrm{CRC}} = \log D_\star$. The correctable public-record definition and conditional operational closure theorem close its operational public-section, correctable-code, capacity-bound, approximate-stability, and greatest-fixed-point implications under named premises. The full specification and receipt schema are carried by the flagship synthesis. The finite record restrictions, reachability, publicness, global coupling, carrier, scalarization, extension, refinement, and sewing receipts are exact at $D = 24$ inside their declared finite source category. A bounded all-rung counterfamily has different exact fixed sets under shared base, positivity, carrier, and executable finite controls. This proves nonidentifiability for the bounded completion class. Universal all-rung membership in the complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent. Direct N is not evaluable on the incomplete source antecedent, and the bounded result does not determine the complete source class. A positive cosmic value requires a complete source antecedent, one physical zero, a universe-level carrier attachment, and the horizon–record identification. The electroweak relation is an independent comparison after F and does not construct it. SI curvature values additionally use the selected scale certificate.*

A different typed local route carrying trace-sensitive or non-null overlap observables is not excluded and would require its own source and physical receipt.

The capacity readout of Λ, its observed-age and flatness benchmarks, the local/global capacity closure and the conditional screen-spectrum theorem are cosmology-layer branches [27]; this paper uses only the null-data blindness above.

Spatial flatness has a separate holonomy readout.

Lemma A.13 (FLRW curvature as visible scalar holonomy)**.** *On a homogeneous-isotropic spatial slice with constant sectional curvature K, small spatial loop holonomy obeys*

$$\mathrm{Hol}_{\square_{uv}} = \exp\Big(K A_\square J_{uv} + O(A_\square^{3/2})\Big)\,.$$

For oriented nondegenerate loops small enough that the logarithm branch is fixed, define the normalized scalar readout

$$\kappa_\square := \frac{\langle J_{uv}, \log \mathrm{Hol}_{\square_{uv}} \rangle}{A_\square \langle J_{uv}, J_{uv} \rangle}.$$

Then $\kappa_\square = K + O(A_\square^{1/2})$. On a visibly separated OPH refinement system, the limit of this normalized readout vanishes if and only if $K = 0$. The unnormalized holonomy itself tends to the identity for every finite K, so it is not a flatness test. Thus a flat FLRW branch is the zero-normalized-spatial-curvature-holonomy branch.

Flatness boundary. This holonomy statement names a conditional cosmology bridge only. It does not add a capacity-theorem output and does not solve the inflationary flatness or horizon problems. Selecting $K = 0$ requires an additional theorem or premise: a direct flatness theorem, a conditional cosmological-minimal-holonomy selector on a clocked FLRW boundary, or an explicit flat-branch assumption. No gauge- or matter-sector result supplied in this packet selects $K = 0$. Any gauge-, matter-, or dynamics-based flatness route needs its own theorem and physical receipt.

B Causal order, counts and clocks on source populations

The theorems of this appendix concern supplied source populations, read laws and model clocks. They prove the causal, count and clock limits under those premises; the native process must produce the population and law before the limits apply to it.

B.1 Causal order on a source-metric population

The continuous rank-three repair carrier admits finite populations chosen by its own metric. A covering property of such a population also supplies a constructive causal approximation, with an explicit local read law and model clock. Let Ω be a convex subset of a real normed space, and let $S \subset \Omega$ satisfy

$$\forall x \in \Omega \quad \exists s \in S : \quad \|s - x\| \le h, \qquad h \ge 0.$$

For an edge radius $a > 2h$, put an event at every (j, s), $j \in \{0, \dots, K\}$, $s \in S$. An event at $(j+1, t)$ reads every (j, s) with $\|t - s\| \le a$. This includes the same-site read, so waiting is permitted. The event order is the reflexive transitive closure of these reads. Reading every permitted neighbour, keeping the full population at each layer, and assigning a duration $\Delta = a/c$ with $c > 0$ are hypotheses of this construction. They are substantive choices of dynamics and clock.

Proposition B.1 (Covering constructs causal paths)**.** *For $s, t \in S$ and $k \ge 1$, with both layers present,*

$$\|t - s\| \le k(a - 2h) \quad \Longrightarrow \quad (j, s) \preceq (j + k, t) \quad \Longrightarrow \quad \|t - s\| \le ka. \tag{1}$$

If S is finite and nonempty, this layered poset has width exactly $|S|$. Every event at layer j has height j, counting edges from layer zero.

Proof. Divide the segment from s to t into k equal pieces, with division points $x_i = s + i(t - s)/k$. Convexity gives $x_i \in \Omega$. Keep the endpoints $p_0 = s$, $p_k = t$ and choose $p_i \in S$ with $\|p_i - x_i\| \le h$ at every interior division point. Then

$$\|p_{i+1} - p_i\| \le \|t - s\|/k + 2h \le a.$$

These are actual permitted reads. For any permitted path the triangle inequality gives the opposite bound in (1). A layer is an antichain, and the same-site waiting chains partition all events into $|S|$ chains. This proves the width statement. Every read increases the layer by one, and a waiting chain reaches every layer, proving the height statement. □

The path construction and outer displacement bound hold in arbitrary real normed spaces; `Lean/Geometry/SourceNetCausalCone.lean` formalizes them. The partial order, its equality with the reflexive transitive closure of the reads, the graph-distance characterization, and the width and height statements are formalized in `Lean/Geometry/SourceNetLayeredOrder.lean`. The volume estimates below concern the three-dimensional Euclidean repair carrier. The vertical weighted

bound and its family limit are formalized below; the separate general moving-tip finite estimate is analytic.

For model time $\tau = k\Delta$, the two radii in (1) are $c(1-2h/a)\tau$ and $c\tau$. Consider a family with

$$a_n \longrightarrow 0, \qquad h_n/a_n \longrightarrow 0. \tag{2}$$

The read law agrees with the speed-c continuum causal order away from its null boundary. More explicitly, take two continuum events with positive time separation whose positions lie in Ω. Sample positions within h_n and times on the layer grid within Δ_n. A strict timelike margin eventually exceeds the endpoint errors, the time rounding errors, and $2ch_n/a_n$ times the sampled duration; the inner implication then applies. A strict spacelike margin eventually exceeds the endpoint and time errors, and the outer implication forbids a path. This statement requires the sampled layers to exist in the chosen time window. It makes no assertion of exact order reflection for null-separated events at a finite resolution.

Volume from a controlled population measure. Suppose $\Omega \subset \mathbb{R}^3$ is bounded and convex, and S is finite. In addition to the h-covering population S, supply a measurable partition $(C_s)_{s\in S}$ of Ω, up to null sets, with positive masses $v_s = \mathrm{vol}(C_s)$ and assignment error

$$y \in C_s \quad \Longrightarrow \quad \|y - s\| \le H \quad \text{for almost every } y. \tag{3}$$

Clipped Voronoi cells allow $H = h$. Equal-volume cells with a different assignment bound are also admitted; no Voronoi hypothesis is required. For $x \in S$ and $T = K\Delta > 0$, let I be the inclusive discrete interval between $(0, x)$ and (K, x), and assign it spacetime mass

$$V_{S,\Delta}(I) = \Delta \sum_{(j,s)\in I} v_s.$$

The spatial assignment rule and the temporal factor are part of the measure. The limiting volume convention is $dt\, d^3x$. For the metric $c^2dt^2 - |dx|^2$, its metric volume is c times this coordinate volume.

Proposition B.2 (Weighted Alexandrov-volume limit)**.** *Assume the preceding hypotheses and $B(x, cT/2 + H) \subset \Omega$. Then*

$$\left| V_{S,\Delta}(I) - \frac{\pi c^3 T^4}{24} \right| \le 4\pi(T+\Delta)(cT/2+H)^2 \left(H + \frac{cTh}{a} \right) + \frac{\pi c^3 T^3 \Delta}{2}. \tag{4}$$

Thus $H_n \to 0$ and (2) imply the continuum Alexandrov-volume limit on aligned, bounded interior time windows. If all cells have a common mass v, this is the raw event-count statement $v\Delta\,|I| \to \pi c^3 T^4/24$.

Proof. Write $u_j = \min(j\Delta, T - j\Delta)$ and $r_j = cu_j$. The causal sandwich puts every population point within $c(1-2h/a)u_j$ of x in the interval, and puts every interval point within r_j. At the endpoint layers the interval contains precisely the single endpoint. The partition assignment therefore gives, at every layer,

$$\frac{4\pi}{3}\big(c(1-2h/a)u_j - H\big)_+^3 \le \sum_{(j,s)\in I} v_s \le \frac{4\pi}{3}(r_j + H)^3.$$

For the lower estimate, a point in the smaller ball is assigned to a population point inside the inner causal radius. For the upper estimate, the union of cells assigned to interval points lies in the larger

ball. The stated interior buffer keeps these balls in the spatial window. The deviation of either radius from r_j is at most $H + cTh/a$. Since the derivative of $(4\pi/3)r^3$ is $4\pi r^2$, the error per layer is bounded by $4\pi(cT/2 + H)^2(H + cTh/a)$.

Finally, $F(t) = (4\pi/3)[c\min(t, T-t)]^3$ has Lipschitz constant $\pi c^3 T^2$, vanishes at both endpoints, and integrates to $\pi c^3 T^4/24$. Its left Riemann sum differs from its integral by at most $\pi c^3 T^3 \Delta/2$. The inclusive right endpoint contributes zero to this reference sum. Summing the spatial errors over $K + 1$ layers proves (4). □

The complete finite estimate, with almost-everywhere cell coverage, disjointness and assignment, is formalized by `SourceNetVolumeError.weighted_alexandrov_error` in `Lean/Geometry/SourceNetVolumeError.lean`. The zero-hop endpoint cases and all $K + 1$ layers are included. The general limit for varying, possibly unequal cell masses is proved by `SourceNetVolumeError.weighted_alexandrov_tendsto`: it assumes $a_n, H_n, h_n/a_n \to 0$, $K_n(a_n/c) \to T$ and the eventual interior buffer, and derives convergence of the actual weighted generated intervals. `Lean/Geometry/GoldenSourceVolumeLimit.lean` instantiates the bound with the actual golden sites and equal cell masses, proves its exact raw count interpretation, and derives the aligned weighted-volume limit as $K_n\Delta_n \to T$, under the stated interior buffer.

Proposition B.3 (Interior diamonds with moving tips)**.** *Let $x, y \in S$, $\|y - x\| < cT$, and $T = K\Delta > 0$. Suppose the continuum diamond between $(0, x)$ and (T, y) has a positive spatial buffer inside Ω. Write*

$$D(t) = B(x, ct) \cap B(y, c(T - t)), \qquad V(D) = \int_0^T \mathrm{vol}(D(t))\, dt.$$

For the inclusive discrete interval I between these tips,

$$|V_{S,\Delta}(I) - V(D)| \le 8\pi(T + \Delta)(cT + H)^2 \left(H + \frac{2cTh}{a}\right) + 4\pi c^3 T^3 \Delta. \tag{5}$$

In particular, the same refinement conditions give weighted interval-volume convergence for arbitrary fixed interior timelike diamonds.

Proof. At layer j, the union of assigned cells contains the intersection of the two balls with radii $(c(1 - 2h/a)j\Delta - H)_+$ and $(c(1 - 2h/a)(T - j\Delta) - H)_+$, up to null sets. It is contained in the intersection with radii $cj\Delta + H$ and $c(T - j\Delta) + H$. Zero-radius balls contribute zero volume. Changing either reference radius changes the intersection only within that ball's spherical shell. Each radial error is at most $H + 2cTh/a$, and each outer radius is at most $cT + H$. The sum of the two shell bounds gives the spatial term in (5). The reference section volume has Lipschitz constant at most $8\pi c^3 T^2$: over a time increment, each of the two radii changes at speed c, and each ball has surface area at most $4\pi c^2 T^2$. Both endpoint sections have zero volume. The same Riemann-sum argument gives the temporal term. □

A conservative-record family with a raw count law. The population and equal-mass hypotheses have an explicit realization in conservative source records. In the exact source-axis presentation, write $\varphi = (1 + \sqrt{5})/2$ and $L = 2/\sqrt{\varphi + 2}$. For consecutive Fibonacci numbers $q = F_n$, $p = F_{n+1}$, with $n \ge 4$, define

$$\xi_b = b\varphi - \lfloor b\varphi \rfloor, \qquad 0 \le b < q.$$

For a triple $b = (b_1, b_2, b_3)$, put $m_i = -\lfloor b_i\varphi \rfloor$ and take the integer record

$$z(b) = (b_2 - m_1, b_2 + m_1, b_3 - m_2, b_3 + m_2, b_1 - m_3, b_1 + m_3). \tag{6}$$

Its coordinate sum is $2(b_1+b_2+b_3)$. It therefore lies in the exact conservative seam-current image, whose integer section supplies a finite source word. Direct contraction with the source axes gives the readback

$$s(b) = L(\xi_{b_1}, \xi_{b_2}, \xi_{b_3}) \in \Omega = [0, L]^3.$$

These are actual source-record images. A regular mesh is used below only to prove a volume estimate; it does not replace the event positions.

Proposition B.4 (Source-record quadrature and four-dimensional counting)**.** *The population $S_q = \{s(b) : 0 \le b_i < q\}$ has exactly q^3 distinct sites. It admits a partition of Ω into cells of common volume $v_q = L^3/q^3$, assigned to the sites with error*

$$H_q = \frac{2\sqrt{3}L}{q}. \tag{7}$$

Consequently, for every Lipschitz function f on Ω,

$$\left| v_q \sum_{s \in S_q} f(s) - \int_\Omega f(x)\, d^3x \right| \le L^3 H_q \,\mathrm{Lip}(f).$$

Apply the preceding complete-neighbour read law on S_q, with $a_q = L/\sqrt{q}$ and $\Delta_q = a_q/c$. Its density in the declared $dt\, d^3x$ convention is

$$\rho_q = \frac{q^3}{L^3 \Delta_q}. \tag{8}$$

Normalized raw event counts converge to volume on fixed bounded regions with volume-null boundary. On intervals whose endpoints converge to a strictly timelike pair and whose limiting diamond has an interior buffer in the spatial and temporal windows, the generated order intervals converge to that same diamond in count measure. In particular, for vertical tips separated by model time T,

$$\frac{|I_q|}{\rho_q} \longrightarrow \frac{\pi c^3 T^4}{24}.$$

The local limiting order and measure are those of flat $1+3$ spacetime. This conclusion concerns this declared source-record family and update law.

Proof. The Fibonacci recurrence gives $\gcd(p, q) = 1$ and $p - q\varphi = (-1)^n \varphi^{-n}$, hence $|\varphi - p/q| < q^{-2}$. Let k_b be the residue of bp modulo q. These residues permute $0, \dots, q-1$. For $b = 0$ the two floors below are zero. For $0 < b < q$, bp/q is at least $1/q$ from either adjacent integer, whereas $|b(\varphi - p/q)| < 1/q$. Therefore

$$\lfloor b\varphi \rfloor = \lfloor bp/q \rfloor, \qquad \xi_b = \frac{k_b}{q} + b(\varphi - p/q), \qquad \left| \xi_b - \frac{k_b}{q} \right| < \frac{1}{q}.$$

Irrationality of φ makes the orbit points distinct; the source readback then distinguishes all q^3 records.

Assign to $s(b)$ the rectangular cell $\prod_i [Lk_{b_i}/q, L(k_{b_i}+1)/q]$, resolving cell-boundary ties arbitrarily. This is a partition up to null sets. A site need not belong to its assigned cell, but every coordinate displacement from a point of that cell is at most $2L/q$. This proves (7), the covering property and the Lipschitz estimate by integrating the displacement bound over each cell. Taking

products with time cells $[j\Delta_q, (j+1)\Delta_q]$ proves convergence of the normalized event measure: for a Lipschitz test function on an aligned time window of length T, the error is at most

$$TL^3(H_q \operatorname{Lip}_x(f) + \Delta_q \operatorname{Lip}_t(f)).$$

An extra endpoint layer contributes at most $L^3\Delta_q\|f\|_\infty$. Continuous approximation then gives the count statement for regions with volume-null boundary.

Use $h_q \le H_q$ in the causal sandwich. Here $a_q \to 0$ and $H_q/a_q = 2\sqrt{3}/\sqrt{q} \to 0$; the sufficient positive-inner-cone condition from this bound holds when $q > 48$. For converging timelike tips, the order and sampling discrepancies are confined to shrinking neighbourhoods of the two limiting null boundaries. Their volume is zero. The uniform cell displacement and temporal error also tend to zero, so the normalized number of events in these shrinking neighbourhoods tends to zero. This proves interval-count convergence; for vertical tips it also follows directly from (4). The formula for the flat diamond is its elementary ball-section integral. □

The spatial arithmetic, partition and quadrature statements are formalized in `Lean/Geometry/GoldenSourceAssignment.lean`; normalized event-count convergence on fixed measurable continuity sets for the declared time grid, with final-layer discrepancy at most $L^3\Delta_q$, in `Lean/Geometry/GoldenSourceCountLimit.lean`. Causal-frontier and pairwise null-separation nullity are proved by Fubini in `Lean/Geometry/SourceCausalBoundary.lean`, and eventual order agreement is derived from actual covering paths in `Lean/Geometry/SourceNetOrderLimit.lean`. Count convergence for moving generated intervals is proved in `Lean/Geometry/GoldenSourceCausalLimit.lean`; its reference volume is restricted to the supplied time and spatial window, and an uncut diamond requires the containment hypothesis. Strict generated-pair convergence to the geometric product measure with the $N_q(N_q-1)$ normalization is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`, the explicit vertical volume error and the aligned weighted count-volume limit in `Lean/Geometry/GoldenSourceVolumeLimit.lean`, and the uncut interval volume together with the ordering-fraction limit of Proposition B.5, with the window containment hypothesis explicit, in `Lean/Geometry/FlatDiamondNormalization.lean`. The moving-tip finite error estimate (5) is an analytic estimate.

Proposition B.5 (The four-dimensional ordering fraction)**.** *For the contained timelike intervals in Proposition B.4, let $N_q = |I_q|$ and let C_q count their strict causally ordered pairs. Then*

$$\frac{2C_q}{N_q(N_q-1)} \longrightarrow \frac{1}{10},$$

the four-dimensional Myrheim–Meyer fraction [8, 9, 10].

Proof. The cell assignment gives convergence also for the product event measure. Interval-membership boundaries, the pairwise null-relation boundary and the coincidence diagonal have zero limiting measure. The causal sandwich confines disagreement to shrinking neighbourhoods of these boundaries. Hence the normalized strict-pair count tends to its continuum double integral; $N_q \to \infty$ removes the distinction between N_q^2 and $N_q(N_q-1)$.

In units $c = 1$, place the limiting tips at $(0,0)$ and $(T,0)$ and write $V = \pi T^4/24$. The volume to the future of (t,x) within the interval is $\pi[(T-t)^2 - |x|^2]^2/24$. Thus the strict oriented-pair integral is

$$\frac{\pi^2}{6}\int_0^T \int_0^{\min(t,T-t)} r^2\big((T-t)^2 - r^2\big)^2\, dr\, dt = \frac{\pi^2 T^8}{11520} = \frac{V^2}{20}.$$

The ordering fraction counts both orientations of distinct comparable pairs, giving 1/10. Constant changes of time and volume units cancel in the ratio. This is an interval statistic; the same value is not asserted for a whole spatial box times a time window. □

The strict-pair count converges to the geometric double-indicator measure, with the $N_q(N_q-1)$ normalization retained. The product endpoint clipping is bounded by $L^6\Delta_q(2T+\Delta_q)$ in `Lean/Geometry/SourceCountTransport.lean`, and the pair limit is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`.

The spherical-section, Lorentz-volume and pair-integral modules derive the geometric measure by Fubini and radial integration. Their explicit Lorentz boost has determinant one and preserves the causal relation. The normalization theorem in `Lean/Geometry/FlatDiamondNormalization.lean` gives the $1/10$ limit for supplied $c>0$ and timelike limiting tips whose diamond lies in $[0,T]\times[0,L)^3$. Measure equality retains the null final-time boundary. These theorems select no population, read law, physical clock or physical measure, and they do not count additional instrument or routing operations.

A sharper finite covering constant is available without changing this construction. Sort the one-dimensional orbit as $0=\eta_0<\cdots<\eta_{q-1}<1$ and put

$$r_q=\max\left\{\frac{1}{2}\max_{i<q-1}(\eta_{i+1}-\eta_i),1-\eta_{q-1}\right\},\qquad h_q=\sqrt{3}Lr_q.$$

The product population has exactly this covering radius in the closed cube. The path condition $2h_q<a_q$ is the exact quadratic-field inequality $12qr_q^2<1$. This sharper h_q concerns paths; the assignment estimate H_q in the count proof is separate.

The source-word section has coefficients growing at most linearly in q; the finite implementation records their actual primitive lengths. Counting one distinguished event per site and layer is part of the stated law. Intermediate seam-word operations, read instructions and bookkeeping records are not silently included in the same density. The spatial populations are nested along increasing Fibonacci cutoffs. Their changing layer duration and edge radius do not make the spacetime histories induced prefixes; regulator convergence uses the common metric carrier and controlled cell assignments.

Source populations and physical interpretation. The finite-word source construction supplies arbitrarily fine metric nets in bounded convex windows with nonempty interior in the rank-three completion. A maximal separated subset retains a controlled fill distance and three-dimensional packing bounds. These statements supply the spatial hypotheses of Proposition B.1 under a specified resolution and population rule. Three-dimensional packing bounds alone do not supply equal-volume sampling. The count conclusion requires a partition satisfying (3) and, for unweighted counts, equal cell masses or a separate counting approximation.

The construction gives a causal approximation on source-metric points without inserting a Cartesian spatial table. The complete neighbour read law, repeated population, cell measure, and model duration are supplied. An implementation by accepted observer repairs must realize these inputs with its actual records and operation costs. A scalar action on the same population can have a different finite propagation relation; the causal cone proved here is not an identification of that action's stencil cone or its continuum response speed. Agreement with a field action and a physical clock is an additional common-model condition.

B.2 Source-population quadrature and detector errors

Cell masses connect a finite source population to an integral on its metric carrier. A covering radius controls nearest-point geometry; an assignment radius controls quadrature for a specified measure. These radii can differ. For example, a golden-orbit point need not lie in its equal-volume permutation cell. The estimates below use the actual cell assignment and its measure.

Let Ω be a metric window with a finite positive Borel measure μ. Choose a finite measurable partition C_i, representatives $s_i \in \Omega$, and masses $v_i = \mu(C_i)$. A measurable convention assigns shared boundaries; null modifications do not affect the statements. Representatives need not belong to their cells. Let f take values in a real Banach space and be integrable on every cell. Set

$$Qf = \sum_i v_i f(s_i), \qquad D_i = \int_{C_i} d(s_i, x)\, d\mu(x).$$

Proposition B.6 (Integrated cell-transport estimate)**.** *Suppose $\|f(s_i) - f(x)\| \le K_i d(s_i, x)$ for almost every $x \in C_i$, where $K_i \ge 0$ and the displacement is integrable. Then*

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \sum_i K_i D_i. \tag{9}$$

If $d(s_i, x) \le H_i$ almost everywhere on C_i, the right-hand side is at most $\sum_i v_i K_i H_i$. In particular, for a globally K-Lipschitz function and a uniform assignment bound H,

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \mu(\Omega) K H. \tag{10}$$

For a sequence of such partitions of the same measured window, $H_n \to 0$ therefore gives $Q_n f \to \int_\Omega f\, d\mu$ for each integrable Lipschitz test function. Neither nesting nor a fixed number of representatives is required.

Proof. The difference equals $\sum_i \int_{C_i} [f(s_i) - f(x)]\, d\mu(x)$. The norm of a Bochner integral is bounded by the integral of the norm. Apply the local Lipschitz bound and sum the resulting inequalities. The uniform estimate uses $D_i \le v_i H_i$ and $\sum_i v_i = \mu(\Omega)$. Its right-hand side tends to zero under the stated refinement. The first-moment coefficient is sharp: on $[0, H]$, with Lebesgue measure, one representative at zero and $f(x) = x$, the error and displacement integral both equal $H^2/2$. □

The Lean module `SourcePopulationQuadrature` proves the actual Bochner-integral estimates for finite cell measures, their specialization to measurable disjoint partitions, and convergence under a vanishing assignment bound. It also checks that a constant detector exposes any error in total assigned mass. The source coordinates, partition geometry and their displacement bounds must be established for each population. These hypotheses are not replaced by an assumed quadrature limit.

For the golden permutation cells of side L/q, the established assignment bound $H_q = 2\sqrt{3}L/q$ and equal masses L^3/q^3 instantiate (10). Product time cells of length Δ_q contribute the independent temporal error $TL^3\Delta_q \operatorname{Lip}_t(f)$ on an aligned window of duration T. The spatial term is $TL^3 H_q \operatorname{Lip}_x(f)$. An unmatched endpoint layer contributes at most $L^3\Delta_q \|f\|_\infty$. This application retains the declared event population and time-cell convention; extra routing or bookkeeping events cannot be inserted into its density without another count estimate. The module `GoldenSourceAssignment` proves the Fibonacci approximation and floor agreement, constructs the residue-permuted rectangular partition, and derives its equal Lebesgue masses and Euclidean assignment bound. For each fixed globally Lipschitz Banach-valued detector, it proves the actual golden-population quadrature estimate and its refinement limit; the closed and half-open cube integrals agree almost everywhere.

Proposition B.7 (Actual event counts on continuity sets)**.** *Fix $T \ge 0$, $L, c > 0$, $q_n = F_{n+2}$ and $\Delta_n = L/(c\sqrt{q_n})$. Let $C_n(B)$ count the distinct points $(j\Delta_n, s_n(b)) \in B$ with $0 \le j \le \lfloor T/\Delta_n \rfloor$*

and $0 \le b_i < q_n$. If B is measurable and its frontier is null for coordinate volume restricted to $\Omega = [0,T) \times [0,L)^3$, then

$$\Delta_n \frac{L^3}{q_n^3}\, C_n(B) \longrightarrow \mathrm{vol}(B \cap \Omega).$$

The discrepancy between equal event weights and the corresponding clipped time-cell weights is at most $L^3\Delta_n$, including an aligned final endpoint layer whose clipped cell is empty.

Proof. On each cell, replace the indicator of B by its value at the assigned event. Away from the null frontier these indicators converge pointwise, because the assignment distances vanish. They are bounded by one, so dominated convergence gives the clipped-cell count limit. The sum of the nonnegative equal-weight excesses is at most $L^3\Delta_n$; this also bounds the excess over any selected cells. The golden and time-coordinate maps are injective, so the label sums are point counts. □

These steps, the actual Δ_n specialization and the endpoint control are kernel-checked in `GoldenSourceCountLimit`. Causal frontiers have zero volume by the Fubini argument in `SourceCausalBoundary`. The actual count measure converges on moving generated-order intervals; the finite window is part of the reference volume. Strict-pair counts converge to the geometric product measure with the pair endpoint error retained. The corresponding proofs are in `GoldenSourceCausalLimit` and `GoldenSourcePairLimit`.

The Lean module `FlatDiamondPairIntegral` evaluates the geometric pair measure by Fubini and radial integration. For a timelike diamond contained in the supplied window, the ordering fraction tends to $1/10$; this normalization is proved in `FlatDiamondNormalization`. The explicit vertical Alexandrov error and the general weighted-family limit are proved in `SourceNetVolumeError`, and the golden specialization in `GoldenSourceVolumeLimit`. Extra routing events require another count estimate. These theorems select no physical measure or clock.

Proposition B.8 (Localized radial-kernel quadrature)**.** *For $\epsilon > 0$, write*

$$w_x(y) = (1 - d(x,y)/\epsilon)_+, \qquad g_x(y) = w_x(y)[f(x) - f(y)].$$

If f is a real K-Lipschitz function, then g_x is K-Lipschitz and vanishes outside the closed radius-ϵ ball about x. With an assignment map $P(y) = s_i$ on C_i satisfying $d(P(y), y) \le H$, where $H \ge 0$, its quadrature error obeys

$$\left| \sum_i v_i g_x(s_i) - \int_\Omega g_x(y)\, d\mu(y) \right| \le K \int_{\Omega \cap B(x,\epsilon+H)} d(P(y), y)\, d\mu(y) \le KH\, \mu(\Omega \cap B(x, \epsilon + H)). \quad (11)$$

The same statement holds with closed balls; their boundaries are irrelevant for Euclidean volume.

Proof. The tent weight lies in $[0,1]$ and satisfies $|w_x(y) - w_x(z)| \le d(y,z)/\epsilon$. Order the two weights so that $w_x(y) \le w_x(z)$. If both vanish the claim is immediate; otherwise $d(x,z) < \epsilon$. The identity

$$g_x(y) - g_x(z) = w_x(y)[f(z) - f(y)] + [w_x(y) - w_x(z)][f(x) - f(z)]$$

is bounded by $Kd(y,z)[w_x(y) + d(x,z)/\epsilon]$. The bracket is at most one because $w_x(y) \le w_x(z) = 1 - d(x,z)/\epsilon$. The same proof with y, z exchanged covers the other order. If $d(x,y) \ge \epsilon + H$, both kernel terms $g_x(y)$ and $g_x(P(y))$ vanish. Integrate the pointwise bound over the remaining region. The tent Lipschitz estimates have Lean proofs; the displayed support-restricted integral applies the cell-transport argument. □

For three-dimensional volume, multiplying (11) by $45/(\pi\epsilon^5)$ gives the quadrature contribution $60KH(\epsilon+H)^3/\epsilon^5$ in the scalar operator estimate, and hence $480KH/\epsilon^2$ when $H \le \epsilon$. Actual local cell masses or integrated displacements can give smaller bounds than the enclosing-ball volume. They must be computed or bounded for the same assignment. A small fill radius does not justify replacing the equal-volume assignment radius. This estimate alone yields no vanishing operator error when $H = O(q^{-1})$ and $\epsilon = O(q^{-1/2})$.

Dual cells for a local tensor action. A different supplied action can use sorted source coordinates $0 = x_0 < \cdots < x_N = L$, with interval lengths $\ell_j = x_{j+1} - x_j$. Half-interval endpoint weights and interior weights $(\ell_{j-1} + \ell_j)/2$ are the masses of the one-dimensional dual cells. Their tensor products satisfy the preceding transport theorem. For $f \in C^2([0, L])$, linear interpolation on each interval also gives the sharper analytic trapezoidal bound

$$\left| Qf - \int_0^L f(x)\,dx \right| \le \frac{\|f''\|_\infty}{12} \sum_j \ell_j^3 \le \frac{L\ell_{\max}^2}{12} \|f''\|_\infty.$$

Indeed the pointwise interpolation error is at most $\|f''\|_\infty (x-x_j)(x_{j+1}-x)/2$; integrate it on each interval. Successive application in three coordinates gives the tensor bound $L^3\ell_{\max}^2 \sum_{a=1}^3 \|\partial_a^2 f\|_\infty/12$. This applies to detector products and sampled mode Gram entries with their stated derivative bounds. Dual-cell weights and equal-volume source-count weights are different finite measures, each with its own comparison to volume. The trapezoidal argument does not improve the radial action by changing its weights implicitly, and it supplies no source-selection law, primitive timing rule or physical calibration.

B.3 Operational selection of the causal cone

The response-selected source Gram space E is three-dimensional. Its unit directions form a sphere, and adjoining a scalar line gives the quadratic form $Q(t, x) = t^2 - \|x\|^2$ on $\mathbb{R} \oplus E$. The finite proper carrier rotations act isometrically on E. Under natural feasibility and objective data with a unique normalized minimizer, the sixty directed seams have equal weight and select the corresponding translation average on the conservative record carrier. These are spatial and algebraic results. The following analytic theorem states an additional condition that selects an operational causal cone on that same carrier.

A displacement cone C contains zero and is closed under addition and nonnegative scaling. Suppose it represents a homogeneous operational order by $p \preceq q$ if and only if $q-p \in C$. Pointedness, $C \cap (-C) = \{0\}$, expresses antisymmetry. Operational covariance means that transformations of admissible histories, interventions and readouts preserve possible influence, with the corresponding linear action on event displacements. Merely changing a frame or relabeling celestial directions does not establish this condition. If C is instead the closed cone generated by finite operations, the conclusion concerns that closure; it does not assert that every displacement is a finite direct read.

For $u \in E$, define

$$X_u(t, x) = (\langle u, x\rangle, tu).$$

For a unit vector n, $B_n(s) = \exp(sX_n)$ is the boost of rapidity s: writing $x = x_\parallel n + x_\perp$,

$$t' = \cosh(s)t + \sinh(s)x_\parallel, \qquad x'_\parallel = \sinh(s)t + \cosh(s)x_\parallel, \qquad x'_\perp = x_\perp.$$

Theorem B.9 (Finite rotations and one boost direction)**.** *Let $G \subset \mathrm{SO}(E)$ be finite and irreducible, as for the proper icosahedral representation. Let $C \neq \{0\}$ be a closed convex pointed displacement*

cone invariant under $\mathrm{diag}(1, R)$ *for* $R \in G$*. Suppose, for one unit* n*, that* $B_n(s)C = C$ *for every real* s*. Then*

$$C = \{(t, x) : t \geq \|x\|\} \quad \text{or} \quad C = \{(t, x) : t \leq -\|x\|\}.$$

A positive time-axis vector in C *selects the first alternative. It is enough to assume boost invariance for a sequence of nonzero rapidities tending to zero.*

Proof. Irreducibility gives $\mathrm{span}\{Rn : R \in G\} = E$. Conjugation by $\mathrm{diag}(1, R)$ gives invariance under every $\exp(sX_{Rn})$. Write $u = \sum_j a_j R_j n$. Since $u \mapsto X_u$ is linear,

$$\exp(sX_u) = \lim_{m\to\infty} \left(\prod_j \exp\left(\frac{sa_j}{m} X_{R_j n} \right) \right)^m . \tag{12}$$

Indeed, the one-step product and $\exp(sX_u/m)$ have equal constant and linear Taylor terms, so their difference in matrix norm is $O(m^{-2})$. A telescoping sum over m factors bounds the total difference by $O(m^{-1})$; all intermediate products have a uniform exponential norm bound. Every finite product preserves C, and closedness passes this inclusion to the limit. Repeating with $-s$ gives equality. Thus every boost preserves C.

The finite average $|G|^{-1} \sum_{R\in G} R$ projects onto the fixed vectors of G, which are zero by irreducibility. Convexity therefore gives $(t, 0) \in C$ whenever $(t, x) \in C$. If $\|x\| > |t|$, boosts parallel to x can make the time coordinate positive or negative. Averaging these two images puts both signs of the time axis in C, contradicting pointedness. Hence every nonzero vector in C is causal.

A nonzero future causal vector has positive time coordinate. Its finite rotation average supplies a positive time-axis vector. Boosts and positive scaling generate every future timelike vector; closedness adds the null boundary. A past vector would similarly supply the opposite time axis, contradicting pointedness. This gives the future cone. Starting with a past vector gives the past cone. Nontriviality ensures one case occurs.

Finally, if $s_k \neq 0$ tends to zero, for every $s \in \mathbb{R}$ choose integers m_k with $m_k s_k \to s$. Integer powers and inverses of $B_n(s_k)$ preserve C. Continuity and closedness give invariance under $B_n(s)$, reducing the last assertion to the preceding proof. □

The theorem uses only two finite representation facts: the boost-axis orbit spans E, and the group average on E vanishes. It does not infer continuous spatial covariance from the finite carrier action. Its conclusion is a restriction: rotation covariance alone permits every cone $\{(t, x) : t \geq 0,\ \|x\| \leq vt\}$ with $v > 0$, and the stationary time-axis cone. Operational boost covariance excludes those alternatives except the unit-speed cone in the specified scalar coordinate.

The scalar coordinate requires an additive clock readout on actual histories. Once that attachment and the normalized boost action are fixed, the cone slope is fixed relative to them. Constructing the clock attachment is a separate dynamical requirement. The theorem does not select a layer tick, read weights, signal amplitudes or a field equation. The count construction in Section B.4 supplies a separate geometric duration readout on the declared order/count family.

At finite resolution, a bounded event set cannot carry a nontrivial continuous boost action by permutations. A physical use of the theorem therefore needs an operational continuum description or controlled comparisons across a varying family. Finite energy cutoffs and bounded windows must be transformed or restricted consistently. The existing algebraic Lorentz and celestial maps alone supply none of these intervention or clock attachments. The cone classification and product limit above are analytic proofs, rather than Lean formalizations of operational covariance.

B.4 A clock read from source interval counts

The equal-mass source populations of Proposition B.4 admit a clock defined by their records. Suppose an observer can read the complete ancestry of an upper event and identify a lower anchor. Its inclusive interval is $I(p,q) = \{e : p \preceq e \preceq q\}$. Choose a nondegenerate reference interval J in the same history and define

$$\mathcal{T}(I;J) = \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{13}$$

This readout uses order and cardinality. It needs no numerical event labels, spatial coordinates, timestamps, absolute density or oscillator frequency. Complete ancestry access is a substantive readout hypothesis: a hash of an unavailable record does not provide its contents. The cost of collecting and counting the records is separate from the duration reconstructed by the readout.

Theorem B.10 (Count clocks and proper-time ratios)**.** *Consider the declared source histories with their controlled order and equal-mass count limit. Let I_n, J_n approximate fixed interior timelike diamonds, with one common event weight $w_n > 0$, and let their limiting proper-time separations be $\tau_I, \tau_J > 0$. Then*

$$\mathcal{T}(I_n; J_n) \longrightarrow \frac{\tau_I}{\tau_J}.$$

The reference interval fixes the time unit; the clock origin is free.

Proof. For coordinate measure $dt\, d^3x$ and metric $c^2dt^2 - |dx|^2$, a flat diamond of proper-time separation τ has volume $K\tau^4$, where $K = \pi c^3/24$. The vertical formula follows by integrating its ball sections. A Lorentz transformation has determinant one and sends a timelike tip displacement to its rest form, so the same formula holds for moving tips. The count-volume theorem gives $w_n|I_n| \to K\tau_I^4$ and $w_n|J_n| \to K\tau_J^4$. Taking their ratio and its positive fourth root proves the claim. Both the event weight and the volume constant cancel. □

The supplied relation $\Delta = a/c$ enters the proof of the order and volume limit. It is absent from the clock's definition. This distinction allows a retrospective clock to be read from the same history without assuming an oscillator Hamiltonian. It does not select the population, read law or laboratory unit from accepted microscopic repairs.

Proposition B.11 (Finite clock enclosure)**.** *Suppose $A = wN$, $B = wM$ and independently justified volume bounds obey $|A - V_I| \le E_I$, $|B - V_J| \le E_J$, with $B > E_J \ge 0$, $E_I \ge 0$ and $V_J > 0$. Then*

$$\left(\frac{(A - E_I)_+}{B + E_J}\right)^{1/4} \le \frac{\tau_I}{\tau_J} \le \left(\frac{A + E_I}{B - E_J}\right)^{1/4}. \tag{14}$$

The same formula uses counts N, M and errors $E_I/w, E_J/w$.

Proof. The true volumes belong to their stated intervals. Division by a positive denominator and the increasing fourth root preserve the two inequalities. The error bound in (5) supplies such volume errors when its geometric hypotheses hold. Counts alone do not supply those errors. □

For relative volume errors $A = V_I(1 + d_I)$ and $B = V_J(1 + d_J)$, with $|d_I| \le \eta_I < 1$ and $|d_J| \le \eta_J < 1$, the measured clock divided by the exact proper-time ratio lies in

$$\left[\left(\frac{1 - \eta_I}{1 + \eta_J}\right)^{1/4}, \left(\frac{1 + \eta_I}{1 - \eta_J}\right)^{1/4}\right]. \tag{15}$$

Proposition B.12 (Unit freedom within the volume clock class)**.** *A positive elapsed-time function $F(V)$ depending only on diamond volume and additive for successive collinear inertial intervals is $F(V) = \kappa V^{1/4}$ for some $\kappa > 0$.*

Proof. Set $g(t) = F(Kt^4)$ for $t > 0$. Inertial additivity gives $g(t+s) = g(t) + g(s)$. Positivity implies strict monotonicity. Additivity first fixes g on positive rationals; monotonicity and rational approximation give $g(t) = g(1)t$ for all positive real t. This is uniqueness within the stated class, not a premise about every physical pointer. □

Accumulating duration along a curve. Let $\gamma(t) = (t, x(t))$ be C^2 on $[0, T]$, with $|x'(t)| \le v < c$, $|x''(t)| \le A_0$, and a common spatial buffer for its small diamonds. Segment durations are comparable to $\delta_n \to 0$. Suppose

$$\frac{\Delta_n}{\delta_n} \to 0, \qquad \frac{H_n}{\delta_n} \to 0, \qquad \frac{h_n}{a_n} \to 0,$$

and sampled tips have spatial and temporal errors $O(H_n + \Delta_n)$. Summing (13) over successive segment intervals, using the same fixed reference interval, converges to

$$\frac{1}{\tau_J} \int_0^T \sqrt{1 - |x'(t)|^2/c^2}\, dt. \tag{16}$$

Indeed, the two-shell estimate is uniformly $O(\delta_n^4 h_n/a_n + \delta_n^3(H_n + \Delta_n))$ per segment. The strict speed margin bounds each exact diamond volume below by a positive constant times δ_n^4. Tip perturbations have the same relative error order. Equation (15) therefore controls all segment clocks by one vanishing relative error.

The continuum polygon error has an explicit bound. For $f(u) = \sqrt{1 - |u|^2/c^2}$ on $|u| \le v$, its Hessian norm is at most $M_0 = [c^2(1 - v^2/c^2)^{3/2}]^{-1}$. Taylor expansion about each segment's average velocity has zero integrated linear term. Since the velocity differs from that average by at most $A_0\delta_i$, the chord and curve proper times differ by at most $M_0 A_0^2 \delta_i^3/2$. The total is at most $M_0 A_0^2 T(\max_i \delta_i)^2/2$. This proves (16). Fixed one-layer intervals do not satisfy these resolution hypotheses.

Finite records and calibration scope. The finite executable control parses writer identities, immutable register versions and values in the $q = 5$ history. Its 500 events give the three centered inclusive counts $2, 41, 80$. The first two continuum diamonds lie inside the source window; the third does not. Exact dyadic fourth-root bounds, schedule permutation and opaque event relabeling check the readout. These coarse intervals provide no finite proper-time accuracy certificate. The source-pinned expected hash authenticates the replayed content relative to that commitment; it is not an external signature or a proof of physical provenance. Unrelated serial audit links are excluded from the semantic read order.

Lean formalizes the finite interval and root inequalities. The count-volume and curve limits above are analytic proofs. In this layered law every causal pair k layers apart has a path of exactly k reads, and every strict chain has at most k edges. Longest-chain height consequently reads coordinate duration independently of displacement. It is not a proper-time clock for moving tips in this family. The count construction concerns geometric duration reconstructed from accessible records; it neither sets the time cost of primitive seam words nor identifies physical clock readings without a common physical interpretation of the event law.

B.5 Observations of the causal manifold

An observer inside the source-record family of Section B.1 has access to the event order and to event cardinalities. This section records what such an observer reads from those two inputs at finite regulator, what the carriers produce of the family itself, and which of the finite statements are theorems.

The finite poset. For a Fibonacci level q the events are the pairs (j, s) of a round $j \le K_q = \lceil \sqrt{q}\, \rceil$ and a site $s \in S_q$, and $(j, s) \preceq (j', t)$ exactly when $j \le j'$ and the graph distance on the neighbour graph satisfies $d(s, t) \le j' - j$. This relation is a partial order, it equals the reflexive transitive closure of the one-round reads, every round is an antichain, the width equals $|S_q| = q^3$, and every event at round j has height j (`SourceNetLayeredOrder`, theorems `layerPrec_isPartialOrder`, `layerPrec_iff_reflTransGen`, `walk_iff_walkDist_le`, `width_eq_card_sites`, `height_eq_layer`; the concrete read law of the family is the instance `precedes_iff_layerPrec`). The flat reference constants of the ordering fraction are exact: the radial-time double integral of Proposition B.5 equals $T^8/1920$, so the fraction equals $1/10$, and the Myrheim–Meyer values in dimensions 2, 3 and 4 are $1/2$, $8/35$ and $1/10$ [8, 9] (`OrderingFractionFourDimensional`). The finite clock enclosure of Proposition B.11 is formalized in `SourceCountClockEnclosure`. The family limit, beyond these finite constants, is proved in `FlatDiamondNormalization`: actual generated strict pair counts have ordering fraction tending to $1/10$ for an interior timelike limiting diamond, with the supplied clock, population and read law kept explicit. The weighted count-volume limit and its finite error are proved in `GoldenSourceVolumeLimit`.

What the observer reads. On the central vertical diamond of K_q rounds the inverted Myrheim–Meyer dimension reads 2.63, 2.93, 4.15, 3.61, 4.10, 4.09 and 4.07 at $q = 5, 8, 13, 21, 34, 55, 89$, with ordering fractions 0.306, 0.241, 0.088, 0.138, 0.092, 0.093 and 0.094 against the flat value $1/10$; levels with even K_q approach the value from below, 0.088, 0.092, 0.093, 0.094 at $K_q = 4, 6, 8, 10$, and levels with odd K_q from above. The outermost diamond of K_q rounds extends past the sampling window at every level, by 9.2%, 0.4%, 0.9%, 0.4%, 0.02%, 0.2% and 0.1% of its continuum volume, so these values read clipped diamonds; the largest interior diamonds, of $K_q - 1$ rounds, read 4.04, 5.02, 2.83, 4.38, 3.55, 3.84 and 3.92. The moving-tip diamonds of Proposition B.3 read 0.250, 0.135, 0.121, 0.115, 0.108 and 0.102 at $q = 8, 13, 21, 34, 55, 89$, dimension 2.89 to 3.97. The same read law on the two- and one-dimensional golden populations reads dimensions 3.02 and 2.01 at $q = 89$ against the constants $8/35$ and $1/2$, so the readout responds to the dimension of the population. The normalized count of the vertical diamond against $\pi c^3 T^4/24$ reads 0.193 against 0.198 at $q = 13$, 0.141 against 0.147 at $q = 34$, 0.172 against 0.177 at $q = 55$ and 0.163 against 0.165 at $q = 89$. The count clock $(|I|/|J|)^{1/4}$ of two nested vertical diamonds reads 1.705, 2.088, 1.907 and 2.051 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Every pair count through $q = 89$ is an exact all-pairs count; the structural fields at $q = 5, 8, 13$ agree byte for byte with the replay of Section B.1. A sampled level at $q = 144$, with 2,985,984 sites and twelve rounds, reads 0.0955 ± 0.0002 on the outermost diamond, so the even-round ladder continues 0.0600, 0.0803, 0.0887, 0.0932, 0.0955 at $K = 4, 6, 8, 10, 12$ with halving increments; its dimension reads 4.05, the largest interior diamond 3.95, the moving tips 0.102, the count clock 1.959 against 2, and the two- and one-dimensional controls 3.02 and 2.00. These are stratified estimates with standard errors, not exact counts. The bounds of Proposition B.2 exceed the diamond volume at these levels, and the finite values above are readings of the supplied law, not a demonstration of its limit.

What the carriers produce. The family is produced by the carriers themselves. One exact twelve-port carrier per site holds the record $z(b)$ as its port loads on the six antipodal axes and reads its position through the rank-three response $x = 2P_{\mathrm{slow}}N$ of the rank-three source metric. The readback metric equals the source-axis contraction $s(b)$ with scale exactly one, an identity in $\mathbb{Q}(\sqrt{5})$ checked on every pair through $q = 21$ and on 10^6 seeded pairs at $q = 34, 55, 89$. Each carrier decides its neighbours from its own readback; the resulting relation equals the family's neighbour graph edge for edge at $q = 5, 8, 13, 21, 34, 55, 89$. The carriers then read one another for K_q rounds, every read an authenticated read-after-write event, and the precedence generated from those events with no declared parent lists equals the layered order on the centre diamond at every level, with strict pair counts 968, 4,244, 102,990, 2,904,889, 47,881,819, 2,083,062,313 and 55,108,871,450. A unit intervention at the centre in round zero changes exactly the future cone in every round. The operation costs are recorded: reads per round $2E + n$ for E edges and n carriers, at $q = 34$ a total of 157,922,880 reads and 275,128 writes, at $q = 89$ a total of 21,845,550,850 reads and 7,754,659 writes by 704,969 carriers over ten rounds.

Inference and boundary. Order and count determine conformal geometry and scale under the continuum causality hypotheses [14, 15]. The finite readings above compare a supplied family with flat reference values; they do not establish those reconstruction hypotheses or a statistical acceptance of flat $1 + 3$ spacetime. The population, the read law, the round duration and one counted event per carrier and round are supplied. Their selection by accepted native repairs, a physical clock, and the continuum limit are separate conditions rather than results of this section. Two further receipts read the order with more than one number. At $q = 8$ to 34 the Myrheim–Meyer dimension is the same at the centre and at every admissible off-centre tip of the same size to within 0.04; the odd multipoles of the link directions vanish by the two-way reads and the $l = 4$ power falls from 0.036 to 0.0016 over the four levels; the abundance of intervals of 64 to 1024 events matches a flat sprinkling within 14 percent while links and small intervals do not, as a layered order requires. Sampled chain counts and interval spectra are retained at $q = 55, 89, 144$. For the central diamond, the three inverted dimensions, the maximum CDF difference D_{25} on the declared 25-point grid, and the interval-mean deviation from the exact flat value 1/105 are:

q	$d(C_2)$	$d(C_3)$	$d(C_4)$	D_{25}	Mean deviation
55	4.09590	4.03210	4.00605	0.075810	+5.0739%
89	4.09437	4.03413	4.01664	0.024725	+8.5836%
144	4.07556	4.01605	3.97098	0.032342	−20.9353%

Here D_{25} compares the weighted source CDF with the retained seeded Monte Carlo flat reference, which has its own sampling uncertainty. It is not an exact continuum CDF distance or a continuous Kolmogorov–Smirnov statistic. The absolute interval-mean discrepancy grows at both refinements, while D_{25} first falls and then rises. Uniform monotone improvement of the retained point estimates therefore fails. This descriptive result neither establishes nor statistically rejects asymptotic convergence.

The accompanying comparison retains all 96 regions across the three levels and the three spatial dimensions, including every moving and off-centre region, both lower-dimensional controls, and comparisons with the five sampled flat references in spacetime dimensions two through six. At $q = 89$ the moving-interval mean is 23.13% above 1/105; at $q = 144$ it is 15.60% below it. The $q = 144$ moving-diamond dimensions are 3.99045, 3.99998 and 4.03391. Over the fourteen off-centre diamonds and their central reference, the $q = 89$ three-chain dimension has mean 4.078 and

spatial standard deviation 0.016; these are regional dispersion statistics, not an error on a common dimension.

Marginal chain-count standard errors are retained historical estimates; their cross-covariances and the uncertainty of the self-normalized interval mean and CDF are unavailable. They cannot support independent joint significances, an uncertainty-qualified refinement test or a numerical interval-mean error bound. The central four-chain relative standard error is about 7.49% at $q = 89$ and 10.56% at $q = 144$, whose source sampling uses half as many draws. The original large-run draws and the chunk-parallel $q = 144$ driver are not retained. Its neighbour digest agrees with the archived family receipt, with recorded sequential versus chunked identities at $q = 21, 34$; these ties do not reproduce the missing computation. The independently checked arithmetic comparison exposes signed deviations, available marginal errors and missing uncertainty. A statistical decision requires a declared bounded control, retained joint sufficient statistics, reference uncertainty and complete driver custody. Curvature estimators are not evaluated. The supplied population, read law and clock, and the separate small clock controls below, retain their stated boundaries.

Exact temporal quadrature control. For continuum spatial balls sampled at $K + 1$ equally spaced layers, $K > 0$, the layer-summed vertical diamond volume divided by its continuum volume is

$$B_K = \frac{32}{K^4} \sum_{j=0}^{K} \min(j, K - j)^3 = \begin{cases} 1 + 4/K^2, & K \text{ even}, \\ (1 - 1/K^2)^2, & K \text{ odd}. \end{cases}$$

Indeed, for $K = 2m$ the sum is $m^3 + 2\sum_{j=1}^{m-1} j^3 = m^2(m^2 + 1)/2$; for $K = 2m + 1$ it is $2\sum_{j=1}^{m} j^3 = m^2(m + 1)^2/2$. The sum-of-cubes identity gives the displayed formulas. For $J \geq 2$, dividing the raw nested count-clock ratio by $(B_K/B_J)^{1/4}$ isolates this temporal discretization effect. The exact finite sums and both parity factors are proved in `FiniteLayerQuadrature`.

q	Raw relative error	Layer-corrected relative error
55	-4.6395%	-0.6853%
89	$+2.5509\%$	-0.5014%
144	-2.0289%	-0.1007%

The numerical control uses the same count receipts, with no fitted parameter. It is exact for continuum balls at discrete layers, not for finite graph diamonds. Spatial error and clipping are separate: all these outermost diamonds fail the stated finite interior-enclosure hypotheses, including $q = 144$, whose clipped fraction is 8.9748×10^{-8}. At $J = 1$ the continuum layer volume vanishes and the correction is undefined. The residuals therefore establish no physical calibration or uniform finite-graph convergence rate.

Uniform history length and interval clocks. For any related pair K layers apart in a read graph with self-waits, a strict chain has at most K edges. A shortest spatial path of length $d \leq K$, padded by $K - d$ waits, attains K. Hence a uniform positive duration per history edge gives moving/rest clock ratio one at fixed K, whereas the supplied Minkowski ratio is $\sqrt{1 - \beta^2} < 1$ for $0 < |\beta| < 1$. Refinement cannot remove this discrepancy at fixed velocity. This excludes that clock assignment for this layered family; it says nothing about state-dependent durations or independently selected records.

A finite comparison retains all 120 scheduled cases: four golden populations $q = 8, 13, 21, 34$, five boost targets, three directions, and two read laws. There are 89 related pairs and 31 timelike but graph-unrelated pairs. At $q = 34$ the nine primary moderate-boost Euclidean cases have median

absolute relative errors 16.290% for uniform history length, 2.914% for the raw volume clock and 0.813% after temporal quadrature correction. The same corrected instrument on axis-only reads gives 16.149% across six related primary pairs; three further failures are retained. Actual velocities use the rounded endpoints. At target $\beta = 0.9$ the Euclidean axis pair is unrelated, and the other two directions have corrected errors -17.19% and -9.70%. The comparison tests read laws under supplied geometry. Its moving-diamond correction uses continuum intersections of spatial balls, so it is a calibration control rather than an intrinsic order-only clock. These deterministic errors are not confidence intervals. The specifications, witness histories, receipts and an independent verifier of the count and quadrature controls are retained [29].

B.6 Source selection and a sufficient record-transport law

The source-selection question has two different meanings. A unique finite golden population and radius are stronger requirements than a common operational causal/count limit. Here we prove a sufficient source law for the latter, and compare M1 as one regulator of it. The proposed law is *additional* to A1–A3. It is neither adopted as a fourth core axiom nor asserted to hold in the registered simulator. Its several substantive requirements are stated together; no claim of minimality or unique necessity is made.

Why the existing data-access clauses do not supply a channel. There is a scalar-seam construction with a complete port response, rather than a response-free scalar toy. At each vertex of the normalized midpoint sphere S_n put

$$\mathcal{A}_i = \bigoplus_{p=1}^{12} M_6(\mathbb{C}), \qquad \mathcal{R}_i = Z(\mathcal{A}_i) = \mathbb{C}^{12}.$$

Its twelve block units are primitive central ports. Distinct-carrier seams are $\mathbb{C}$, with scalar inclusion and the unital normalized trace $E_i(a) = \sum_p \operatorname{Tr}(a_p)/72$ as visible data restriction. Triple restrictions are identities. Local chart overlaps are distinguished from these inter-carrier seams. Their proper action groupoid implements all sixty carrier rechartings; it does not transmit a private record to another carrier.

An exact response on the same accessible internal $\mathbb{C}^6$ is

$$D_p = \operatorname{diag}\Big(\widehat{v_p/2} + iQ_p, \widehat{\sigma(v_p)/2}\Big), \qquad Q_p = \frac{v_p v_p^{\mathsf{T}}}{2} - \frac{r^2}{6} I + \frac{1}{12} I,$$

where v_p is an icosahedral calculation frame, $r^2 = \|v_p\|^2$, σ is its Galois conjugation and $\widehat{v}w = v \times w$. Even and odd antipodal fields span respectively the six imaginary symmetric directions and the two three-dimensional real skew spaces. Thus the response is injective, commutator closed and has image $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Full M_6 effects observe its relative block phase; restricting to block-diagonal effects loses one observable response direction. For each proper action a, $U_a = \operatorname{diag}(R_a, \sigma(R_a))$ satisfies $U_a D_p U_a^* = D_{ap}$. Executed Cayley factors in the original response span realize these matrices, their covariance and their compositions. The central permutation is the chart map; an inner automorphism of a carrier algebra is not claimed to permute its center.

The admitted instruments are local: central readout, destructive internal effect readout into two central ports, requested-block trace reset, prediction from a supplied preparation and local conditioning, and in-place checkpoint continuation. Only the declared response span generates infinitesimal reversible port operations. Irreversible read/reset channels have kernels and add no

reversible direction. Independent ticks carry no record-dependent controller information between carriers.

The open-star nerve is S_n, the bridge is the identity and its fundamental cycle has degree one. Midpoints coarsen to declared endpoints, transported under relabeling; fundamental chains push forward exactly and spherical mesh contracts. Coarse local states average their descendant fibers. The instrument lift runs independently on every descendant, with each branch's own outcomes, then uses one fixed local tag through continuation. Selective conditioning updates tag weights; longer refinements use the composed weights. The tagged direct-sum experiment is not misidentified with the descendants' joint tensor algebra or with a communication channel.

The feasible local-state family is unconstrained except for normalization and chart covariance. Its complete grammar contains all 431 independent Hermitian state coordinates per carrier. The declared reference is $I_{72}/72$, the cover contains every carrier, and the weight is $1/N_n$. The weighted Umegaki objective has the unique tracial minimizer by nonnegativity and its equality condition. Every omitted coordinate admits two positive trace-one perturbations invisible to the remaining coordinates. This is an explicit A3 problem, not invariance used to choose a reference among all possible source ontologies.

Induction on every admitted local operation shows that changing an independent private source port leaves another carrier's entire local transcript unchanged. The induction permits arbitrary receiver-local memory and stopping rules, so no decoder of that transcript reads both interventions correctly. For a uniform unknown port its success probability is at most $1/12$. Hence the stated typed data-access clauses, with local chart holonomy distinct from inter-carrier record transport, do not entail M1's remote read service. The detailed full-clause interpretation and all-level refinement proof are in `code/source_selection_model/DERIVATION.md`; finite receipts do not replace that analytic model-membership argument. If reversible overlap holonomy is strengthened to require inter-carrier record transport, this particular countermodel is excluded by that additional requirement.

Proposed law RG: response-natural classical record transport. The complete-response theorem for A1–A2 gives $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ at a complete reference carrier. Its unique three-dimensional simple ideal E, with the response trace form, has connected adjoint action $SO(E)$. This is an internal Euclidean response space. RG says that the central-record source admits a complete, intervention-faithful, refinement-compatible realization of the following generated process theory, graded by a source-observable duration:

1. Objects are arbitrary finite independent classical registers at addresses in the affine space modeled on E. Finite tensor products, local preparation, finite deterministic classical maps, nondestructive read/copy, discard and retained versions are realizable at finite cutoff. Their ideal displacement and local duration are zero. Their actual implementations and storage costs are retained.

2. One retained-record flight has displacement e, $\|e\| = 1$, and duration $\tau > 0$. Translation, connected response rotation and positive scaling generate actual flights $(\lambda Re, \lambda\tau)$ for $R \in SO(E)$ and $\lambda > 0$. Waiting preserves the record. These processes transmit retained records between carriers.

3. Sequential and finite parallel composition are realized. Time adds on dependency paths; independent branches share elapsed time. Every native dependency, including controller, timing and metadata channels, factors through the graded local/flight/wait grammar. Conversely every finite generated process has an admitted native realization preserving all independent

record interventions and immutable versions, with every intermediate event and its resource bill retained. Realizations commute with the declared source refinements.

Intervention faithfulness includes local interventions at every designated intermediate record output before publication, with all other registers and cached copies held fixed. Lowering must commute with insertion of these operations and subsequent reads. The version is committed once after its intervention. Merely agreeing on final outputs for all initial preparations is weaker: two versions can both contain $x + 1$, making a substituted read indistinguishable until one version alone is perturbed.

The exact version realizes these ideal grades. The vanishing-overhead version requires, for every complete finite experiment and every $\epsilon > 0$, a finite realization with exact record values and endpoint addresses, no faster native dependence, and each logical event time in $[t, t + \epsilon]$ relative to its ideal schedule. This uniform whole-program requirement is part of RG. Nonnegative primitive overheads summing to at most ϵ on every complete path suffice, by induction on the earliest-time recurrence using addition and maximum. A fixed error at each layer is not a substitute.

RG supplies flat homogeneous operational addresses, record protection, cofinal workspace, parallel composability and a physical connection from the internal response to record motion. None follows just from internal rotation covariance. Translations and scales act on a family of finite experiments, possibly changing cutoff; they are not extra infinitesimal port generators on one carrier. A proposed implementation must obey A1's full response constraint. A3 can select states inside such a source but is not used to derive RG or compel a particular observer program.

Proposition B.13 (RG suffices for the operational M1 limit)**.** *Under the complete-response conclusion of A1–A2 and RG, there exist cofinal finite independent central-record experiments whose native ancestry on their distinguished records has the round* $3 + 1$*-dimensional causal limit, continuity-set volume counts, Alexandrov interval counts and four-dimensional ordering fraction. M1's golden population with* $q = F_n$ *and observation duration* $L/(c\sqrt{q})$ *is one regulator with these limits, where* $c = 1/\tau$*. Its finite radius is not a uniquely selected microscopic length.*

Proof. Each primitive flight has norm c times its duration. The triangle inequality bounds every complete native dependency by $\|y - x\| \le c(s - t)$, including paths through auxiliary events. Conversely, for $\|v\| \le cT$, response transitivity rotates e to $v/\|v\|$. Scaling by $\|v\|$ and waiting for $T - \tau\|v\|$ realizes the read. The zero displacement uses waiting. The exact process has precisely the closed round cone; vanishing overhead attains every strict timelike read and excludes every spacelike read. Finite tensor composition realizes any finite list simultaneously, with protected source versions.

In a calculation frame prepare records at $x_b = Lb/q$, $b \in \{0, \dots, q-1\}^3$. RG supplies their joint finite preparation, rather than a presumed native carrier census. Half-open cells have equal volume $(L/q)^3$ and assignment/covering error $h_q \le \sqrt{3}L/q$. Choose $\delta_q \to 0$ with $q\delta_q \to \infty$. Exhaustively observe the preceding population within duration δ_q, then commit the consumed source versions. The primitive latency calculation derives the complete read menu $\|x_b - x_a\| \le c\delta_q$; this menu is not input to the source law. The covering-path theorem constructs a k-layer read path whenever

$$\|y - x\| \le k(c\delta_q - 2h_q), \qquad \text{and every generated path obeys} \quad \|y - x\| \le kc\delta_q.$$

Independently variable records and retained versions make these actual causal dependencies. Compiling a primitive correctly on every input state extends to every finite word by induction; concatenating the native words retains the sum of their event bills.

For each bounded horizon, lower the entire finite experiment with uniform time error $\epsilon_q \le \delta_q/q$. Commits need not be perfectly simultaneous. Let P_q be the consumed-writer order and N_q full

native ancestry restricted to those commits. For each consumed-writer edge $u \to v$, vary the record at u's output boundary while holding other records fixed. The corresponding tuple coordinate read at v changes. If no native path connected them, the ancestors of v would form a dependency-closed region unchanged by that local intervention. Induction on the native updates would leave its read unchanged, a contradiction. This proves $P_q \subseteq N_q$ using intermediate intervention faithfulness; the value-only compilation induction does not prove it. Complete native grading then gives

$$P_q \ \subseteq \ N_q \ \subseteq \ C_c \quad \text{at the actual native event assignments.}$$

Every fixed strict timelike comparison eventually belongs to P_q, while every fixed spacelike or backwards comparison is eventually excluded from N_q. Moving two positions by at most H and times by at most ϵ changes the causal margin by at most $2H + 2c\epsilon$; this tends to zero.

Space and time cells give distinguished-event weight $w_q = (L/q)^3\delta_q$. For any bounded continuity set A in the window, its pulled-back indicator converges almost everywhere and is bounded by one, hence $w_q \# A_q \to \int_A dt\, d^3x$. A clipped terminal layer has mass error at most $L^3\delta_q$. For intervals, use the order sandwich and the null measure of the endpoint cone boundaries. For strict pairs, apply the same argument on the product window to both interval indicators and the precedence indicator; the time diagonal is also null. Dominated convergence gives the limits for both P_q and N_q, without excluding additional native edges by assumption. An interior rest diamond of duration T has $V = \pi c^3 T^4/24$ and oriented-pair measure $V^2/20$. Consequently $2\,\#\text{strictPairs}/[N(N-1)] \to 1/10$.

For the golden regulator use the proved equal-volume assignment with $h_q \le 2\sqrt{3}L/q$ and $\delta_q = L/(c\sqrt{q})$. RG supplies the finite preparations and transmissions, and all previous bounds apply. The radius $L/\sqrt{q} = c\delta_q$ is the reach of the chosen observation clock. Both regulators thus converge to the same causal/count observables. For a K-Lipschitz observable, assignment errors (H, ϵ) change its weighted integral by at most $K(H + \epsilon)$ times the window volume, plus the clipped endpoint term. Comparing regulators adds their bounds; indicator comparisons use null boundary tubes instead. □

This conclusion concerns a realizable observer probe and its operational limit. It neither selects spontaneous execution of that probe nor equates its distinguished-event volume with a count of all microscopic repair events. It proves no Lorentz covariance of matter, curved geometry or laboratory clock calibration. Finite graphs need not be isomorphic, and unrelated field dynamics do not inherit equivalence from count convergence.

Finite clearance, verification and deletion controls. The golden squared displacements in units of L^2 lie in $\mathbb{Z}[\phi]$. For integer $q > 1$ none equals $1/q$: irrationality would first force the coefficient of ϕ to vanish, leaving an integer equal to $1/q$. Thus every admitted nonself golden flight has positive finite slack, although controller/commit costs remain charged. The rational-grid reference uses $L = c = 1$ and $\delta_q^2 = 2/(2q+1)$; boundary equality would require $(2q+1)m = 2q^2$, impossible since $\gcd(2q+1, q) = 1$.

The reference prepares, forks, flies, waits, adds locally and commits two layers for $q = 2, 3, 5$. Baselines, every single-site initial intervention and every first-layer commit intervention retain 323 executions, totaling 5,344,106 primitive events. Compact commitments are checked against live streams and independent semantic replay, including all consumed writers and intervention path multiplicities. These are executions of RG's reference grammar, not of an asserted native simulator implementation. Source pins and hostile-input tests reject forged timing, values, missing events and filtered intervention denominators. The additive reference diagnostic is distinct from the proof's lossless tuple payload, constructed analytically from RG's finite classical maps. Its intervention

tests certify that each consumed value affects the diagnostic; they do not certify arbitrary payload maps or native programs.

The Lean proofs cover the local information bound, the primitive cone and its covering paths, the lowering induction, the localized-intervention ancestry reduction, event bills, population size and margin stability. The full model interpretation and the join from RG to native ancestry and counts are analytic. Detailed proofs appear in `code/source_selection_model/RECORD_GLUING.md`.

Deleting transport gives the scalar-seam control; keeping only axis flights gives an ℓ^1 cone; an ungraded metadata channel defeats the upper cone bound; destructive transfer or absent parallel workspace defeats the finite experiment. These controls explain the content of the proposed law instead of concealing it in a successful M1 schedule.

C Fields, volume and entropy on source populations

C.1 Spatial populations and a common scalar continuum action

The completed conservative source-record carrier is a Euclidean three-space under the stated response and metric hypotheses. Its whole dense record group is not a locally finite population. A finite-resolution selection and a transport law can be stated on that carrier without inserting a Cartesian lattice. This construction keeps both choices explicit.

Fix $r, \delta, \eta > 0$, a response-metric ball $\Omega = \overline{B}(0, r)$, and a finite directed seam-word endpoint menu $M \subset \Omega$ that is an η-net of Ω. Such menus exist: density approximates a slightly inward-shifted point by a source endpoint, compactness extracts finitely many approximants, and a finite maximum word budget contains all of them. This is algebraic endpoint availability, not a proof that a repair law accepts those words as moves. Choose a maximal δ-separated subset $S \subset M$, and set $h = \delta + \eta$. The resolution, window and tie rule are supplied; maximal means inclusion-maximal, not maximum cardinality or a selected energy minimum.

Proposition C.1 (Finite spatial population and metric transport)**.** *The selected points form an h-net, and*

$$(r/h)^3 \le |S| \le (1 + 2r/\delta)^3.$$

For $R \ge h$ and $B(x, R) \subset \Omega$,

$$((R - h)/h)^3 \le |S \cap B(x, R)| \le (1 + 2R/\delta)^3.$$

Connect distinct points whose source distance is at most $a > 2h$, weighting edges by that distance. The graph is connected, degrees are bounded by $(1+2a/\delta)^3 - 1$, and its shortest-path metric satisfies

$$|s - t| \le d_G(s, t) \le \left(1 + \frac{2h}{a - 2h}\right) |s - t| + 2h.$$

Consequently $a \to 0$ and $h/a \to 0$ give uniform metric convergence on the fixed window. With $\eta \le \delta$ and $a = 8\delta$, degrees instead stay at most 4912 and $d_G \le 2|s-t| + 4\delta$; this latter estimate alone does not imply a Euclidean metric limit.

Proof. Maximality puts each menu point within δ of S, proving coverage. Disjoint radius-$\delta/2$ balls give the upper counts; covering the window, or $B(x, R-h)$, by radius-h balls gives the lower counts. For the graph, divide the segment from s to t into $k = \lceil |s - t|/(a - 2h) \rceil$ pieces. Convexity retains the segment in Ω; approximate its interior division points within h, keeping its endpoints exact. Consecutive approximants are at most a apart and the path length is at most $|s - t| + 2hk$. The triangle inequality gives the lower bound. These estimates are independent of the choice among maximal sets. □

The metric fixes relative three-volume. Let C_i be the Voronoi cell of $x_i \in S$, clipped to Ω, and $v_i = \mathrm{vol}(C_i) > 0$. Thus $\sum_i v_i = V = \mathrm{vol}(\Omega)$, and almost every point of C_i is at distance at most h from x_i. These are metric weights, not a physical event-count density. For $0 < h \le \epsilon$ supply

$$a_{ij} = \frac{45 v_i v_j}{\pi \epsilon^5}(1 - |x_i - x_j|/\epsilon)_+, \qquad (Lq)_i = v_i^{-1} \sum_j a_{ij}(q_i - q_j).$$

The weighted inner product is $\langle p, q\rangle_v = \sum_i v_i p_i q_i$. Symmetry gives $\langle q, Lq\rangle_v = \frac{1}{2}\sum_{ij} a_{ij}(q_i - q_j)^2 \ge 0$. The common finite action is

$$S_h[q] = \frac{1}{2}\int \left[\|\dot q\|_v^2 - c^2\langle q, Lq\rangle_v - m^2\|q\|_v^2\right] dt, \qquad c > 0, \quad m \ge 0. \tag{17}$$

Its equation is $\ddot q + c^2 Lq + m^2 q = 0$. The coefficient m has inverse-time units in this convention; neither physical mass nor time units are inferred.

Proposition C.2 (Uniform scalar and detector limit)**.** *Let $u_{tt} - c^2\Delta u + m^2 u = 0$ be a smooth reference on the full carrier. On $[0, T]$, its spatial support has distance greater than ϵ from the complement of Ω. Write $M_1 = \sup|\nabla u|$ and $M_4 = \sup\|D_x^4 u\|_{\mathrm{op}}$, with suprema over the full carrier and time window. Then*

$$\left|(Lu|_S)_i + \Delta u(x_i)\right| \le \frac{15}{112}\epsilon^2 M_4 + \frac{120 h(\epsilon + h)^3}{\epsilon^5} M_1 \le \frac{15}{112}\epsilon^2 M_4 + \frac{960 h}{\epsilon^2} M_1.$$

Put $B = c^2\sqrt{V}[(15/112)\epsilon^2 M_4 + 960 h\epsilon^{-2} M_1]$. Initialize the finite action with the sampled reference position and velocity. Then

$$\|\dot q - u_t|_S\|_v \le tB, \qquad \|q - u|_S\|_v \le \tfrac{1}{2}t^2 B.$$

For a fixed C^1 detector b, the field readout $D_h = \sum_i v_i b(x_i) q_i$ and its continuum counterpart $D = \int_\Omega bu$ obey

$$|D_h(t) - D(t)| \le \tfrac{1}{2}\sqrt{V}\|b\|_\infty t^2 B + Vh\|\nabla(bu(t))\|_\infty.$$

Thus $\epsilon \to 0$ and $h/\epsilon^2 \to 0$ yield a uniform finite-time observable limit. In normalized carrier units $\epsilon = h^{1/4}$ gives an $O(\sqrt{h})$ bound. Nested populations are unnecessary.

Proof. Compare L with the integral operator having kernel $45(1 - |z|/\epsilon)_+/(\pi\epsilon^5)$. Odd Taylor terms cancel and

$$\int (1 - |z|/\epsilon)_+ z_1^2\, dz = \frac{2\pi}{45}\epsilon^5, \qquad \int (1 - |z|/\epsilon)_+ |z|^4\, dz = \frac{\pi}{14}\epsilon^7.$$

The second moment gives $-\Delta$; the fourth-order Taylor remainder, bounded by $M_4|z|^4/24$, gives $15\epsilon^2 M_4/112$. For nearest-cell projection P, compare $g_x(y) = (1 - |x - y|/\epsilon)_+(u(x) - u(y))$ with $g_x(P(y))$. Splitting the difference at the larger kernel value uses a point within ϵ of x; kernel and field Lipschitz bounds give $|g_x(P(y)) - g_x(y)| \le 2hM_1$. This difference vanishes outside $B(x, \epsilon + h)$, giving the stated quadrature term. If $u(x) \ne 0$, the support buffer retains the whole kernel ball in Ω; otherwise the integrand outside Ω is zero. No boundary term is omitted.

The sampled residual in the weighted norm is at most B. The positive operator $A = c^2 L + m^2 I$ has $\|\cos(t\sqrt{A})\| \le 1$ and $\|\sin(t\sqrt{A})/\sqrt{A}\| \le t$, with the zero eigenspace interpreted by continuity. Duhamel integration gives the two trajectory bounds. Cauchy–Schwarz and nearest-cell quadrature for bu give the detector bound. □

The same cell argument gives $\|L\|_v \le 960/\epsilon^2$, using twice the largest weighted degree. `MetricKernelEnergy` formalizes the finite positive-energy and mass-weighted degree inequalities; the geometric, integral and continuum arguments above are analytic proofs. A local discrete-action time integrator with a strict Courant bound may consume this operator and add its own time and initialization errors.

Smooth compact initial data admit the reference until their speed-c domain of influence reaches the window boundary. Local energy flux for the Klein–Gordon equation gives that speed; the nonnegative mass term changes energy density but not the characteristic cone. Applied to a smooth local preparation difference, the detector estimate bounds the finite response outside that cone. A nonzero continuum signal is resolvable when its lower bound exceeds the approximation and detector errors. Instantaneous small tails of the finite graph are not identified with detectable superluminal signals. This is an interior effective limit, not a theorem about arbitrary boundary conditions or exact microscopic order reflection.

The canonical variables $Q_i = \sqrt{v_i} q_i$ and $P_i = p_i/\sqrt{v_i}$, $p_i = v_i \dot{q}_i$, give the oscillator Hamiltonian $\frac{1}{2}(|P|^2 + Q^T \widetilde{A} Q)$, where $\widetilde{A} = M^{1/2}(c^2 L + m^2 I) M^{-1/2} \ge 0$ and $M = \mathrm{diag}(v_i)$. Supplied canonical quantization on $L^2(\mathbb{R}^{|S|}, dQ)$ gives the same field readouts as multiplication operators. For $m > 0$ every mode has a Gaussian vacuum; at $m = 0$ zero modes remain free particles unless a separate restriction is declared. This finite Hilbert construction is not an interacting quantum continuum or a derivation of the Standard Model.

Coarsening, the volume prescription, radial force law, inertia, action coefficients and accessible observer variables remain added assumptions. The kernel graph may have growing degree. Algebraic source words alone do not implement its couplings as accepted primitive repairs or calibrate their duration. The construction supplies one conditional scalar effective world on the response carrier; its selection and physical interpretation remain separate obligations.

Proposition C.3 (Controlled boost comparison for scalar detectors)**.** *Let Λ be a proper orthochronous Lorentz transformation of $c^2dt^2 - |dx|^2$, and let u be a smooth Klein–Gordon solution on the full spacetime. Put $u^\Lambda = u \circ \Lambda^{-1}$. Suppose both references satisfy the support and derivative hypotheses of Proposition C.2 on the same $[0,T]$ and Ω, with a common positive interior buffer. Choose a real $f \in C_c^1(\mathbb{R}^{1+3})$ such that both f and $f^\Lambda = f \circ \Lambda^{-1}$ are supported in $(0,T) \times \Omega^\circ$. For $w = u, u^\Lambda$, let q_h^w solve the same action (17), with the same L, v_i, c, m, initialized by $w(0, x_i)$ and $\partial_t w(0, x_i)$. Define the spacetime-smeared readouts*

$$\mathcal{D}_h(f; q) = \int_0^T \sum_i v_i f(t, x_i) q_i(t)\, dt, \qquad \mathcal{D}(f; w) = \int_{\mathbb{R}^{1+3}} f w \, dt \, d^3x.$$

Using the preceding spatial derivative bounds $M_1(w), M_4(w)$, put

$$B_w = c^2 \sqrt{V} \left[\frac{15}{112} \epsilon^2 M_4(w) + 960 h \epsilon^{-2} M_1(w) \right],$$

and

$$\mathcal{E}_h(f, w) = \frac{\sqrt{V} B_w}{2} \int_0^T t^2 \|f(t, \cdot)\|_\infty \, dt + Vh \int_0^T \|\nabla_x (f(t, \cdot) w(t, \cdot))\|_\infty \, dt.$$

Then

$$\left| \mathcal{D}_h(f^\Lambda; q_h^{u^\Lambda}) - \mathcal{D}_h(f; q_h^u) \right| \le \mathcal{E}_h(f^\Lambda, u^\Lambda) + \mathcal{E}_h(f, u). \tag{18}$$

The right-hand side tends to zero for $\epsilon \to 0$ and $h/\epsilon^2 \to 0$. The conclusion is uniform over a bounded boost class when the common window, buffer and displayed derivative and detector bounds are uniform over that class.

Proof. Lorentz covariance of the Klein–Gordon equation makes u^Λ a solution with the same c, m. The preceding detector estimate applies at each time with $b = f(t, \cdot)$; its time integral is $|\mathcal{D}_h(f; q_h^w) - \mathcal{D}(f; w)| \le \mathcal{E}_h(f, w)$. Since $\det \Lambda = 1$ and both detector supports lie in the comparison window, change of variables gives $\mathcal{D}(f^\Lambda; u^\Lambda) = \mathcal{D}(f; u)$. The two approximation bounds and the triangle inequality prove the claim. □

The transformed preparations are supplied through the continuum reference on the transformed Cauchy surface; this comparison does not construct a boost map from one finite history or its initial data. The same estimate applies to smooth local preparation differences by linearity. Spacetime smearing is essential to the displayed comparison: a fixed-time detector instead transforms to a tilted hypersurface with its transformed measure. The result controls detector values for one common declared action, not exact finite-graph support, native source selection, or boosts of arbitrary finite records. Time integration and preparation errors, if present in an execution, require their own bounds.

C.2 Action measure, record production and geometric density

A supplied scalar response operator contains more information about its action measure than an unweighted event count. It does not identify that measure with physical volume. The distinction can be tested on the same source-addressed family, with its preparation, protected records, read law and model clock kept explicit.

Proposition C.4 (Identification of normalized action measure)**.** *Let A be a finite real operator whose nonzero off-diagonal support is connected. Suppose a positive diagonal matrix $M = \mathrm{diag}(m_i)$ makes MA symmetric. The equations*

$$m_i A_{ij} = m_j A_{ji}, \qquad \sum_i m_i = 1$$

determine m uniquely. If the two directed coefficients on every support edge are negative, a spanning tree constructs the candidate through $m_j/m_i = A_{ij}/A_{ji}$; the remaining edges test consistency.

Proof. For two positive solutions m, n, multiplying their edge equations and cancelling a nonzero coefficient gives $m_i n_j = m_j n_i$. Thus m_i/n_i is constant along every path. Connectivity makes the constant global, and normalization sets it to one. Tree products produce the candidate; positivity and every remaining edge equation certify its extension to the whole graph. Multiplying both M and the stiffness MA by one positive constant leaves A unchanged, so reciprocity cannot identify an absolute action scale. □

This reconstruction also has a finite conditioning bound. If each directed edge coefficient has relative error at most $\epsilon < 1$, each inferred edge ratio changes by a factor between r^{-1} and r, where $r = (1 + \epsilon)/(1 - \epsilon)$. For a tree of depth d, the root-relative weights change by factors in $[r^{-d}, r^d]$. Normalizing their sum gives factors in $[r^{-2d}, r^{2d}]$. This bounds the tree estimate; noisy cycle disagreement remains a model-consistency test.

On the authenticated $q = 5$ scalar family, the independent certificate recovers all 64 normalized weights from the directed operator alone, checks its 144 undirected edges, and compares the result with the separately reconstructed golden dual-cell quadrature. The tree has depth 9 and 81 remaining cycle edges. The parent mass vector is used for this comparison, not by the reconstruction algorithm. Operator and action selection remain supplied inputs.

A production control on the same finite family. Two preparations, with amplitude multipliers 1/2 and 1 relative to the retained source, give 2,944 independently replayed field writes. They have the same response operator and field ancestry; their initial energies differ by a factor of four. Local observer records read the actual writer, immutable version and value at the same site. Their extra read/write operations are charged separately and have no identified physical durations. The union of these local certificates is not a construction of remote record transport to one observer.

For an interior field-ancestry interval I, let $v_e = \tau m_{i(e)}$, where $\tau = \sqrt{5}/35$ is the retained action tick. Declare a production multiplier $r > 0$ and an integer exposure budget K. The finite local production rule

$$n_e = \lfloor K r v_e \rfloor, \qquad \widehat{V}_r(I) = \frac{1}{Kr} \sum_{e \in I} n_e$$

has the exact deterministic enclosure

$$0 \le \sum_{e \in I} v_e - \widehat{V}_r(I) < \frac{|I|}{Kr}. \tag{19}$$

The proof sums $0 \le K r v_e - n_e < 1$. Four controls cross the two source strengths with $r = 1, 2$ at $K = 4096$. All 324 local observer records are replayed. Changing r changes raw density while the field order and action measure stay fixed; dividing by the declared production law recovers the same finite measure within (19). Neither that production law nor the identification of τm_i with physical four-volume is derived by this test. No Poisson reference or old causal count-clock theorem is substituted for the actual field ancestry.

What a specified conformal action adds. Consider the separate control metric

$$g = u(x)\,(-dt^2 + dx^2 + dy^2 + dz^2), \qquad u > 0,$$

and the canonical massive scalar action in these coordinates. Its kinetic weight scales as u, whereas its metric four-volume scales as u^2. In the finite control, $M_i(u) = u_i M_i(1)$, each undirected gradient coefficient is multiplied by $(u_i + u_j)/2$, the Dirichlet boundary coefficient by its interior u_i, and the unit-mass potential coefficient by u_i^2. These terms define one symmetric quadratic action and reproduce the original operator at $u = 1$.

Reciprocity determines the relative profile u_i/u_j after division by the reference weights. The full operator supplies more: at an interior site with all six neighbours and the specified unit mass potential,

$$\sum_j A(u)_{ij} = u_i.$$

Gradient terms cancel in this row sum. It calibrates the common profile factor; all eight interior sites provide independent consistency checks. Thus no additional value of u is imported into this reconstruction. The unit scalar mass and the conformal action class are essential inputs. The controls $u = 1$ and $u = 1 + x/2$ recover both profiles exactly while retaining the same nonzero support and hence the same layered support order. Their normalized kinetic and four-volume measures differ for the nonconstant profile.

For an identified affine control $u = a + bx$, its specified continuum metric has scalar curvature

$$R = -6\Omega^{-3}\Box\Omega = \frac{3b^2}{2u^3}, \qquad \Omega = \sqrt{u},$$

with signature $(- + ++)$ and the displayed curvature convention. The nonzero control curvature is obtained through the supplied action and metric dictionary, not from an unweighted ordering statistic. No law makes either profile respond to the simulated source strength.

Scalar readouts do not determine tensor response. The same finite action also permits a direct comparison of directional responses. Write its potential as $V = G_x + G_y + G_z + V_m$, including every interior and Dirichlet edge, and consider the retained neighbouring configurations q_-, q. Their discrete action contribution is

$$L_d = T_d - \tau V, \qquad T_d = \frac{(q-q_-)^T M (q-q_-)}{2\tau}.$$

The potential uses the right endpoint q. Summing these contributions gives the recorded centred update through the discrete Euler–Lagrange equation. At fixed configurations, a stretch of coordinate axis k by s changes the contribution to

$$L_d(s) = sT_d - \tau\{s(V - G_k) + G_k/s\}, \qquad \left.\frac{dL_d}{d\log s}\right|_{s=1} = T_d - \tau(V - 2G_k).$$

An x/y permutation commutes with the complete cubic source operator. It preserves T_d, V, the action measure and the isotropic sum of these derivatives, but interchanges unequal directional responses. The certificate uses the two retained configurations and independently reassembles the stretched-grid action at $s = 2$ and $s = 1/2$; two thirds of their difference equals the displayed derivative exactly. Kinetic variation is included. These are finite action responses, with no identification of a physical stress source or gravitational coupling.

The connectedness and normalization theorem, common-scale ambiguities, finite production-error bound and conformal calibration algebra are checked in `SourceActionMeasure`; the source histories, exact controls and an independent verifier accompany it [29]. They identify a relative action measure and a profile within a specified conformal action class. They do not select the population, action, clock, production grammar, storage, source-to-metric update or physical tensor coupling. Those inputs are needed before the recovered quantity can be used as a sourced physical volume or an Einstein response.

C.3 Calibrated clocks and an undetermined light-deflection coefficient

The source action measure and a readable scalar clock do not by themselves select the spatial gravitational response. A constructive comparison uses the grounded massless stiffness $K = M(\mathsf{A} - I) > 0$ on the same prepared golden addresses, with two real fields u, v and an identical source j. The constitutive law, source preparation, update program and physical probe dictionary are declared. They are not consequences of A1–A3.

Proposition C.5 (Two-field clock nonidentifiability)**.** *For every $\kappa > 0$ and $\gamma \in \mathbb{R}$, let*

$$E_{\gamma,\kappa}(u, v; j) = \frac{1}{2} u^{\mathsf{T}} K u + \frac{\kappa}{2}(v - \gamma u)^{\mathsf{T}} K (v - \gamma u) - j^{\mathsf{T}} u. \tag{20}$$

Every positive symmetric two-field edge matrix whose clock Schur complement equals one has this form. The unique stationary fields satisfy $u = K^{-1} j$ and $v = \gamma u$. The entire clock susceptibility, Gaussian clock marginal at any fixed positive inverse temperature, and clock dynamics under the kinetic form $[\dot{u}^{\mathsf{T}} M \dot{u} + \kappa \dot{w}^{\mathsf{T}} M \dot{w}]/2$, where $w = v - \gamma u$, agree throughout this family for matching clock initial data.

Proof. For an edge matrix $S > 0$, set $\kappa = S_{22}$ and $\gamma = -S_{12}/S_{22}$. Its normalized Schur complement gives $S_{11} = 1 + \kappa\gamma^2$. The change to u, w has determinant one and separates two positive quadratic forms. Completing the square gives the common minimum $-j^{\mathsf{T}} K^{-1} j/2$ and a nonnegative excess energy. Stationarity, Gaussian integration and the separated kinetic equations then give the stated identities. □

The same separation works for any common C^2 clock energy $H(u;j)$ with a finite Gibbs partition. Coercivity alone does not imply that partition condition. In particular, the resting-source energy $H = u^{\mathsf{T}}Ku/2 + \sum_i j_i(e^{-u_i} - 1)$, $j_i \geq 0$, has positive Hessian, a unique minimum and a Gaussian lower bound ensuring integrability. Its stationary equation $Ku = je^{-u}$ therefore preserves the spatial ambiguity with the same proper-time coupling used for probe clocks.

Use one metric and one universal matter/Maxwell dictionary for every member:

$$ds^2 = -e^{-2u}c^2dt^2 + e^{2v}d\mathbf{x}^2, \qquad S_{\text{clock}} = -mc^2\int d\tau, \qquad S_{\text{photon}} = S_{\text{Maxwell}}[g]. \tag{21}$$

An identical local gap ΔE has coordinate phase rate $e^{-u}\Delta E/\hbar$, with $\Delta E = c^2\Delta m$ for a rest-mass gap. Static exchange gives $\nu_r/\nu_e = e^{u_r - u_e}$, independent of γ. The optical index is $n = e^{u+v}$. For a calibrated exterior field $u = \mu/r$, $\mu = GM/c^2$, the infinity-to-infinity first-order deflection is $2(1+\gamma)\mu/b$ for fixed finite γ in the weak-field limit $|u|, |\gamma u| \ll 1$. The member $\gamma = -1$ is conformal and has exactly zero bending, while $\gamma = 1$ has the Einstein weak-field coefficient. The linearized exterior tensor is

$$G_{ij} = (\partial_i\partial_j - \delta_{ij}\Delta)(u - v).$$

For a nonzero compact source, $G_{ij} = 0$ selects $\gamma = 1$ in this class. All members have zero linearized exterior scalar curvature and a divergence-free linearized Einstein tensor. Scalar vacuum and Ward balance therefore do not supply this selector. The exponential $\gamma = 1$ metric is not asserted to solve the full nonlinear vacuum Einstein equation.

Sixteen independently replayed local histories cover both source laws, strengths, schedules and transverse seeds. Actual final writers feed clock and optical-index reads. Exact residuals and a maximum-principle barrier enclose the stationary fields, and the conformal and Einstein optical intervals are disjoint in every history. A compact C^1 density with a C^3 potential gives a controlled continuum sequence. For rays whose actual closest approach lies outside the source and $0 \leq p = (1+\gamma)\mu/r_{\min} \leq 10^{-3}$, direct differentiation of the exact optical integral gives $|\theta(p) - 2p| \leq 11p^2$. Numerical quadrature supplies a separate check, not a certified quadrature enclosure or a finite-observer radio error budget.

The retrospective measurement comparison retains the Pound–Snider static redshift result and all four overlapping Fomalont solar-radio reductions, including their measurement conventions and statistical errors. The bounded redshift systematic is kept separate. These compressed published results are not independent repetitions or a likelihood fit of raw observations. The same-input general-relativistic reference is evaluated without a fitted per-channel normalization. The construction identifies a missing physical tensor condition; it supplies no selected constitutive law, source-to-stress identification, physical calibration or autonomous apparatus history.

C.4 Finite energy and entropy on one interacting action

The current field attachment uses the same supplied Weyl/contact CAR Hamiltonian as the bound-pair calculation, $H = d\Gamma[-(1/8)\sigma \cdot \sin k] + \sum_x n_{x0}n_{x1}$. On the open cube of 27 sites, its complete two-fermion space has dimension 1,431. Assign half of each Hermitian hopping bond to either endpoint and the contact energy to its site. Then

$$h_x = UD_x + \tfrac{1}{2}\sum_{b\ni x} H_b, \qquad \sum_x h_x = H, \qquad J_{x\to y} = i[h_x, h_y], \qquad \frac{dh_x}{dt} + \sum_y J_{x\to y} = 0.$$

Here t is Hamiltonian model time with the supplied convention $\hbar = 1$. Hermiticity and antisymmetry of the currents follow directly. Summing the identity over a region cancels all internal currents

and gives its complete outward flux. Overlapping bond stars produce range-two currents; omitting them fails conservation. Exact Gaussian-integer matrices verify all 351 site pairs and 27 local equations, and deleting U changes 54 currents. This is interaction energy accounting, not a number current renamed as stress. The bond-energy division is stated explicitly and is not unique.

On this full finite algebra, choose the faithful density $\tau = e^{-\beta H}/Z$ with declared $\beta = \log(25/8)$. For every trace-zero Hermitian variation δ,

$$DS_\tau(\delta) = \beta \operatorname{Tr}(\delta H), \qquad S(\rho) - S(\tau) = \beta \operatorname{Tr}((\rho - \tau)H) - D(\rho\|\tau).$$

These follow from $-\log\tau = \beta H + \log Z$; no commutation of δ with τ is needed. The retained test uses a localized contact-pair perturbation that does not commute with the reference. This is a concrete noncommutative attachment, not an invocation of the older finite-simplex entropy theorem on an incompatible domain. The full Gibbs reference is supplied, not prepared by native repair.

Signed CAR restriction to two adjacent sites keeps all local sectors $N = 0, 1, 2$ and gives an eleven-dimensional positive support. It does not make the state faithful on the full sixteen-dimensional local Fock space. For the restricted density, $K_A = -\log\tau_A$ has nonzero commutator with the bare internal H_A and fails even the best fit by $aI+bH_A+\mu N_A+\nu N_A^2$. Thus the global Gibbs identity is not silently transferred to a bare regional clock. This is a finite numerical countercontrol for the declared region and reference, not a universal regional modular obstruction.

An exact test strengthens the boundary for one specified regional algebra. On the fixed-N direct sum $\bigoplus_n(\mathcal{H}_A^n \otimes \mathcal{H}_{A^c}^{N-n})$, a faithful density admits a preserving unital completely positive retraction onto $\bigoplus_n(\mathcal{B}(\mathcal{H}_A^n) \otimes I)$ exactly when it has blocks $r_n \otimes \tau_n$, with $\operatorname{Tr}\tau_n = 1$. Multiplicative-domain identities remove cross-sector blocks and force this conditional product form. For a nonzero Gibbs parameter this requires block-additive H and hence $[H, N_A] = 0$. The actual adjacent-site region has 36 ordered crossing one-body entries of modulus $1/16$ in 54 modes, so $\|[N_A, H]\|_{\mathrm{HS}}^2 = (54 - 2)36/256 = 117/16$. No dense Gibbs logarithm is needed to exclude that exact retraction or its exact generator intertwining. This does not exclude partial traces, reduced modular clocks, approximate maps or different accessible domains. Where nested algebras do admit expectations preserving one faithful reference, uniqueness supplies their tower law. Independently selected regional references do not supply it.

A weaker observation interface does compose without factorization. For a CPTP state restriction T_{BS} and one faithful reference with $\rho_B = T_{BS}(\rho_S)$, define

$$E_{BS}(X) = \rho_B^{-1/2} T_{BS}(\rho_S^{1/2} X \rho_S^{1/2}) \rho_B^{-1/2}.$$

Its explicit Kraus matrices make it unital and completely positive, and $\operatorname{Tr}(\rho_B E_{BS} X) = \operatorname{Tr}(\rho_S X)$. For nested restrictions and their same reference reductions, cancellation of the intermediate square roots gives $E_{AB}E_{BS} = E_{AS}$. This finite Petz-transpose construction preserves complete POVMs and arbitrary references. It generally neither fixes the embedded local algebra nor intertwines modular flows; its reference-dependent effects are not thereby implemented native local detectors. It supplies a compatible observation tower while leaving those stronger clock and access requirements explicit.

These constructions supply exact finite conservation and a quantum entropy map to the active premise ledger. They do not inhabit the Einstein premise bundle. Needed are a common symmetric Lorentz stress tensor, its geometric Ward identities and null charges, modular diamonds and their edge split, fixed-proper-volume area variation, Bianchi/geometry data and a controlled common refinement with physical normalization. In particular, finite Gibbs stationarity is not generalized-entropy stationarity at fixed proper volume. The complete source records and proof boundaries are

retained in `model/field_stress` and `model/field_regions`; their finite attachments close only the obligations on those declared algebras.

C.5 Entropy stationarity from a continuing native record

The native population interface admits an entropy calculation after record-adaptive repairs, without assuming an unread stationary continuation. Fix the binary total-two carrier and a faithful reference with weight w_q per configuration of initial core charge $q = n_0 + n_1$. The core pair observable is $c = n_0 n_1$. Let b be a complete native tape with bounded stopping length T_b, and let d_b count its compatible endpoints. Native branch inversion identifies each endpoint with an initial state. Assume the tape resolves q, and that any retained flag has probability $\beta > 0$ independent of the initial state.

Proposition C.6 (Counted native-record entropy envelope)**.** *For an initial population p, let p_B be its tape law conditional on that flag. Define*

$$F(p) = H(p_B) + \mathbb{E}_p \log d_B - \mathbb{E}_p T_B \log 2 - \log \beta.$$

Writing σ for the reference population, one has

$$F(p) - S(\sigma) = h_1(\langle q\rangle_p - \langle q\rangle_\sigma) + h_2(\langle c\rangle_p - \langle c\rangle_\sigma) - D(p_B \| \sigma_B),$$

where $h_1 = \log(w_0/w_1)$ and $h_2 = \log(w_1^2/(w_0 w_2))$. Hence the fixed-moment first variation vanishes at σ, and its Hessian is the negative semidefinite Fisher form of the tape response.

Proof. Injectivity and the retained fair coins give $\sigma_B(b) = d_b w_{q(b)} 2^{-T_b}/\beta$. Substituting this law in $D(p_B\|\sigma_B)$ leaves $-\sum_b p_B(b) \log w_{q(b)}$. State-independent flag probability makes its initial-state marginal p. Finally $-\log w_q = -\log w_0 + h_1 q + h_2 c$. The second variation of relative entropy is the Fisher form; directions invisible to this record map can remain flat. □

The charge-resolution hypothesis has a native readout. For E, the contrasts on seams $(0, 1)$ and $(0, 5)$ determine the three binary occupations at the start of that census: a common all-one offset is impossible at total load two. Retained prefix currents then recover the initial core, including after crossing repairs. No success conditioning is needed. For C, waiting for both coin values on each census seam gives a bounded service with state-independent success; all timeouts stay in the complete instrument. Unresolved timeout fibres have counted-entropy excess bounded by $(1 - \beta) \log(w_{\max}/w_{\min})$. The expected coin cost must vary with p when stopping depends on records. Cap completions are marginals of the same conditional endpoint measure, so their shared predictions agree without independently resetting the source.

This is a population-record entropy theorem. It identifies neither the completion count nor the coin-cost subtraction with gravitational area, nor the core moments with physical volume or stress. Its stationarity is a fixed-moment record stationarity, not the fixed-proper-volume generalized-entropy variation of (G4); joining the two requires an area, volume and stress attachment on the same process.

C.6 Finite modular covariance and compressed responses

Geometric modular covariance requires a comparison beyond the existence of a finite modular clock. The following finite identity states a necessary boundary of that comparison, independently of the repair scheduler.

Proposition C.7 (Finite modular dilation and its compression boundary)**.** *Let G, P be Hermitian matrices on a finite Hilbert space, let $U(s) = e^{isG}$, and let $\kappa \in \mathbb{R}$. Then*

$$\|i[G,P] - \kappa P\|_{\mathrm{HS}}^2 = \|[G,P]\|_{\mathrm{HS}}^2 + \kappa^2 \|P\|_{\mathrm{HS}}^2, \qquad \|U(s)PU(s)^* - e^{\kappa s}P\| \geq |e^{\kappa s} - 1|\,\|P\|.$$

For an orthogonal projection Q, write $G_Q = QGQ$, $P_Q = QPQ$ and $B_Q = i(QG(1-Q)PQ - QP(1-Q)GQ)$. Then

$$Q\,i[G,P]Q = i[G_Q, P_Q] + B_Q, \qquad \|Q(i[G,P] - \kappa P)Q\|_{\mathrm{HS}} + \|B_Q\|_{\mathrm{HS}} \geq |\kappa|\,\|P_Q\|_{\mathrm{HS}}.$$

Proof. Cyclicity gives $\mathrm{Tr}(P\,i[G,P]) = 0$, so the cross term in the squared Hilbert–Schmidt norm vanishes. Unitary invariance of the operator norm and the reverse triangle inequality give the finite-parameter bound. Inserting $Q + (1-Q)$ into each product gives the compressed identity. Apply the first identity to the compressed matrices, then the triangle inequality, for the last bound. □

For the native faithful reference with per-configuration core weights $(2/225, 1/36, 2/45)$, take $G = \log\sigma$; thus s is the dimensionless modular parameter. Choose configurations a, b with core occupation zero and two, respectively. On their two-state corner declare

$$P = \begin{pmatrix} 1 & i/2 \\ -i/2 & 1 \end{pmatrix}, \qquad Q = |\psi\rangle\langle\psi|, \qquad |\psi\rangle = (|a\rangle + |b\rangle)/\sqrt{2},$$

and extend P by zero. It is positive, with nonzero eigenvalues $1/2, 3/2$. For $\kappa = \log(5)/2$, the projected first derivative is exactly κP_Q, supplied entirely by B_Q; the internal compressed commutator is zero. Nevertheless

$$\langle\psi|U(s)PU(s)^*|\psi\rangle = 1 + \tfrac{1}{2}\sin(s\log 5) \neq e^{s\log(5)/2}$$

in general. At $s = 1/\log 5$ the two values are approximately 1.420735 and 1.648721. The declared probe is not identified with physical null momentum.

Consequently a fixed finite Hilbert algebra cannot supply nonzero global modular dilation. A spectral projection of G has $B_Q = 0$ and retains the same obstruction. Restricted continuum comparisons must specify states, observable domains and finite parameter windows, retain the compression boundary, and control the actual responses on one refining family. Unbounded continuum generators are not excluded by these bounded-operator statements.

D Local clocks, record gluing and information capacity

The proposed record-gluing law RG (Section B.6) connects response directions to classical record transport. It has distinct information, geometry and implementation premises. A simpler sufficient transport principle replaces exact dilation covariance and continuous rotation symmetry by a regular local clock and time-faithful refinement. This is a proposed physical extension, not a fourth adopted core axiom or a property certified for the simulator. Its target is the operational causal/count limit; it does not uniquely select a finite Fibonacci population or a microscopic read radius.

The local-clock interface. Let E be the three-dimensional simple response ideal with its positive trace metric. Its attachment to operational positions is supplied. Assume:

1. Independent classical bits, clean auxiliary registers, retained versions, preparation, discard, waiting and NOT/Toffoli gates admit arbitrary finite composition and parallel realization. All auxiliary work is counted. Local processing has zero ideal duration or arbitrarily small positive overhead for each complete finite program.

2. Every sufficiently small displacement v has a retained-record primitive of duration $\ell(v)$, independent of origin and covariant under carrier rotations, with $\ell(0) = 0$ and

$$\ell(v)^2 = Q(v) + o(\|v\|^2), \qquad Q > 0.$$

The expansion is uniform in direction. Local directional coverage is a premise; finitely many rotated directions do not imply it.

3. Every native dependency, including control, timing and metadata, admits refinement into small primitives and waits with the same endpoints and nonnegative elapsed time converging to the original time. Local operations and waits have zero displacement. A local primitive costs at least $\ell(v)$. Finite programs have exact value-preserving realizations with arbitrarily small whole-program overhead. Local interventions at every designated intermediate output before publication are preserved while all other existing records and caches are fixed.

These are LC1–LC3. They constrain all admitted influences, including those outside a chosen observation program.

Finite symmetry and the limiting cone. The quadratic form inherits the latency symmetry by dividing $\ell(sRv)^2 = \ell(sv)^2$ by s^2 and taking $s \to 0$. In an orthonormal icosahedral frame, invariance of a symmetric quadratic form under $\mathrm{diag}(1,-1,-1)$ and $\mathrm{diag}(-1,1,-1)$ makes its off-diagonal entries zero. The cyclic coordinate permutation equates its diagonal entries. These proper rotations preserve the twelve ports and generate a tetrahedral subgroup. Thus $Q(v) = \tau^2\|v\|^2$ for $\tau > 0$, and uniformly $\ell(v) = \tau\|v\| + o(\|v\|)$.

For any native path of elapsed time T and displacement v, refine until every step satisfies the local lower bound. For any $0 < \epsilon < \tau$,

$$T_n \geq \sum_j \ell(v_j) \geq (\tau - \epsilon) \sum_j \|v_j\| \geq (\tau - \epsilon)\|v\|.$$

Time-faithful refinement and then $\epsilon \to 0$ give $T \geq \tau\|v\|$. Conversely, N equal steps have total time $N\ell(v/N) \to \tau\|v\|$. Finite realization with a sufficiently small overhead meets every strict deadline $T > \tau\|v\|$; waiting fills the remaining time. The resulting access cone has speed $c = 1/\tau$. Exact null access is not asserted.

The example $\ell(v) = \tau r + \kappa r^2$, $r = \|v\|$, $\kappa > 0$, has subdivision time $\tau r + \kappa r^2/N$. Subdivision followed by waiting preserves an old primitive's duration, yet no nonzero finite path attains the null boundary. It violates exact dilation covariance while satisfying the local-clock principle. This separates LC from exact RG geometry; an approximate implementation of RG can have the same limiting behavior. The icosahedrally invariant polyhedral norm $\max_{u \in V} |u \cdot v|$ shows why finite symmetry without quadratic regularity is insufficient. A fast coarse shortcut preserves local Taylor data but violates time-faithful refinement. Both premises bear physical content.

For q^3 distinguished records in a window of side L, spacing L/q, and layers $\delta_q \to 0$ with $q\delta_q \to \infty$, finite independent programming realizes inner read menus $a_q = (1 - \eta_q)c\delta_q$, $\eta_q \to 0$, with sufficient deadline slack. The source-net covering theorem gives inner and outer radii $k(a_q - 2h_q)$ and ka_q, where $h_q = \sqrt{3}L/q$. Localized intermediate interventions place consumed-writer paths in complete native ancestry; the clock theorem excludes native spacelike shortcuts. The orders converge away from the null boundary. Cell assignment supplies count-volume weight $(L/q)^3\delta_q$ for the distinguished records. The M1 choice $\delta_q = L/(c\sqrt{q})$ is one regulator; auxiliary work has a separate bill.

Classical compilation and finite algebra coherence. Every Boolean map has a clean reversible implementation: toggle an output for each enabled minterm, compute its multi-input conjunction using a Toffoli ladder, and reverse the ladder. NOT gates implement negative controls. CNOT is a derived Toffoli gate with a constant control; its clean control register, preparation and reset are counted. The input is retained, the desired value is XORed into the output, and all auxiliary bits return to zero. This construction is exponential in general. It proves finite programmability, not an efficiency or energy bound.

On a finite graph take private factors $\mathbb{C}_i^{12} \otimes Q_i$ and shared edge buffers $B_e = M_{12}$. The carrier algebra consisting of its private factors and incident buffers has center exactly $\mathbb{C}_i^{12}$. The channel with Kraus operators

$$K_{p,b} = e_p \otimes |p\rangle\langle b|, \qquad \sum_{p,b} K_{p,b}^\dagger K_{p,b} = I,$$

retains central port p and copies it into a quantum buffer's fixed basis. Reading that basis and preparing a target port completes a classical copy; unknown quantum states are not cloned. Off-diagonal buffer effects remain usable measurements. Private response actions commute with these copying maps. Buffer access, controller transport and readback receive their actual local-clock grades.

An explicitly included joint observer and its marginal constraints provide a nonempty finite state family. Choose the local references as restrictions of the same faithful joint tracial state. These compatible references and an injective positive-weight entropy cover give the joint tracial state as the unique unconstrained minimum. Independent preparations are interventions or constraints, not that unconstrained minimum's selected records. Adding blank factors and lifting old channels gives value-level refinement. This is a finite consistency witness. It does not furnish the all-level source-bound support, degree-one bridge, operational deletion tests or proof that the private response exhausts all admitted reversible source responses. A dormant support added to a circuit would not meet those obligations. Compatibility matters: on a shared classical bit with two equal-weight references $(1/2, 1/2)$ and $(3/4, 1/4)$, both faithful and tracial, the summed entropy derivative at probability $1/2$ is $-\log 3$, so the joint reference does not minimize that incompatible-reference problem.

There is also a finite process-to-state connection. For g gates on R bits, n of them independent input bits, let $D = R(g+1)$ and admit all 2^D ambient histories. Initial blanks and actual gate equations define the feasible family by linear indicator constraints. Each input x has exactly one feasible history $h(x)$. A faithful input prior $p(x)$ with fair references on all other history coordinates gives $Q(h(x)) = p(x)2^{-(D-n)}$. Hence every feasible P satisfies

$$D(P\|Q) = D(P_{\text{input}}\|p) + (D-n)\log 2.$$

With the full joint history observer as the injective scored cover, the unique minimum is $P_*(h(x)) = p(x)$, exactly the executed circuit ensemble. Feasibility uses actual transitions, not success at the target truth table: a defective program retains its defective histories with positive weight. This is an explicitly classical finite score problem. Additional scored marginals, the program and its physical clock are not selected by this identity. The native all-level source obligations remain separate.

Independent motivation and its boundary. Reversible classical computation motivates cleaning auxiliary work after copying an answer. Its relation to a transport clock is an extra premise. For a tracial matrix reference $\rho_0 = I/d$ and injective traceless Hermitian linear perturbations $X(v)$, direct expansion gives

$$D(I/d + X(v)\|I/d) = \frac{d}{2}\operatorname{Tr} X(v)^2 + O(\|v\|^3).$$

An equivariant perturbation makes this positive susceptibility isotropic by the same finite-symmetry argument. Identifying $\ell(v)^2$ with a positive multiple of this entropy cost to second order would supply a regular clock, but A3 supplies neither that calibration nor its coefficient. Equal-radius latency ratios and coarse/fine timing comparisons test this proposal without an M1 fit. Entropy curvature alone is not evidence for a physical clock law.

Information and resource consequences. If a receiver decodes m independently intervenable d-ary records from a finite transcript alphabet $\mathcal{T}$, with all caches fixed, the transcript map is injective. Therefore $|\mathcal{T}| \geq d^m$. Payload, timing, control and metadata all count. Additive accumulation cannot decode arbitrary tuples; preexisting correlations do not evade fresh independent interventions.

For $q = n^2$ and $n \geq 2$, put $m = \lfloor n/2 \rfloor$. In the M1 grid, all offsets in $[-m, m]^3$ lie strictly within the radius $L/\sqrt{q}$. At least $(n^2 - 2m)^3$ interior receivers each consume $(2m+1)^3$ records. Thus there are at least $n^9/8 = q^{9/2}/8$ read incidences per layer and $n^{10}/8 = q^5/8$ in a horizon L/c. Each interior receiver needs at least $(2m+1)^3 - 1 \geq n^3/2$ incoming binary transcript bits per layer under fresh independent interventions, hence bandwidth at least $cq^2/(2L)$. This excludes its own locally available record from the communication cut. Multicast can share transport paths; the aggregate statement counts output incidences rather than unicast packets. For inner menus, the same cube is included when $\eta_q < 1 - \sqrt{3}/2$, giving the asymptotic bounds; the all-n constants above concern the full-radius experiment. The rate bound presumes independent new interventions with prior caches fixed. It does not infer a fresh entropy rate from an unperturbed correlated history.

Cofinal workspace is consequently substantive. A fixed per-receiver bandwidth cannot support literal dense lossless observation at all refinements under these conditions. A single central $\mathbb{C}^{12}$ record has only twelve alternatives; an unbounded lossless tuple therefore requires a growing composite observer/register bank. The logical receiver cannot be identified with one unchanged primitive carrier. The same transport law also permits identity, parity and other local updates, so it selects no matter equation, energy law or Lorentz covariance of dynamics. Its consequences beyond M1 are communication, refinement timing and information-capacity constraints.

The complete analytic argument and controls are in `code/rg_principle/DERIVATION.md`. The compact receipt independently verifies the full named catalog and replays 263 finite circuits, including 7,880 localized interventions and 40 paired intermediate-writer comparisons, with every reference gate and work register counted. Coordinated circuit/table substitutions and cached-writer substitutions fail verification. Exact algebra controls cover all 1,728 basis units of the copying channel. Twelve kernel-checked results and a transitive standard-axiom audit are in `Lean/Geometry/RecordGluingPrincipleAxiomAudit.lean`. Finite controls do not replace the analytic all-size arguments or certify a native source realization.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Affine Lorentz atlases, Poincare cocycles, celestial null directions, observer hyperboloids, and Lorentz-orthogonal rest spaces [180, 188, 185].
- Axiomatic reconstruction programs (AQFT, causal sets, holography) [145, 1, 106].
- Bisio-D Ariano-Perinotti QCA [97].

- Borchers-Wiesbrock theory, Araki-Zsido, Guido-Longo-Wiesbrock, Wiesbrock modular intersections, HJPW Markov structure [103, 216, 92, 144, 217, 151].
- Cauchy-Binet and total nonnegativity [157, 133].
- Causal-set faithful embeddings and order-plus-number reconstruction, causal-conformal reconstruction, and Lorentzian limit diagnostics [1, 14, 15, 2, 7, 5].
- Charge conservation and Gauss-law preservation for discrete Maxwell schemes on cell complexes [212, 105, 218].
- Classical store-and-forward channels and finite dataflow refinement [110, 95].
- Classical store-and-forward networks and dataflow refinement [110, 95].
- coercive finite differences [207, 131].
- Combinatorial surface classification, Rado theorem, Mobius cross-ratio geometry, and modular GNS theory [136, 190, 177, 93, 209].
- Consistent positive finite differences and CFL stability [179, 169, 119, 120].
- Deterministic floor discrepancy bounds and conformal scalar action identities [167, 213].
- Discrete Poynting theorems on cell complexes [208, 105].
- Disjoint projection monoids, finite observation kernels and quantum phase sensitivity [149, 100, 109].
- Doob conditioning and relative entropy [127, 168].
- Dual-rail arithmetic and finite-precision error propagation [202, 154].
- Dual-rail arithmetic, register lifetime allocation and finite-precision numerical composition [202, 189, 113, 154].
- Exact classical readback/restoration and interval arithmetic [178].
- Explicit models of axiom systems and satisfiability witnesses for typed premise bundles [155].
- Fibonacci Diophantine approximation and deterministic equal-cell quadrature [150, 182].
- finite conditioned KL and Dyck path combinatorics [168, 121, 203, 204].
- finite Fourier and graph spectral analysis [210, 115].
- Finite graph invariants and noninjective observation maps [140].
- Finite linear observability and canonical Weyl algebras [161, 135].
- finite linear observability and word counting [161, 173].
- Finite linear recurrence stability and Duhamel error bounds [128, 187].
- Finite null tomography and Ward/Bianchi Einstein-form tensor identities [213, 175].
- Finite operational channel theory [214, 152].
- finite propagation for Klein-Gordon fields [130, 206].
- Finite quantum information projections [37, 214].
- Finite symmetry [197].
- Finite-graph divergence theorems and discrete radial-shell bookkeeping [141, 172].
- free Gaussian fields [139, 124].
- Gaussian oscillator and polynomial preservation identities [135, 116].

- Gaussian oscillator covariance [215, 196].
- Geometric spanners and causal-set order/count limits [181, 1, 2].
- Graph subdivision diameter bounds and register lifetime allocation [125, 189].
- information capacity [198, 121].
- Inverse problems on hyperbolic space, finite frames, residual minimization [153, 114, 129].
- Jacobson 1995, Jacobson 2016, BW modular theory [159, 160, 98, 99].
- Jacobson 1995/2016 small-ball derivations [159, 160].
- Jacobson Clausius bookkeeping, entanglement first laws [159, 101].
- Kraus instruments, Weyl channels and GHZ parity measurements [166, 156, 142, 102].
- lattice quadrature [123, 200].
- Lattice surface tensions and ball-cap crossing integrals [126, 195].
- Linear observability under time-dependent averaging and triangular inversion [161, 201].
- Longest-chain bounds in layered orders, the sum-of-cubes identity and integration of causal-diamond volume [174, 4, 138].
- Lorentz boost identification of rest spaces and the inverse-square shell law [180, 158].
- Marsden and West, Discrete mechanics and variational integrators (2001) [38].
- Mlodinow-Brun quantum-walk Dirac limits [176].
- Modular Hamiltonians of Gibbs states and the Clausius form of the entanglement first law [146, 159, 101].
- Modular temperature and KMS parameter rescaling in finite quantum statistical mechanics [117, 146, 108].
- Nonlocal BV perimeter theory [122, 91].
- Nzongani et al. tetrahedral walks [184].
- Order embeddings of finite causal sets into Minkowski causal cones [220, 1].
- Parameter identifiability and thermodynamic work derivatives [194, 111].
- Partial-information decision rules and absorbing finite random walks [132, 164].
- Positive diagonal symmetrization and edge-ratio cycle consistency [165, 163].
- Primary mathematical and physical sources [160, 183].
- Projective null-cone geometry, PSL(2,C) Lorentz isomorphism, hyperboloid H3 model [188, 192].
- Quantum Gibbs-state mixing, conditional mutual information, approximate quantum Markov chains, Fawzi-Renner recovery [134, 162, 107, 171].
- Ranard-Walter-Witteveen strict causality [191].
- Renes et al. tetrahedral SIC [193].
- reversible classical computation [94, 137].
- Scalar field stress tensors, harmonic Bose ensembles and spectral numerical error bounds [170, 186, 148].
- Sparse geometric graphs [181].

- Spectral graph diffusion, incidence homology, root lattices, finite-group isometries, and metric completion of dense arithmetic modules [115, 96, 118].
- standard Gram inversion and perturbation estimates [157, 205].
- Stress tensor tomography, modular Hamiltonian densities in QFT [112, 99, 199].
- Symplectic Verlet spectral error analysis [211, 147].
- Textbook Gauss-law inverse-square arguments [158, 143].
- Watrous Section 5.2 [214].
- Whitney finite elements, Maxwell stress and classical canonical Gaussian ensembles [104, 158, 186].
- Yoshida higher-order symplectic composition [219].

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